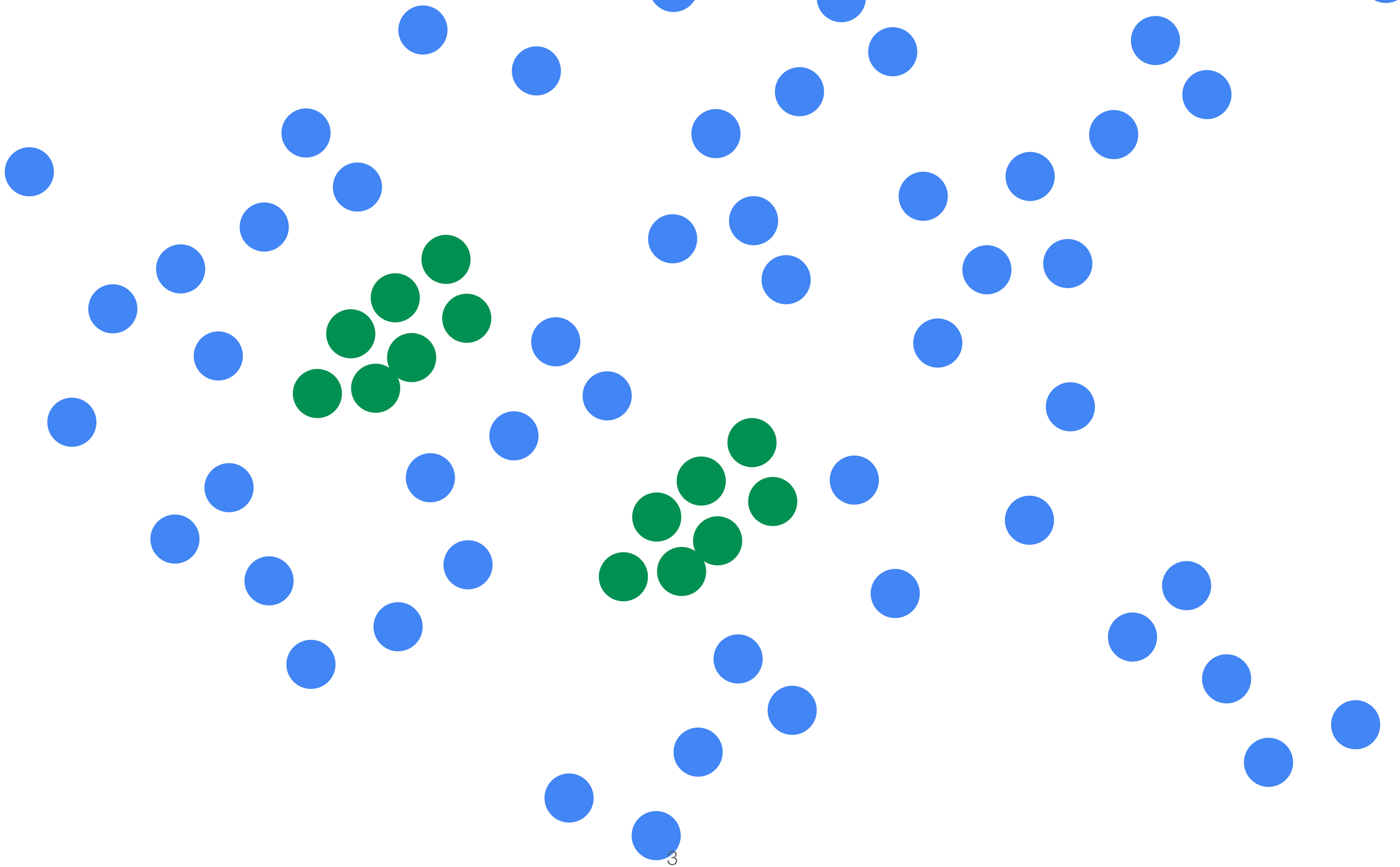
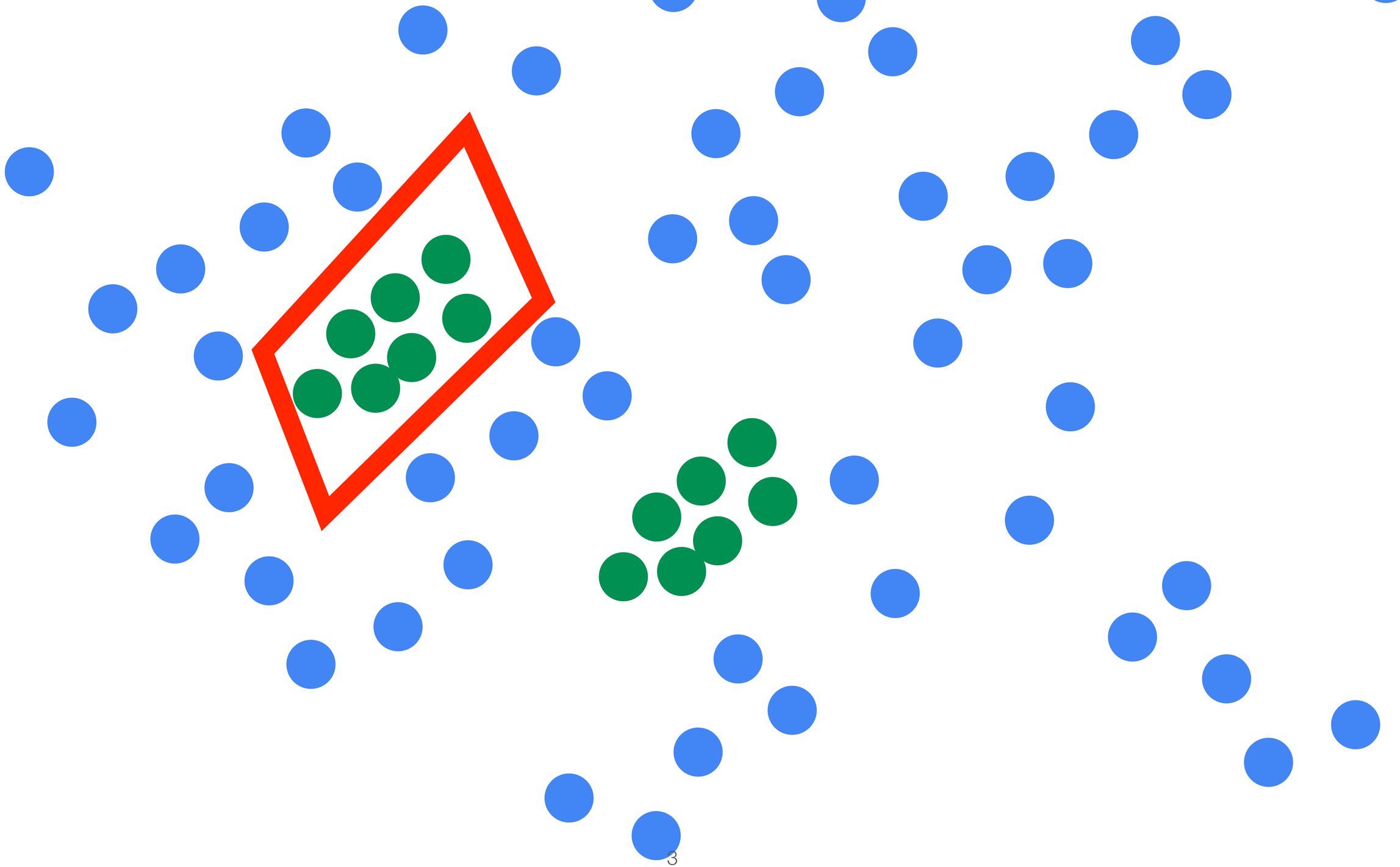
Introduction to Deep Learning

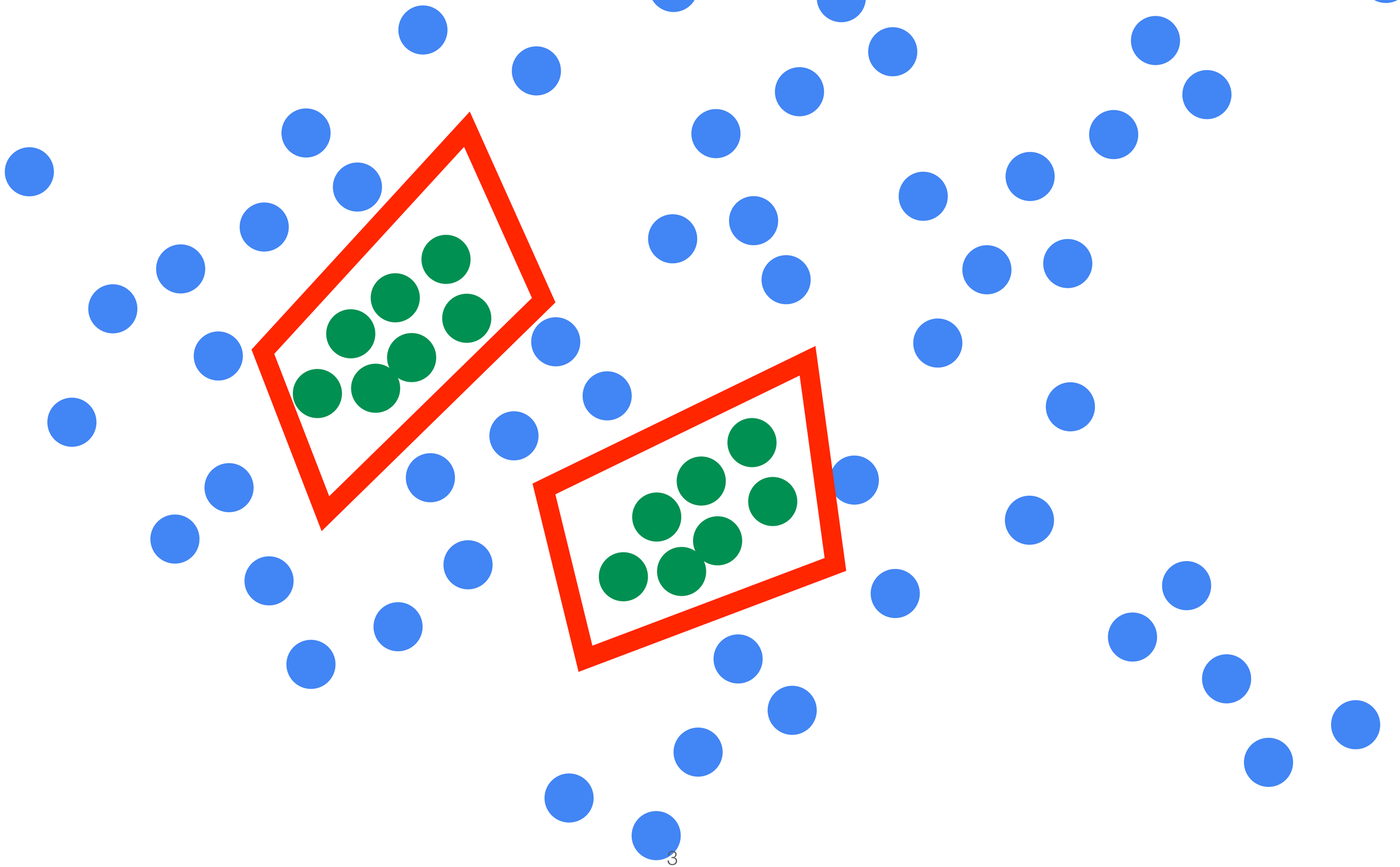
CS/EE/CMS 148a - Winter 2026

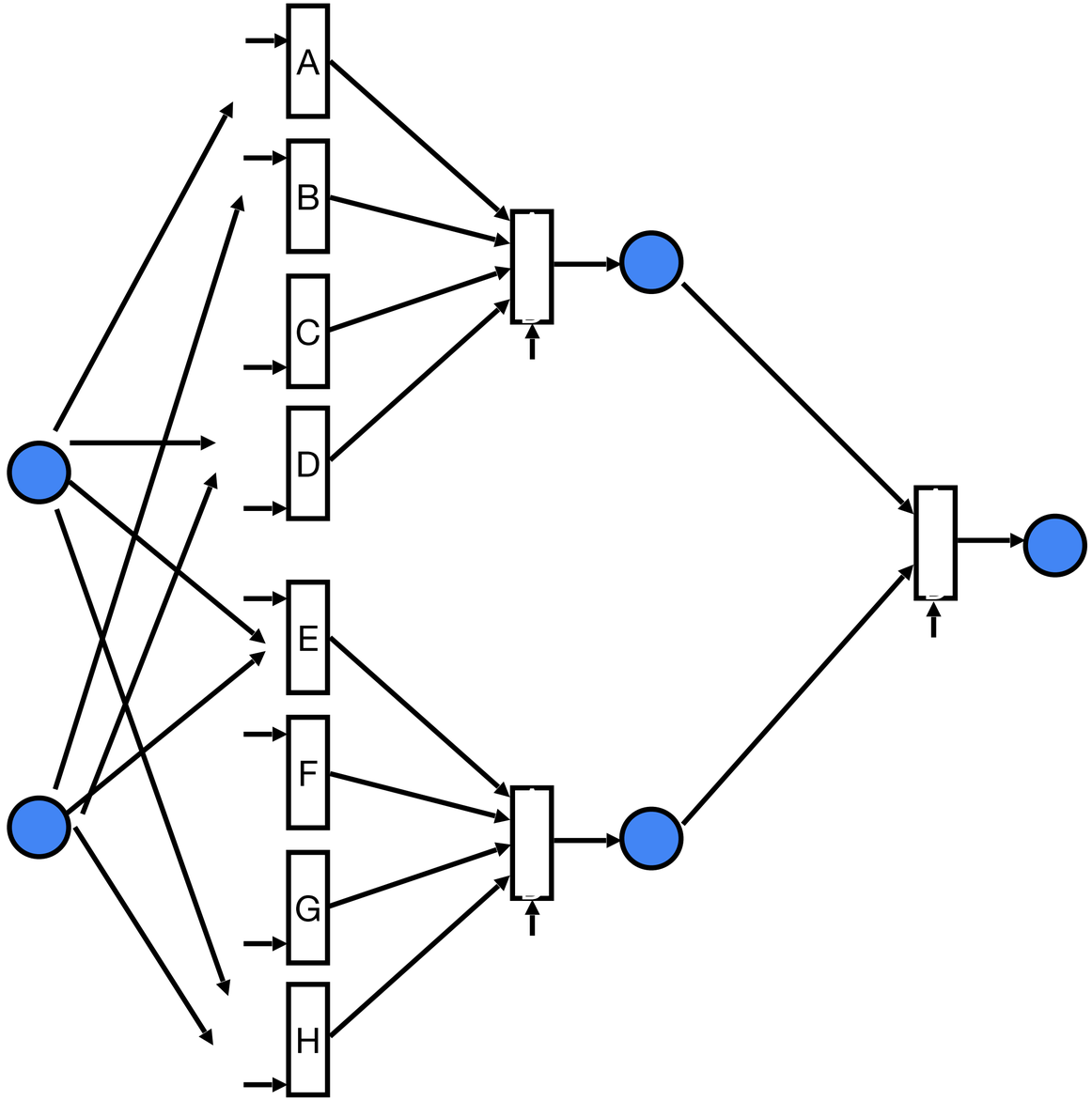
Lecture 7 - Convolutional Neural Networks (CNNs)

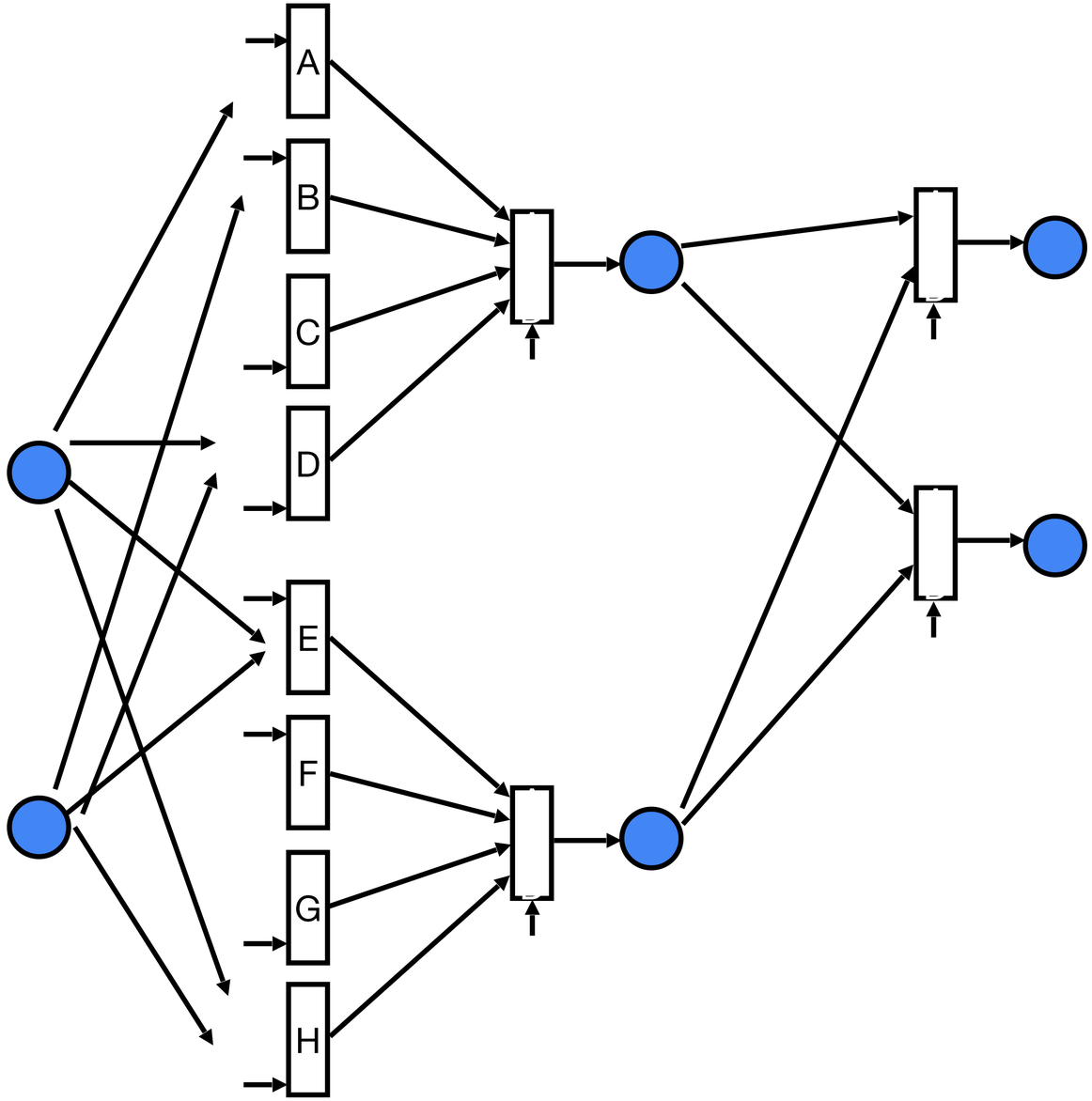
Perceptrons and deep neural networks recap








A
B
C
D
E
F
G
H


A
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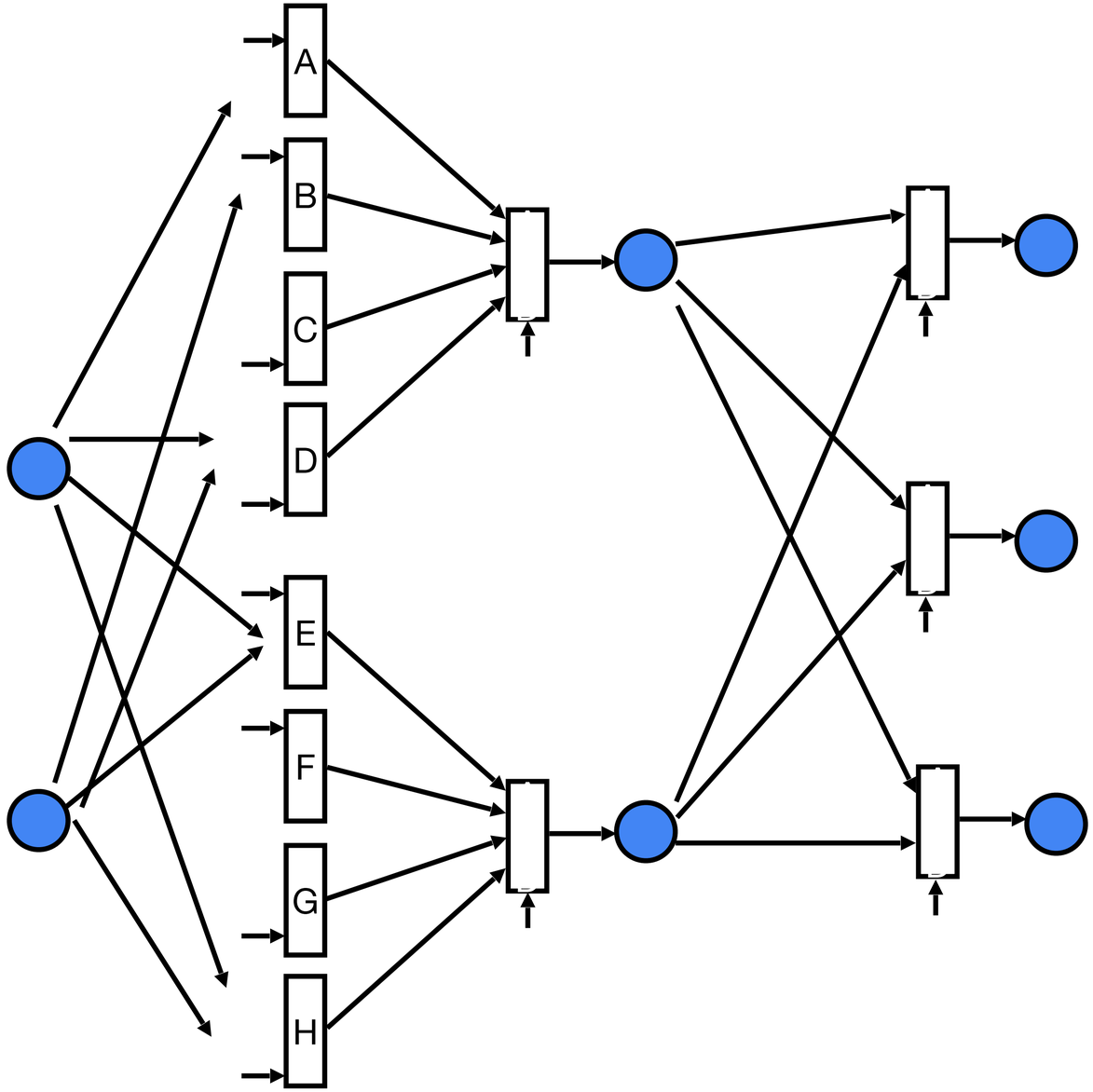
A
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C
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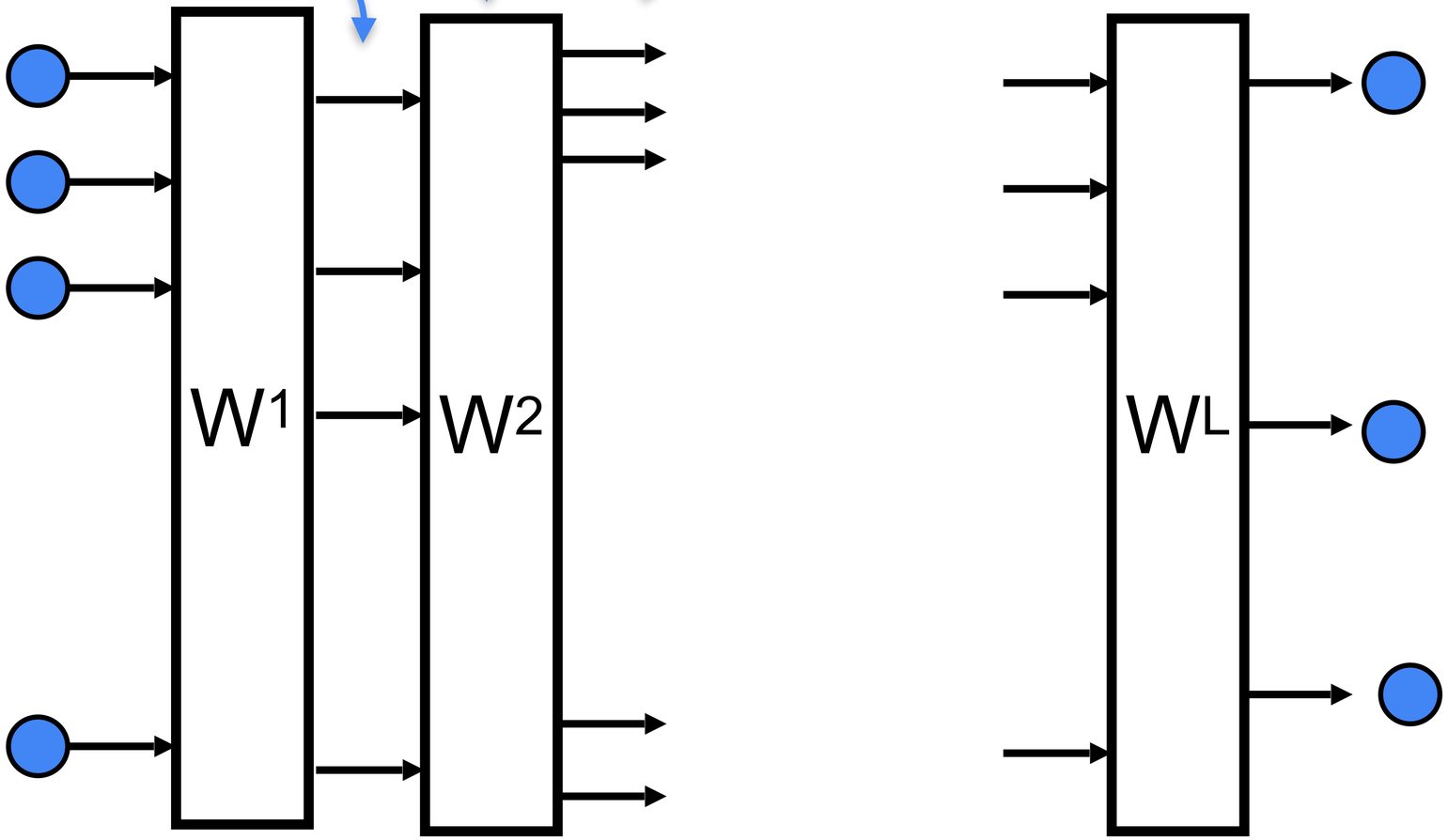
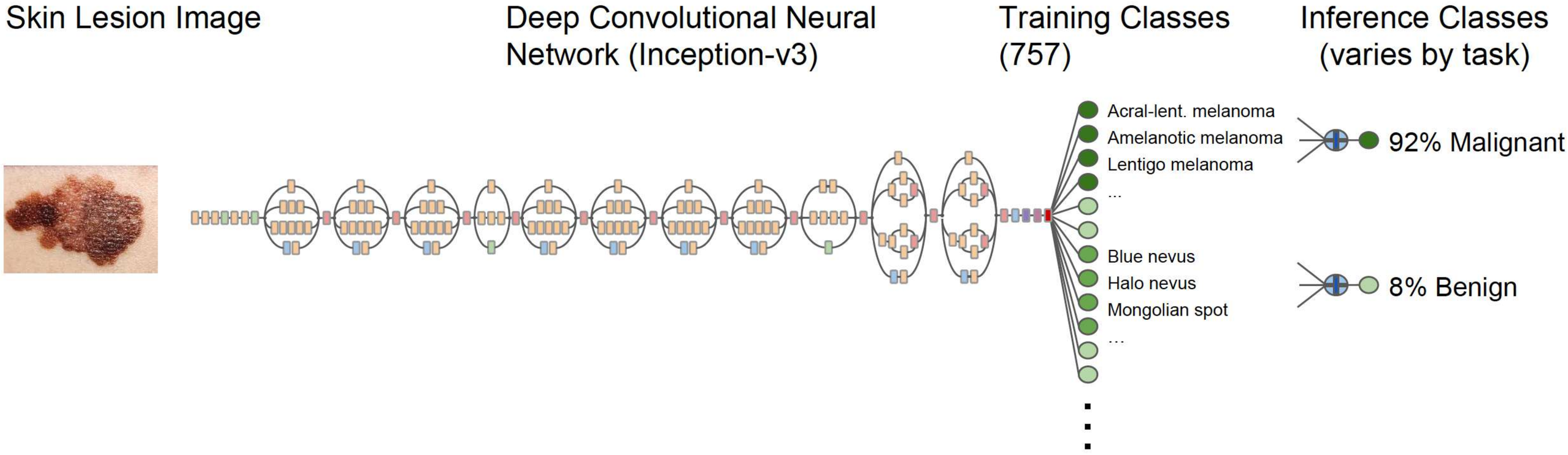
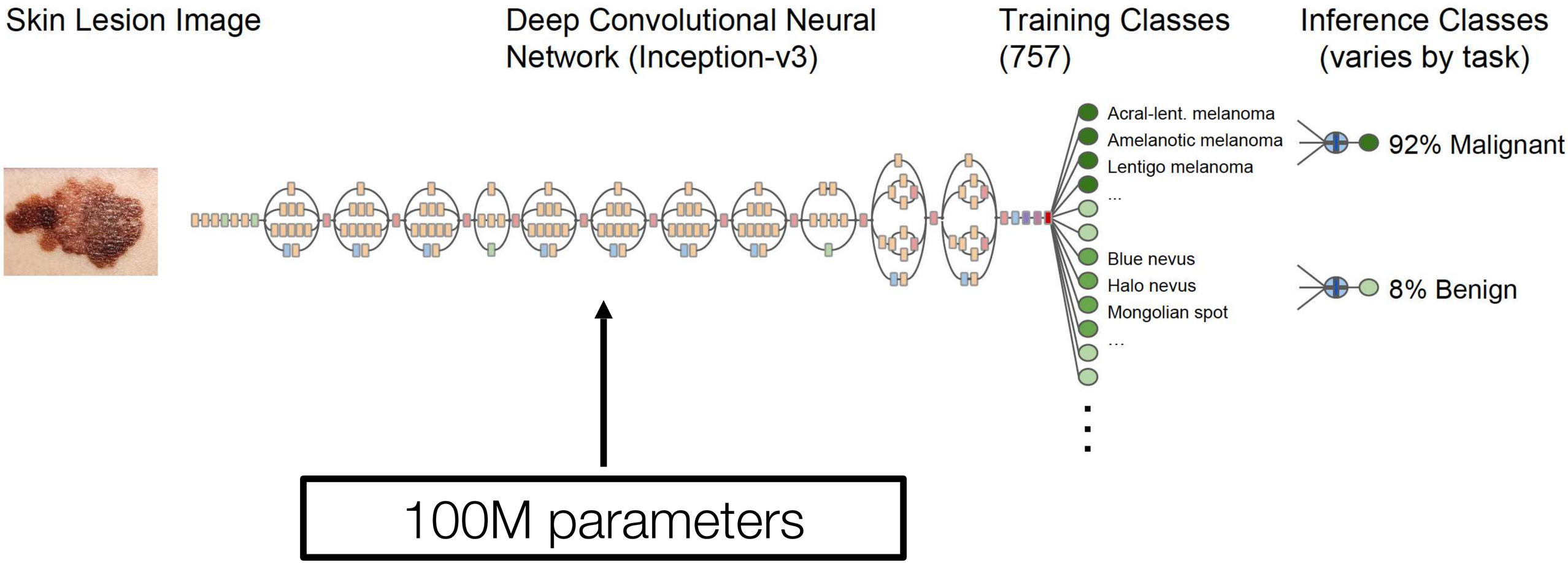
$$x^{(l-1)} \rightarrow W^{(l)} x^{(l-1)} = s^{(l)} \rightarrow h(s^{(l)}) = x^{(l)}$$



x^0 x^1 x^2 x^L



[Esteva et al. 2017]

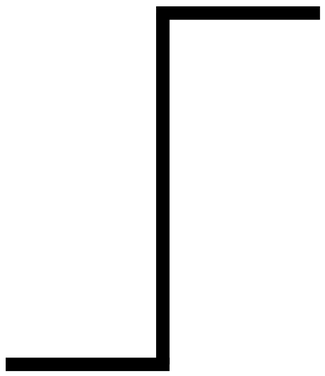


[Esteva et al. 2017]

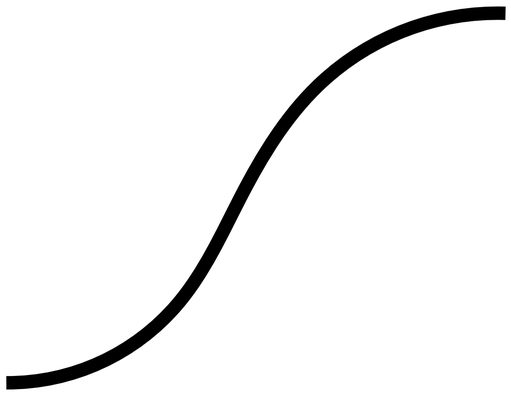
Questions

- Nonlinearity g(.)
- N. of parameters in W
- Architecture (deep, shallow)
- Learning —> stochastic gradient descent, backpropagation ✔

Favorite nonlinearities g(·)

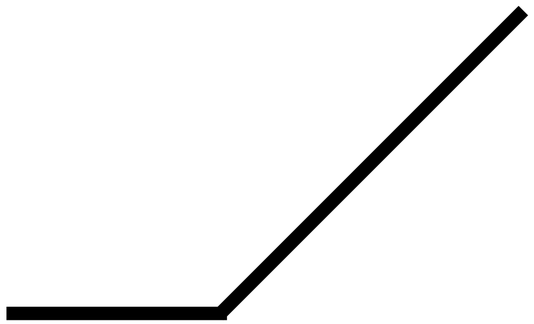


step



sigmoid

- tanh
- logistic



relu

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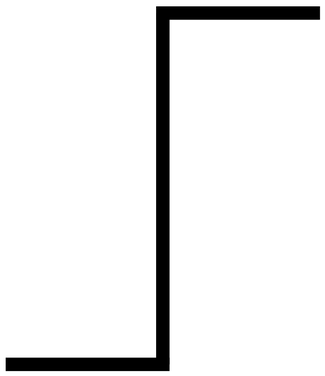

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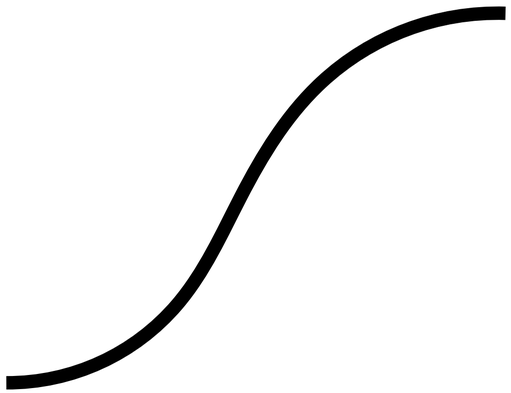

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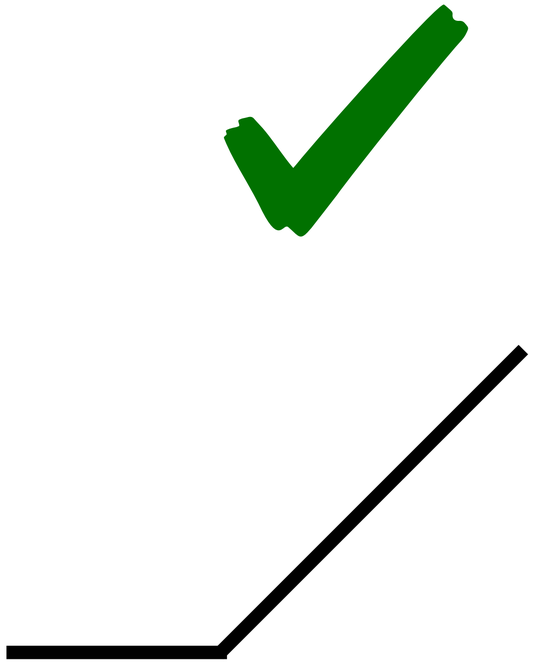
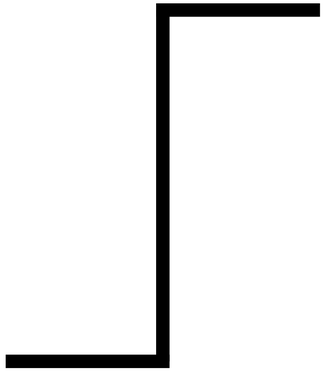


relu

Favorite nonlinearities g(·)



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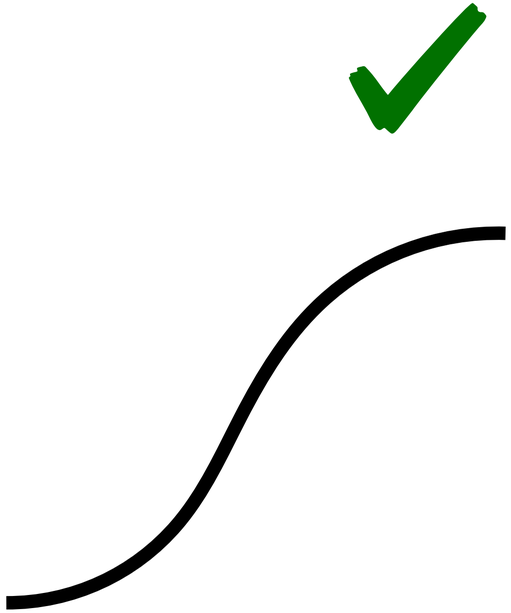

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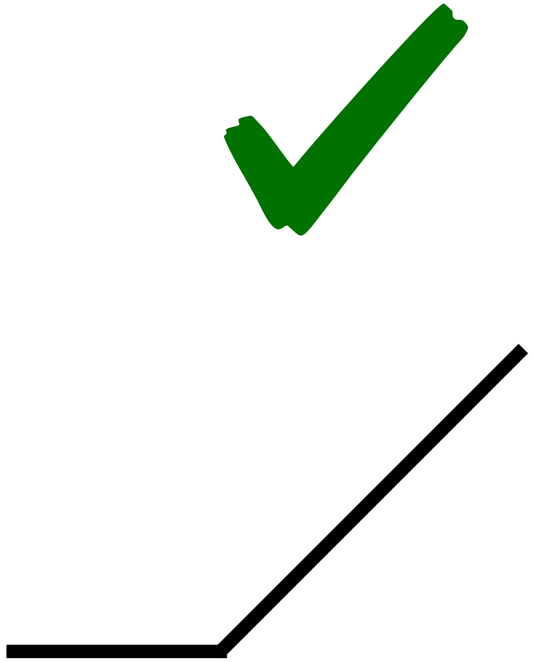

relu

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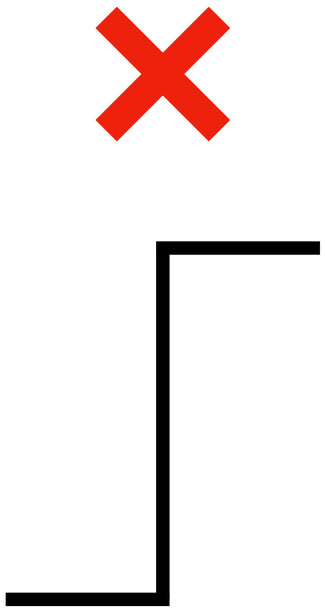

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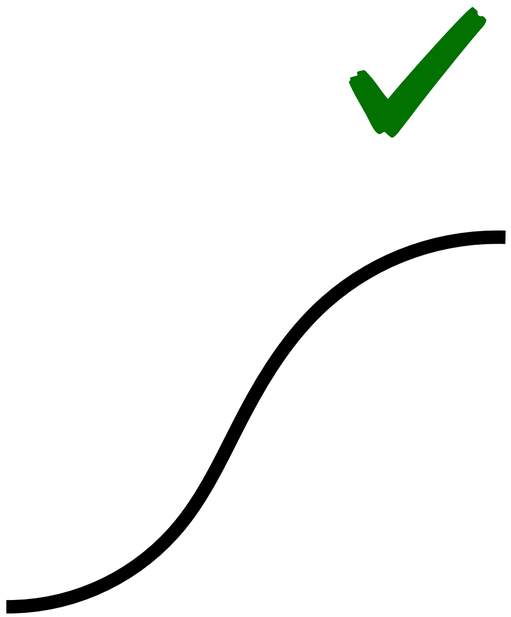

sigmoid

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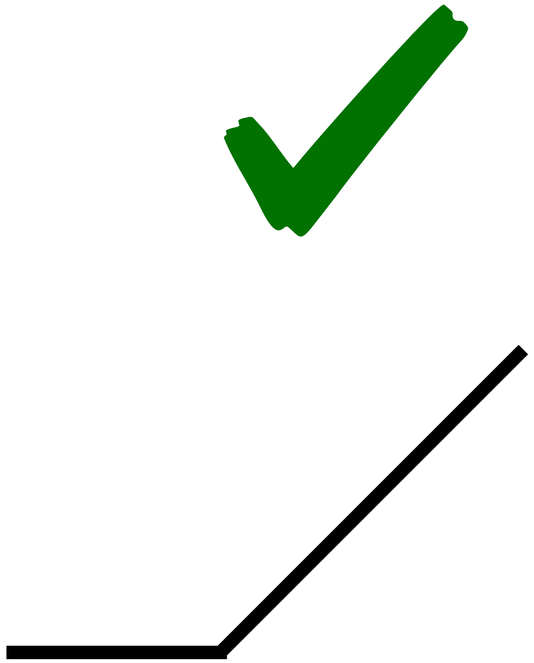
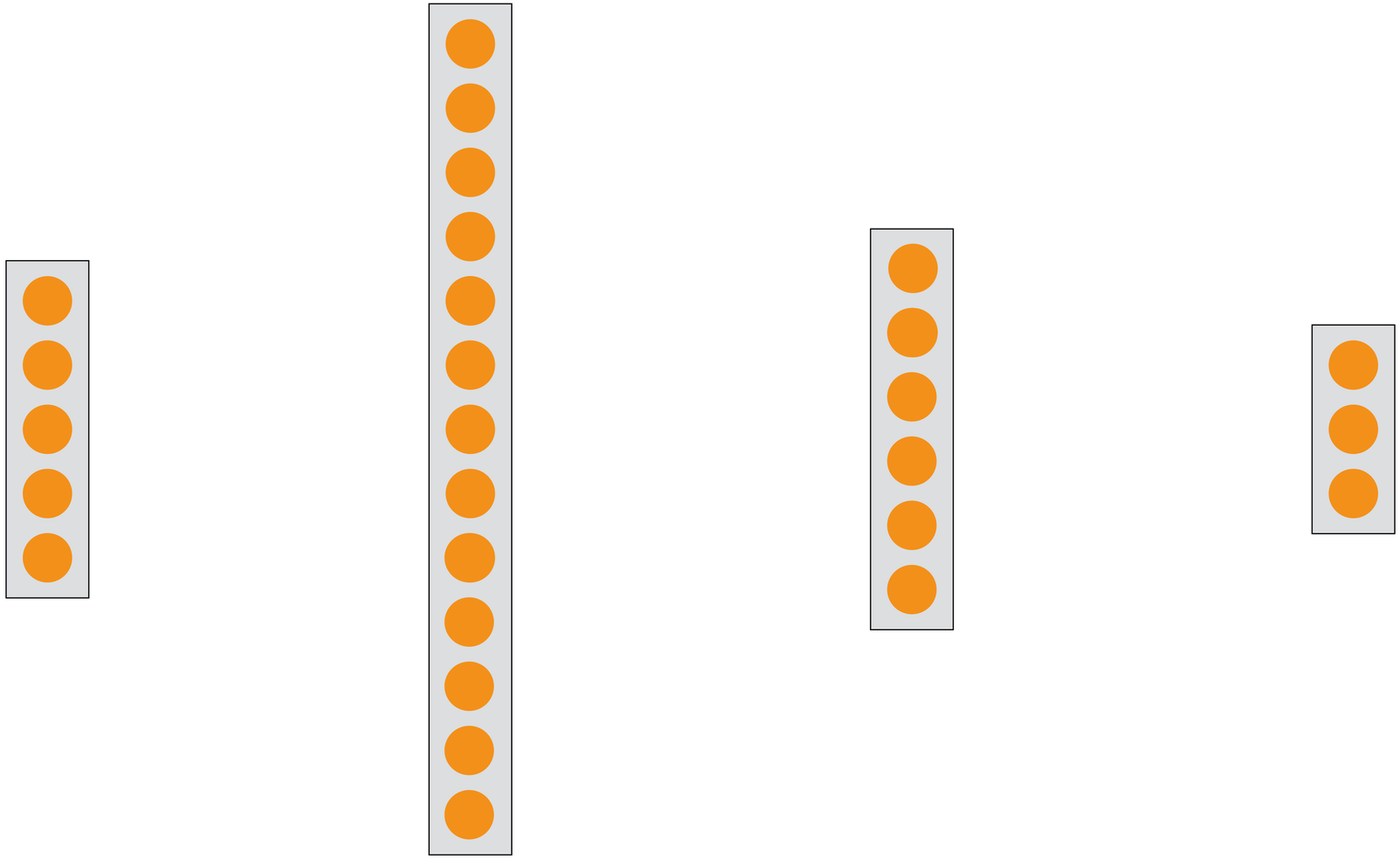


relu

How many weights? Which layer?



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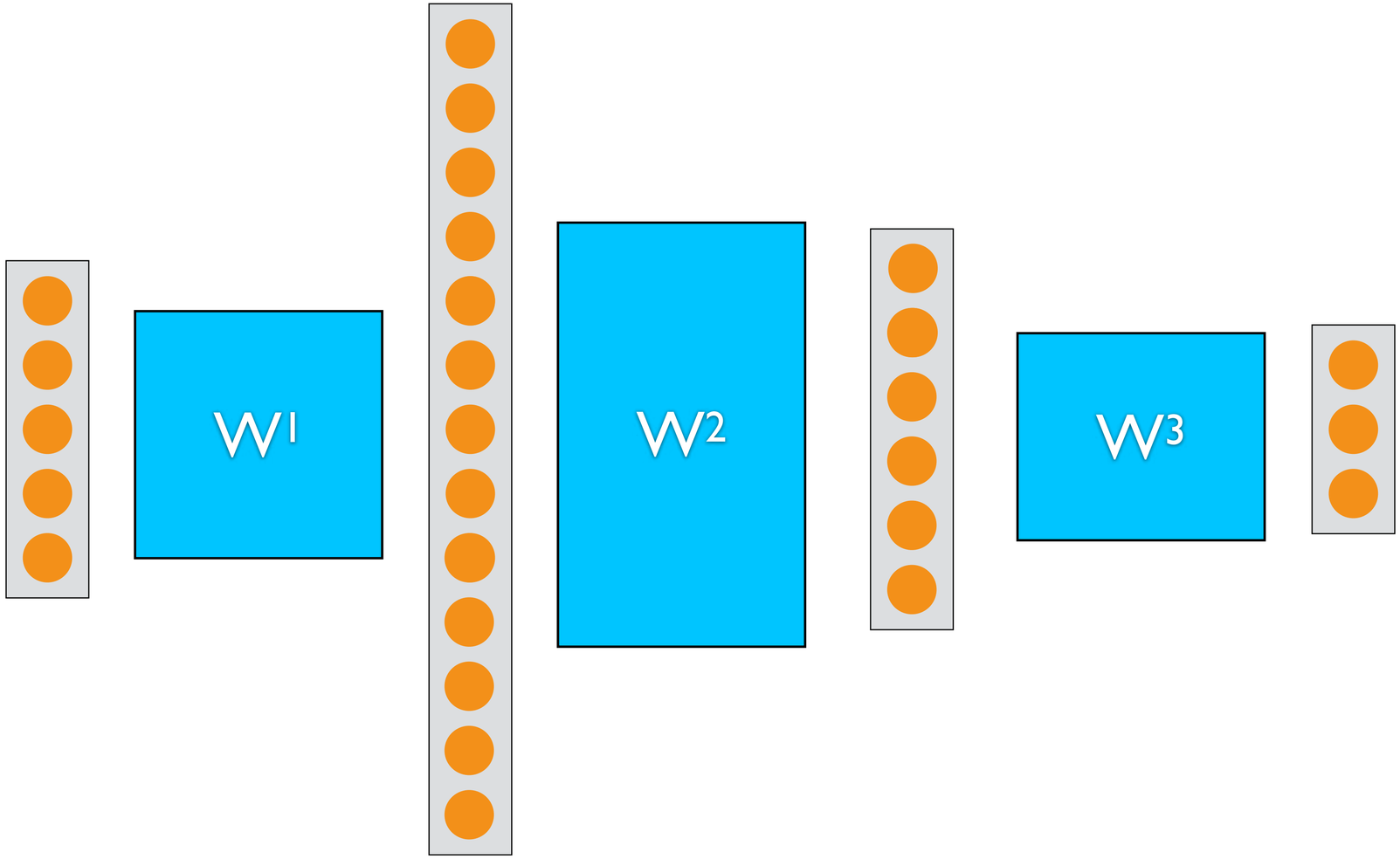
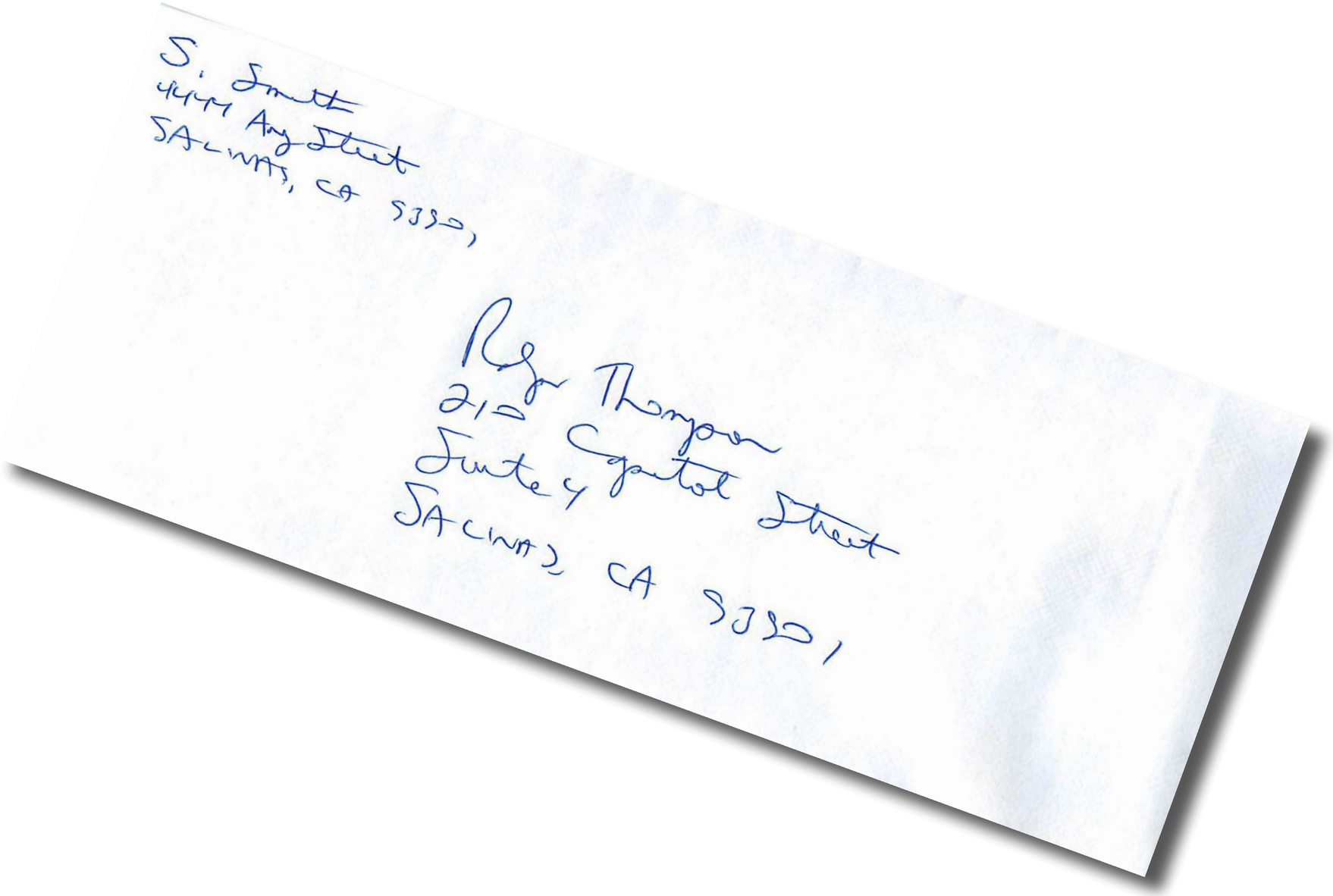
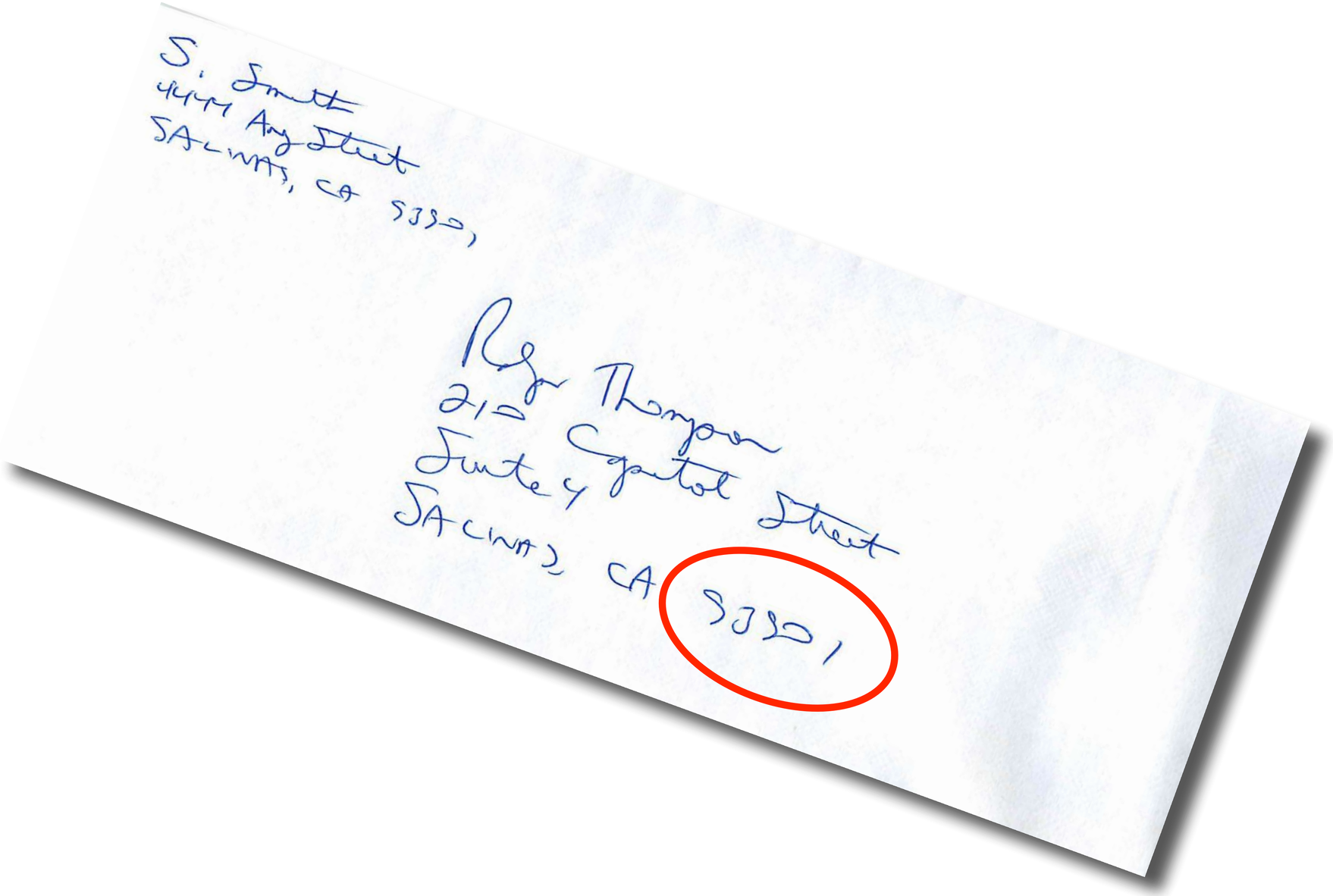



S. Smith
4444 Any Street
SALINAS, CA 93930,
Roger Thompson
210 Capitol Street
Suite 4
SALINAS, CA 93901,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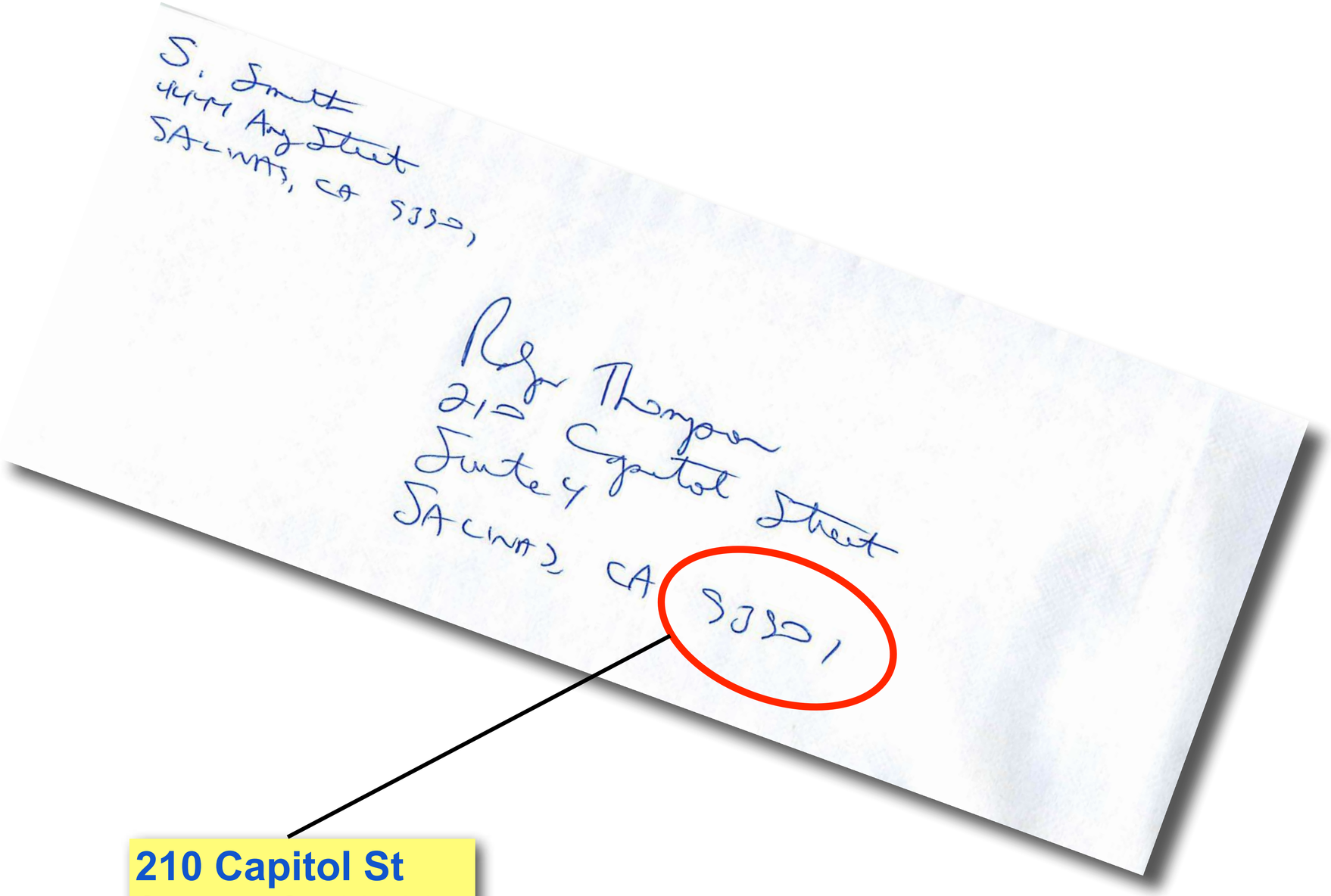
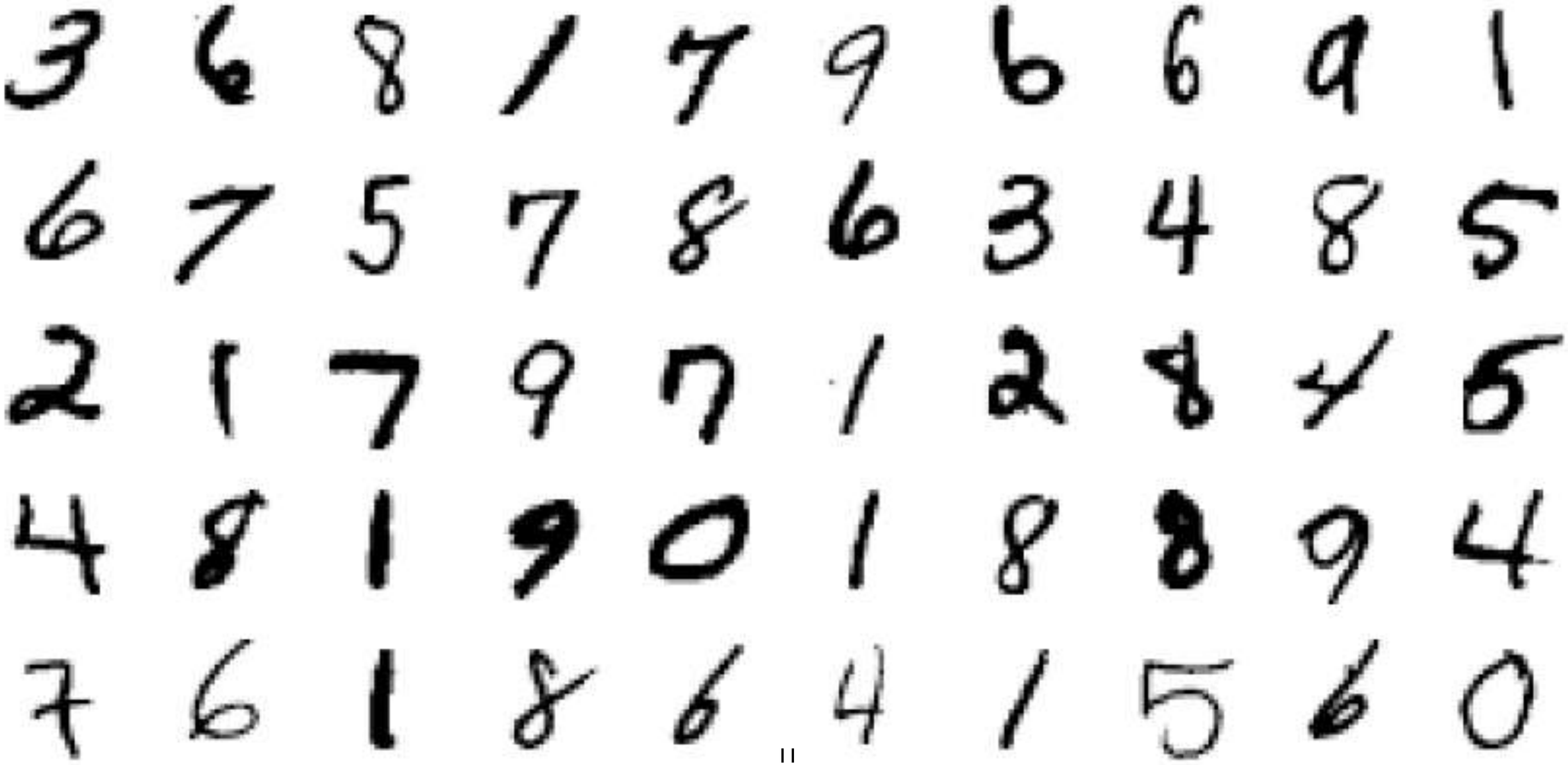
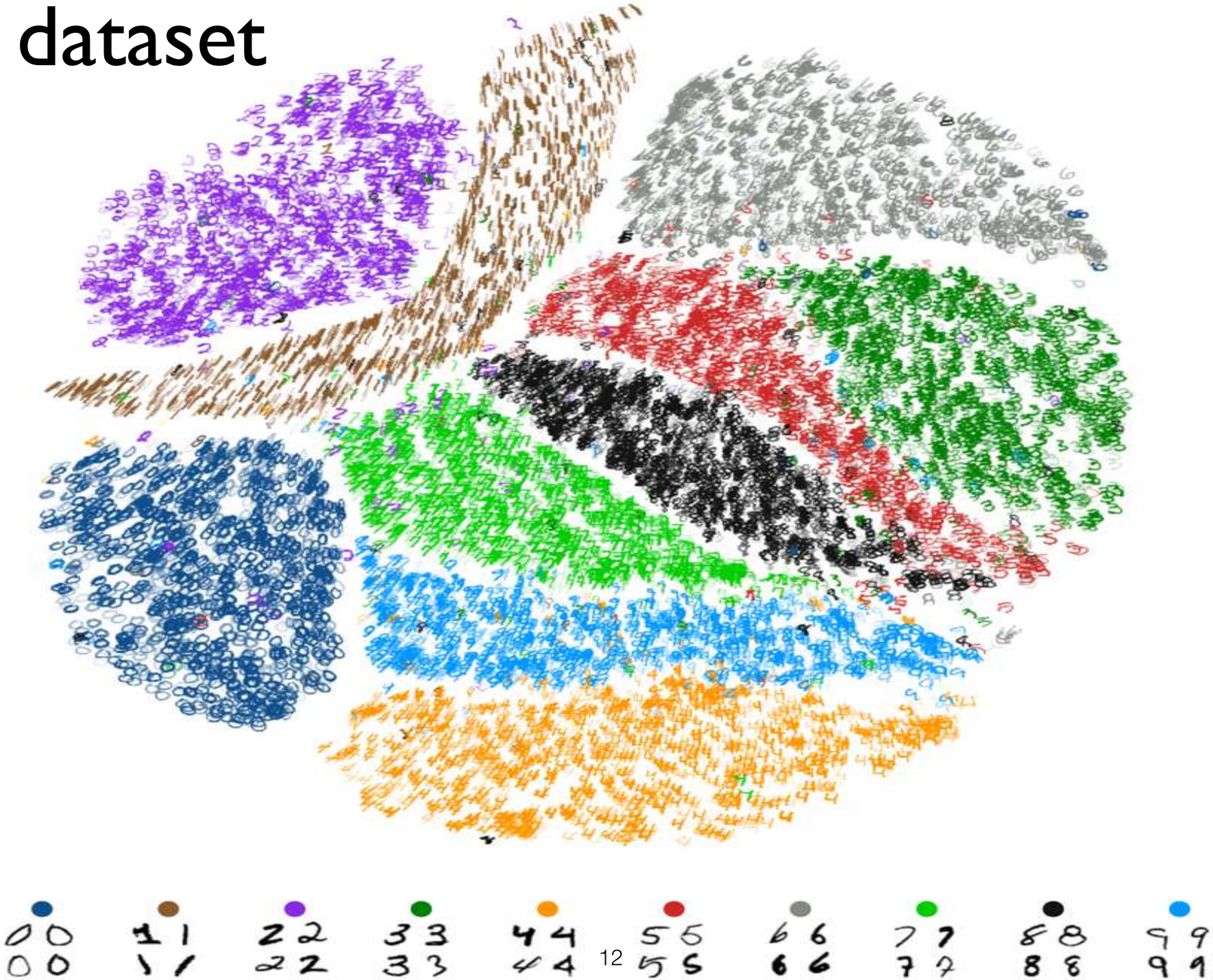


210 Capitol St
Salinas, CA 93901

Handwritten digit classification

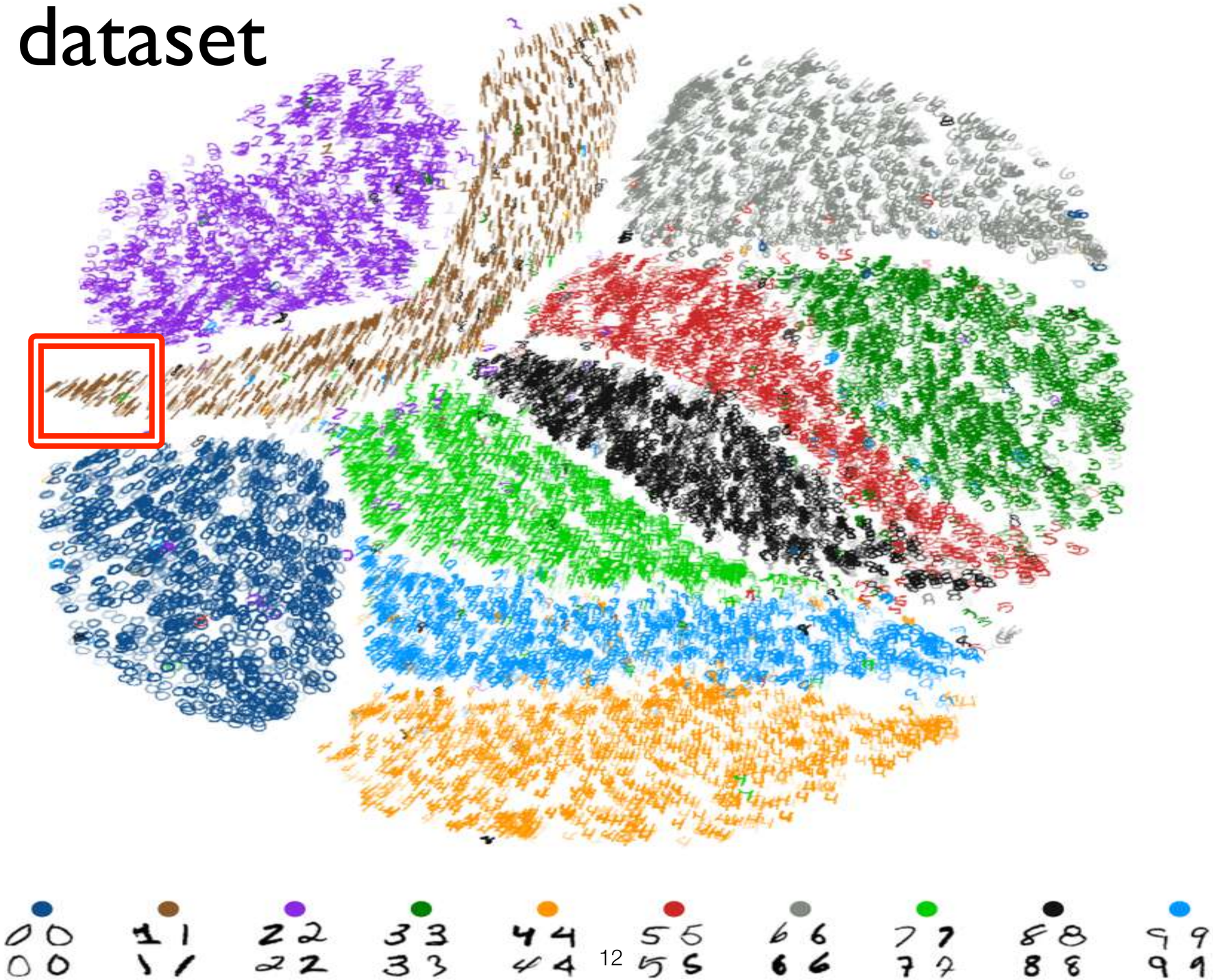


MNIST dataset



[N. Pezzotti, 2019]

MNIST dataset



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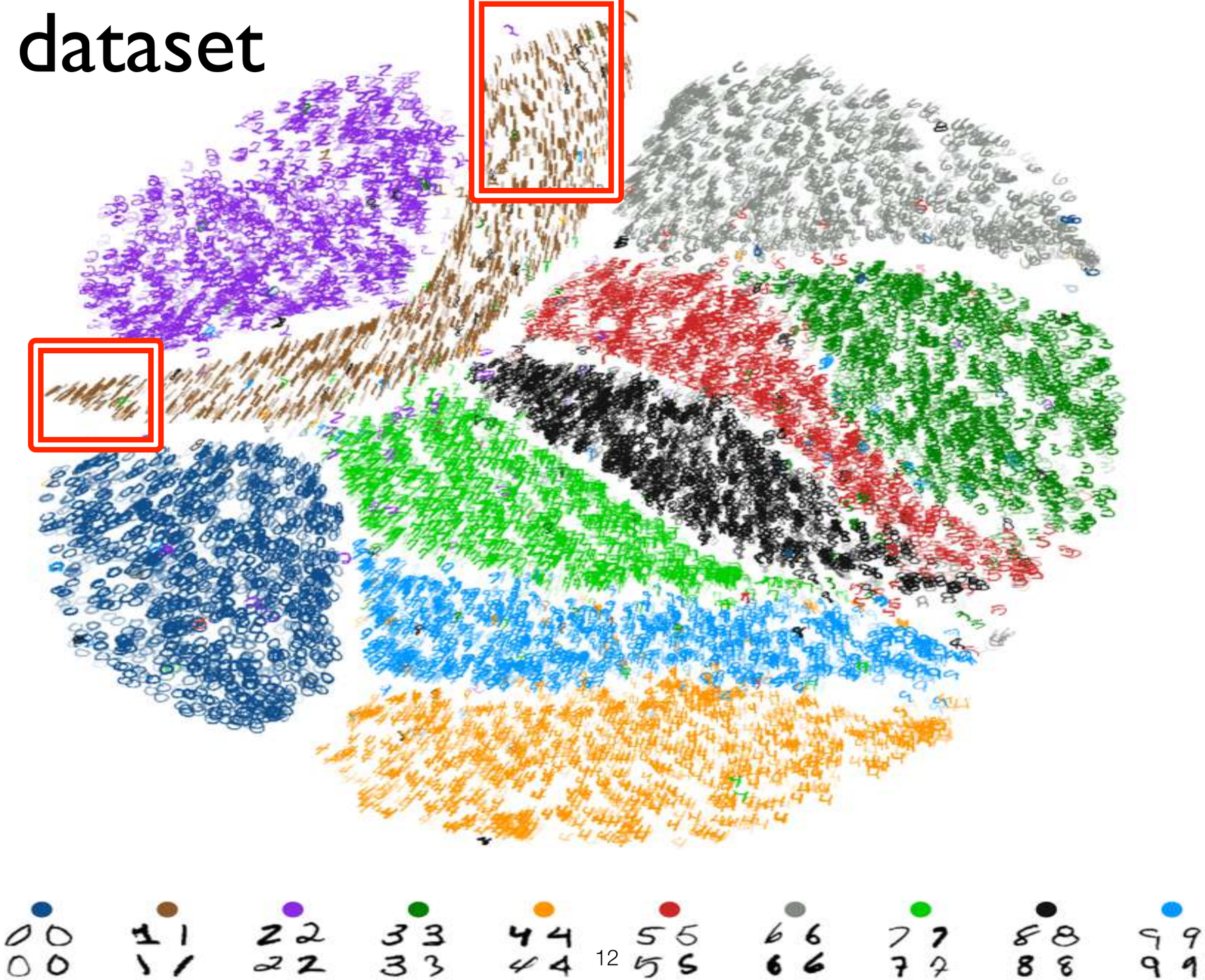

[N. Pezzotti, 2019]

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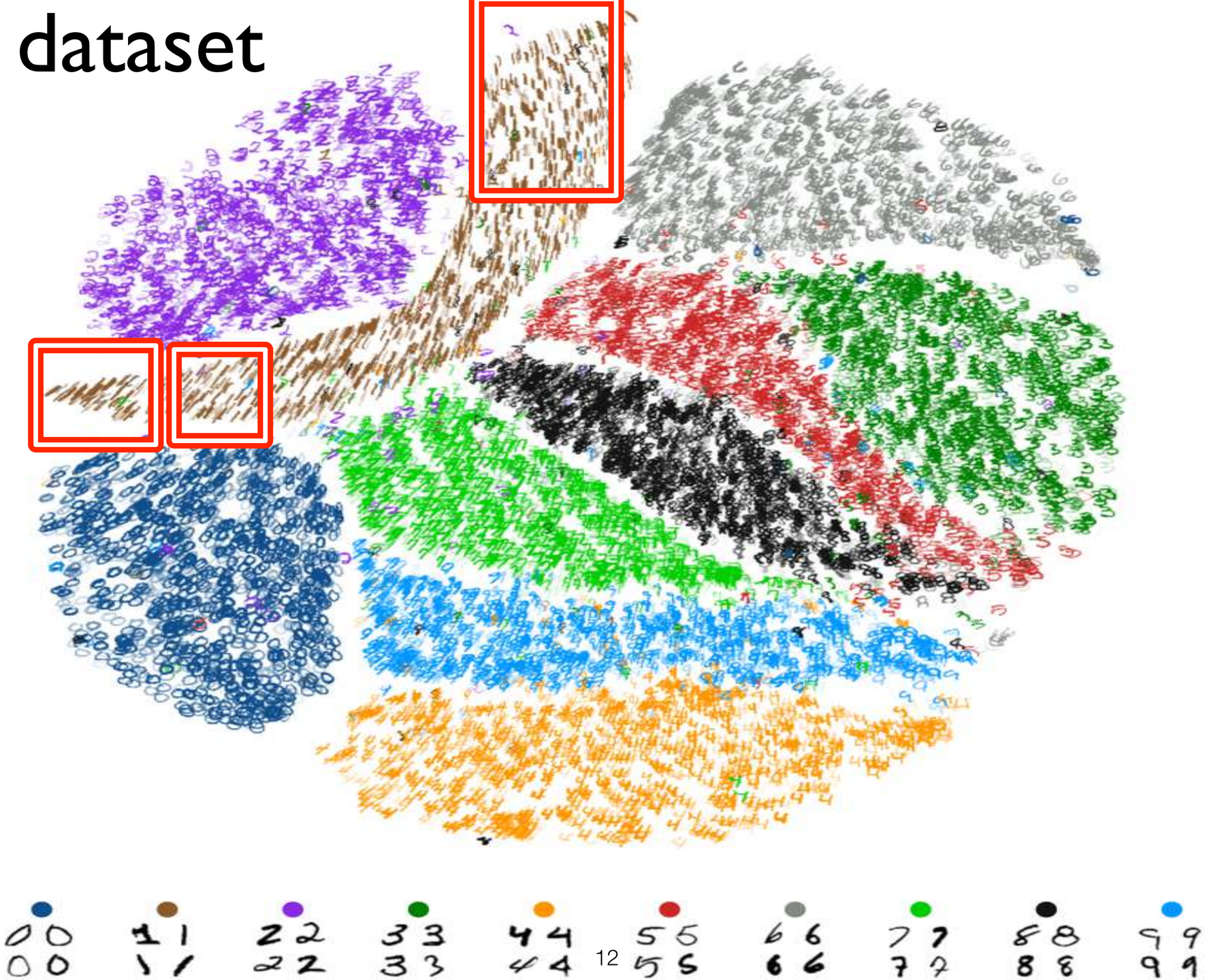
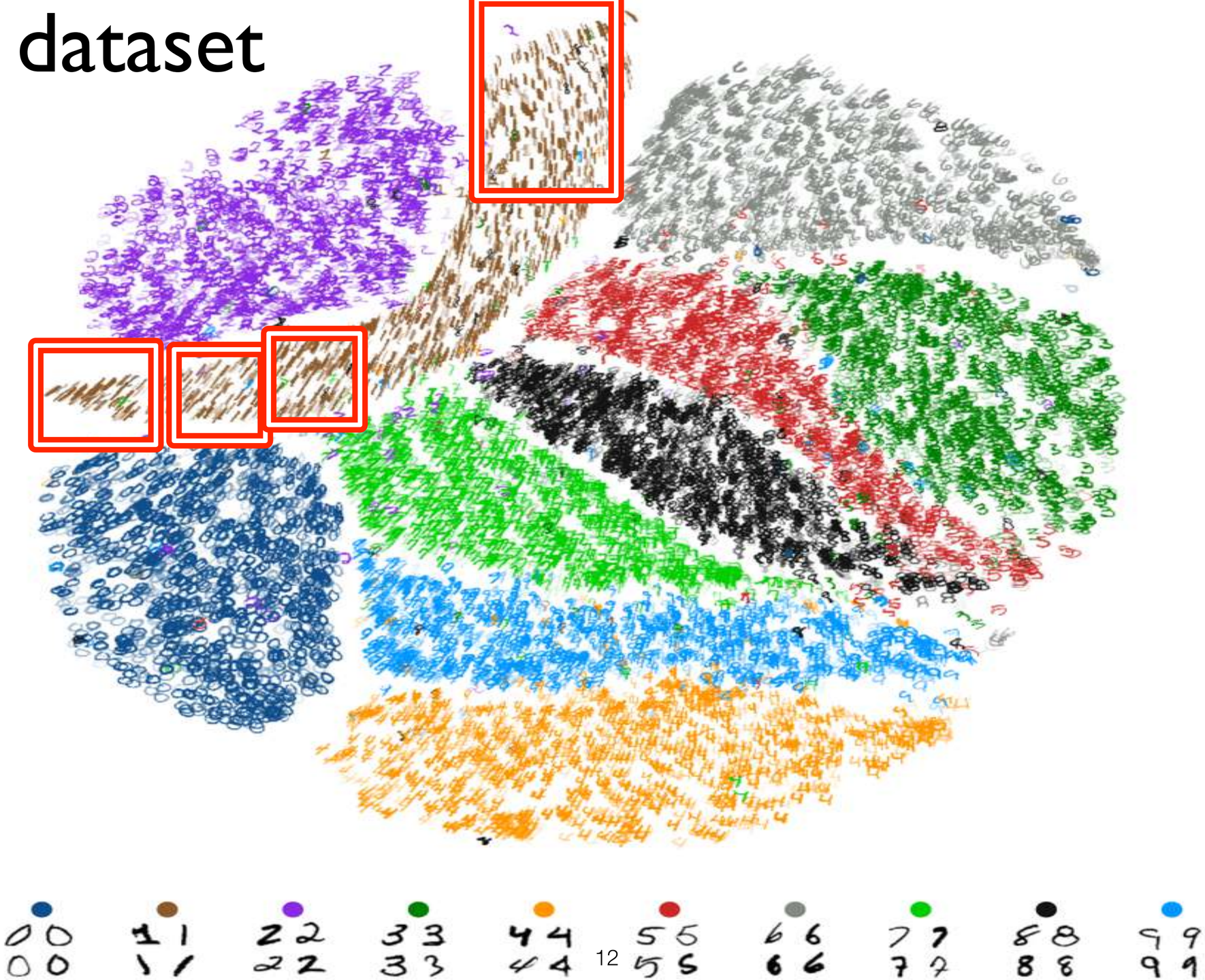


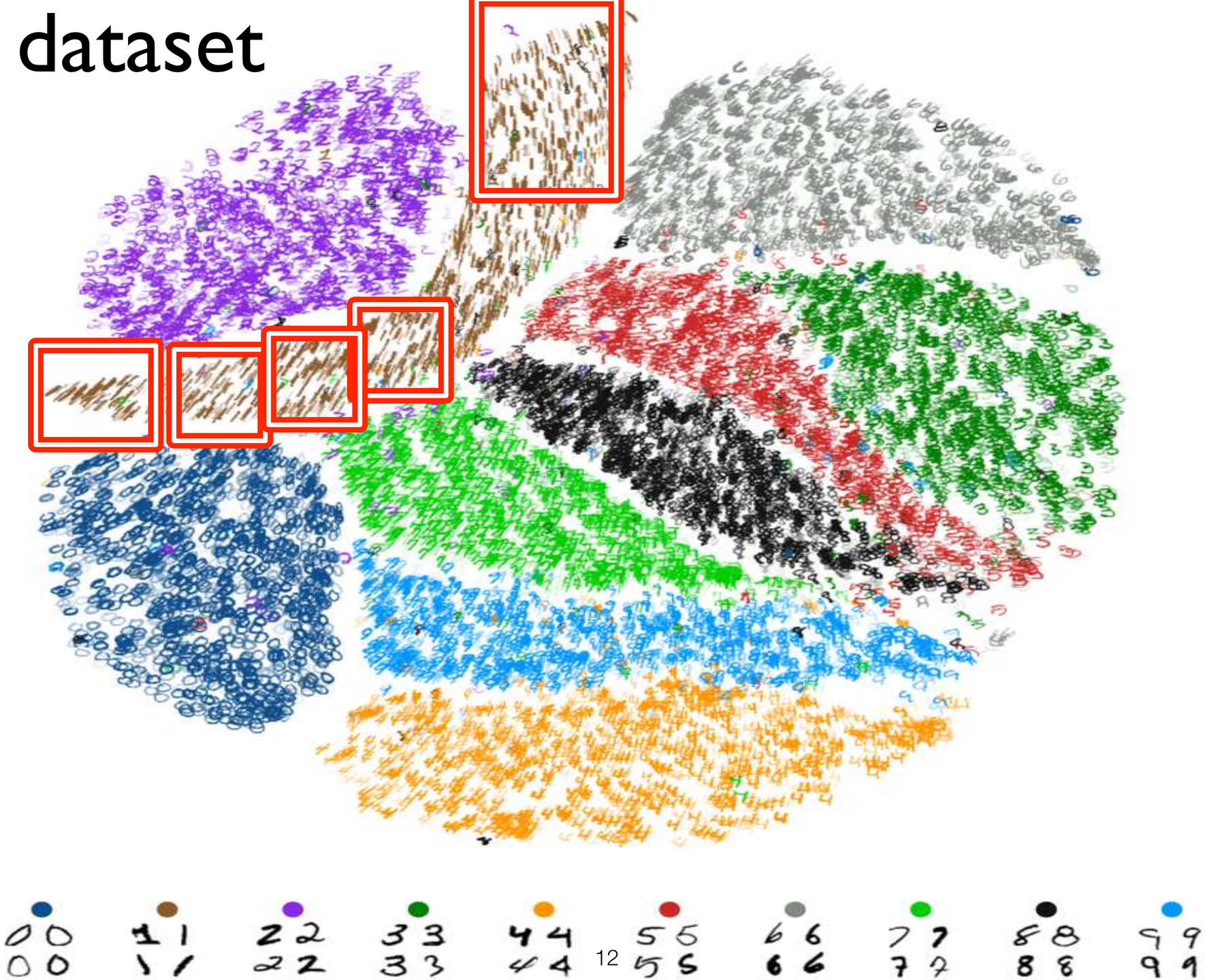
[N. Pezzotti, 2019]

MNIST dataset



[N. Pezzotti, 2019]

MNIST dataset



[N. Pezzotti, 2019]

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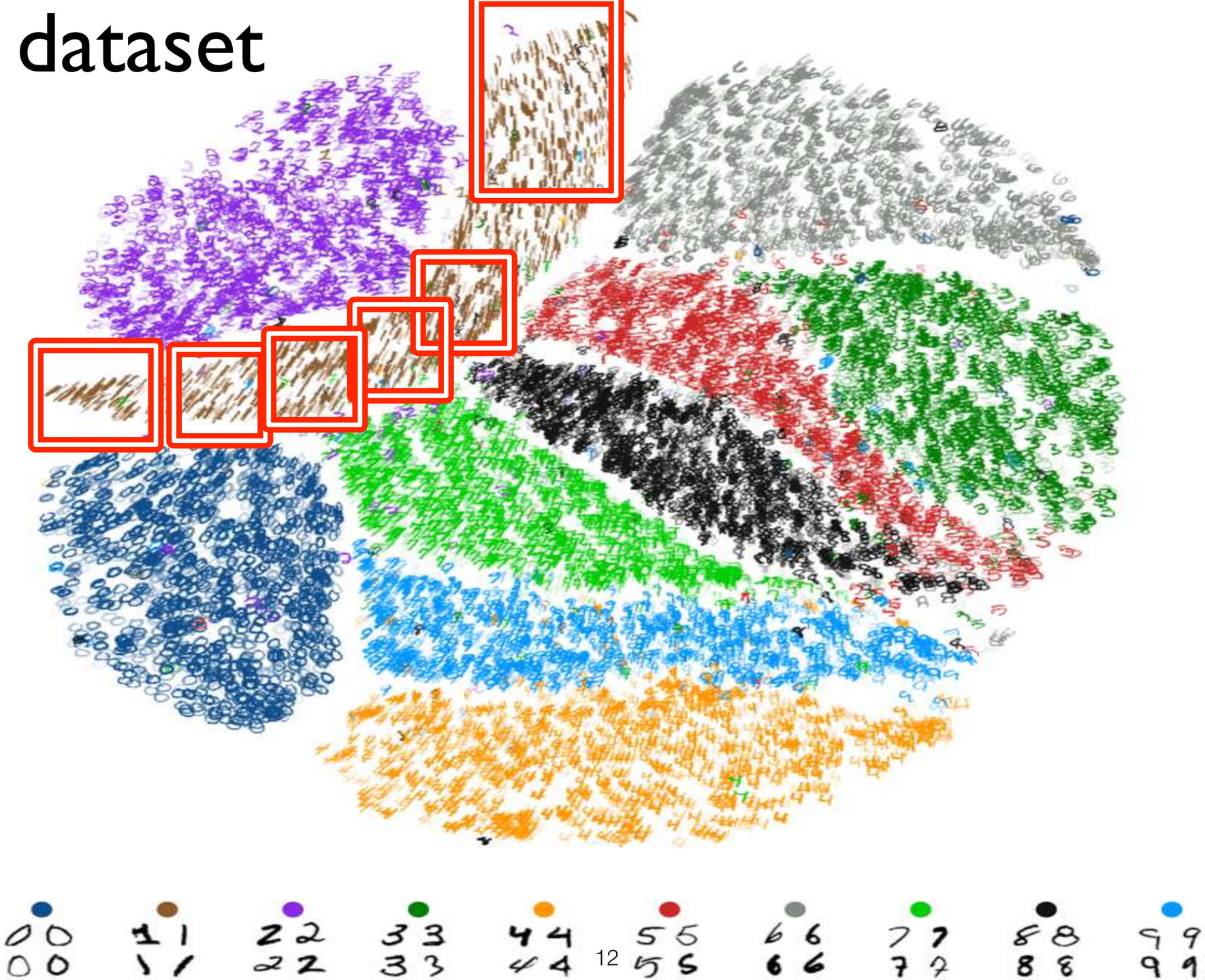

[N. Pezzotti, 2019]

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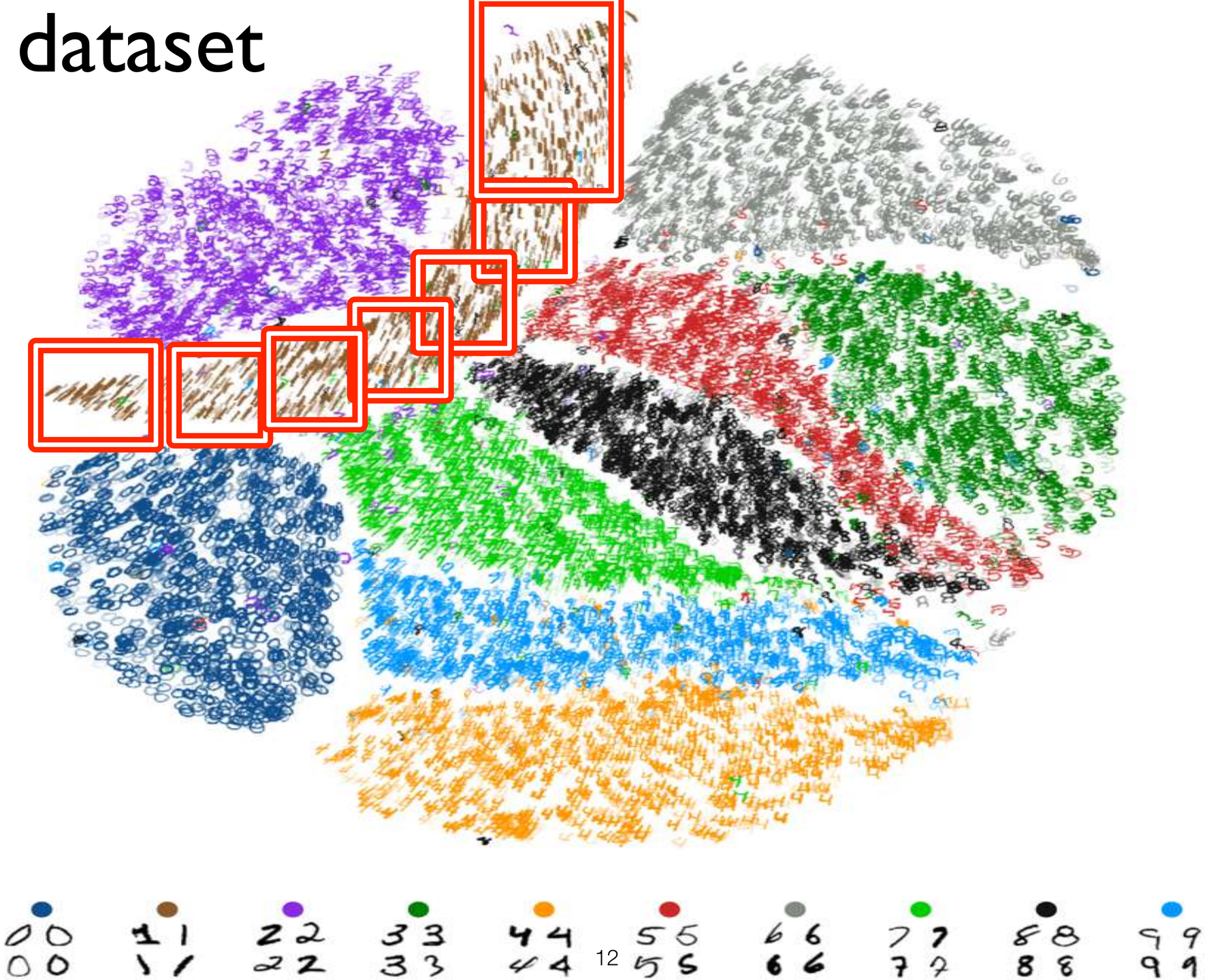
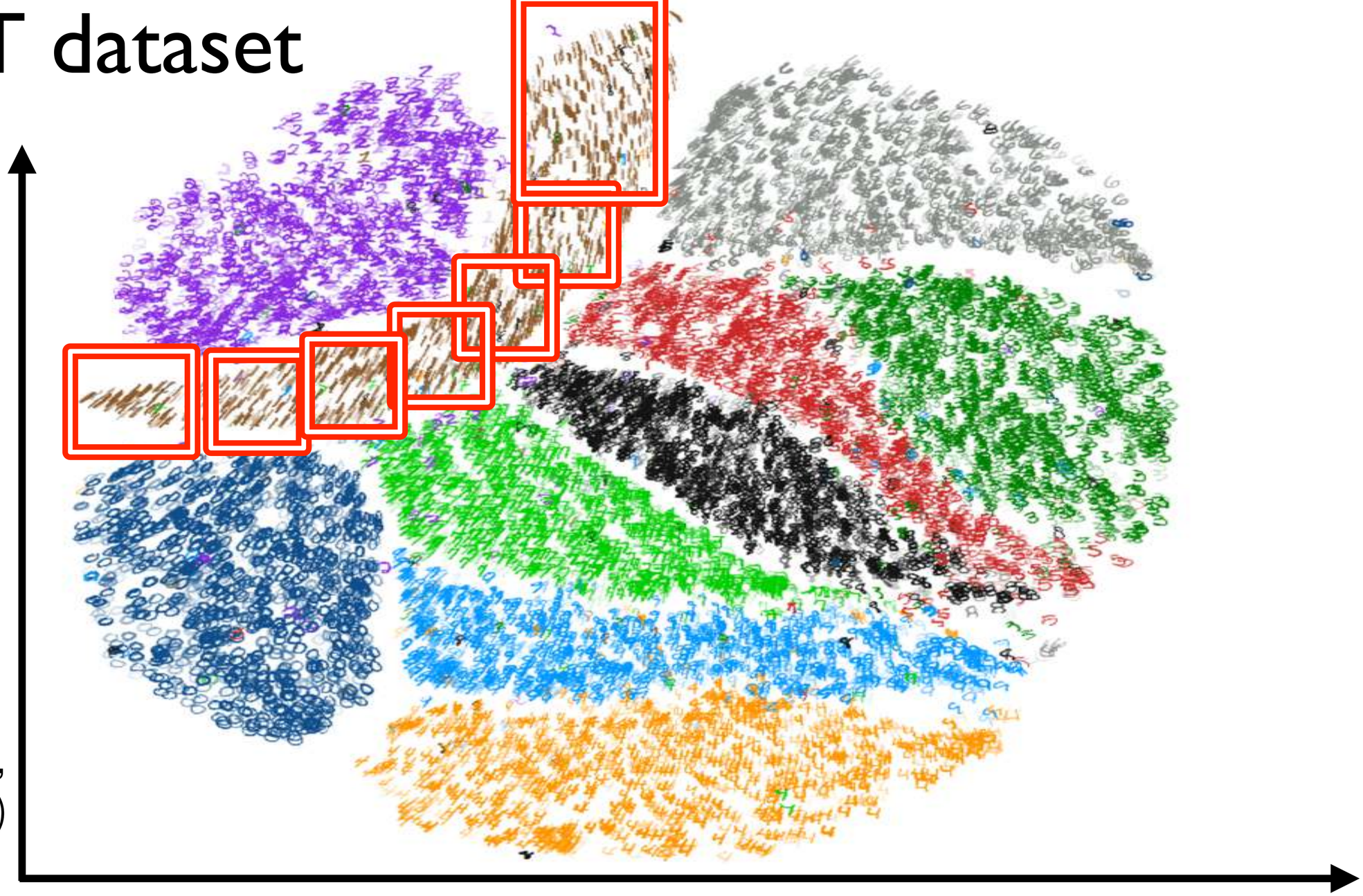
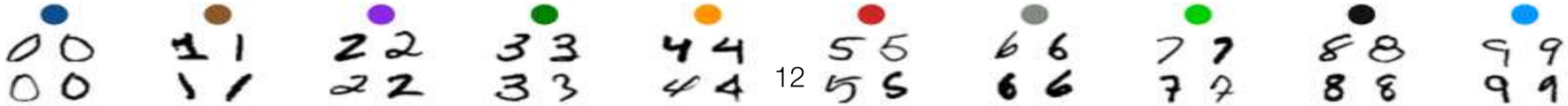


[N. Pezzotti, 2019]

MNIST dataset

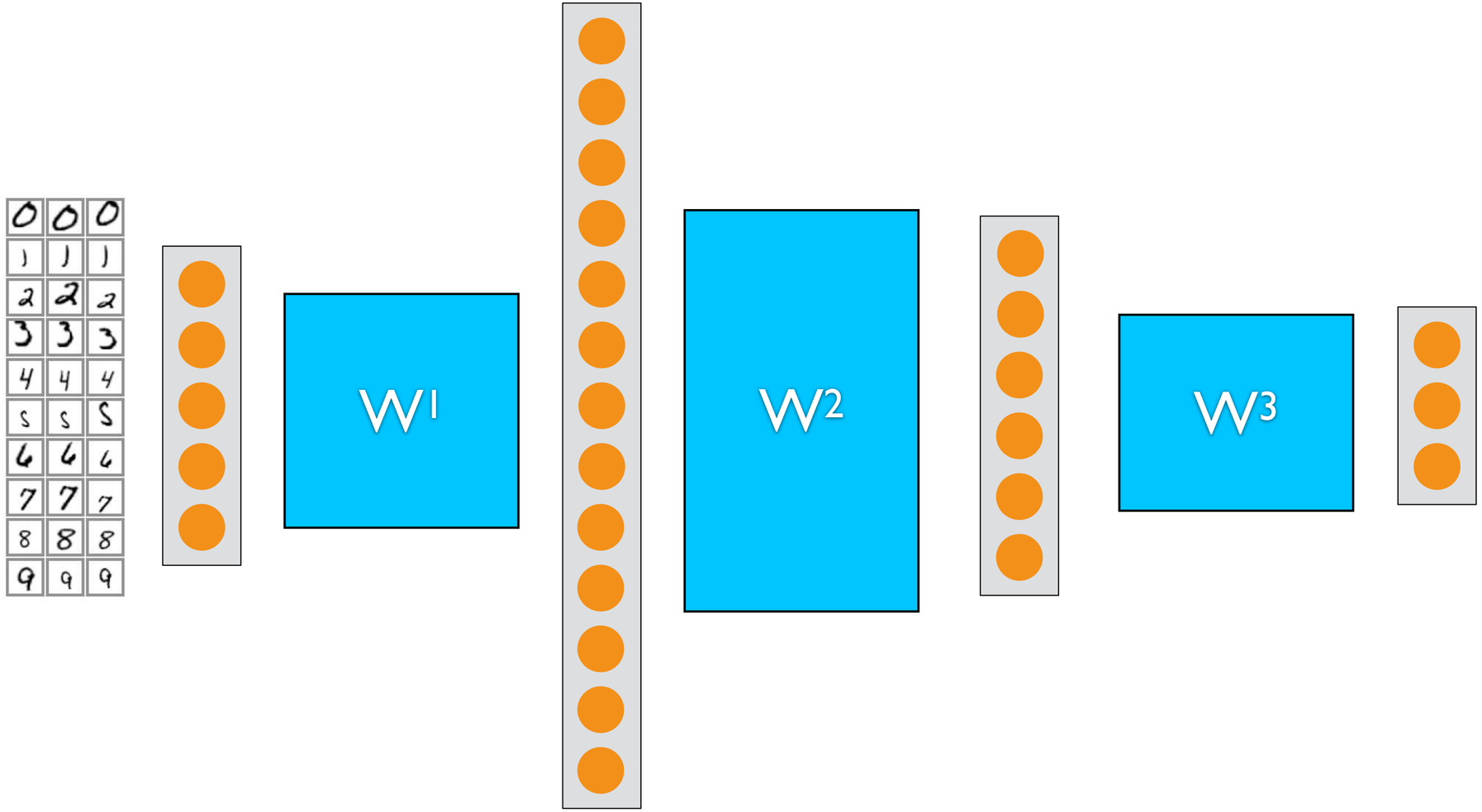
(this is an embedding, details later)





[N. Pezzotti, 2019]

Classifying MNIST: how many parameters?



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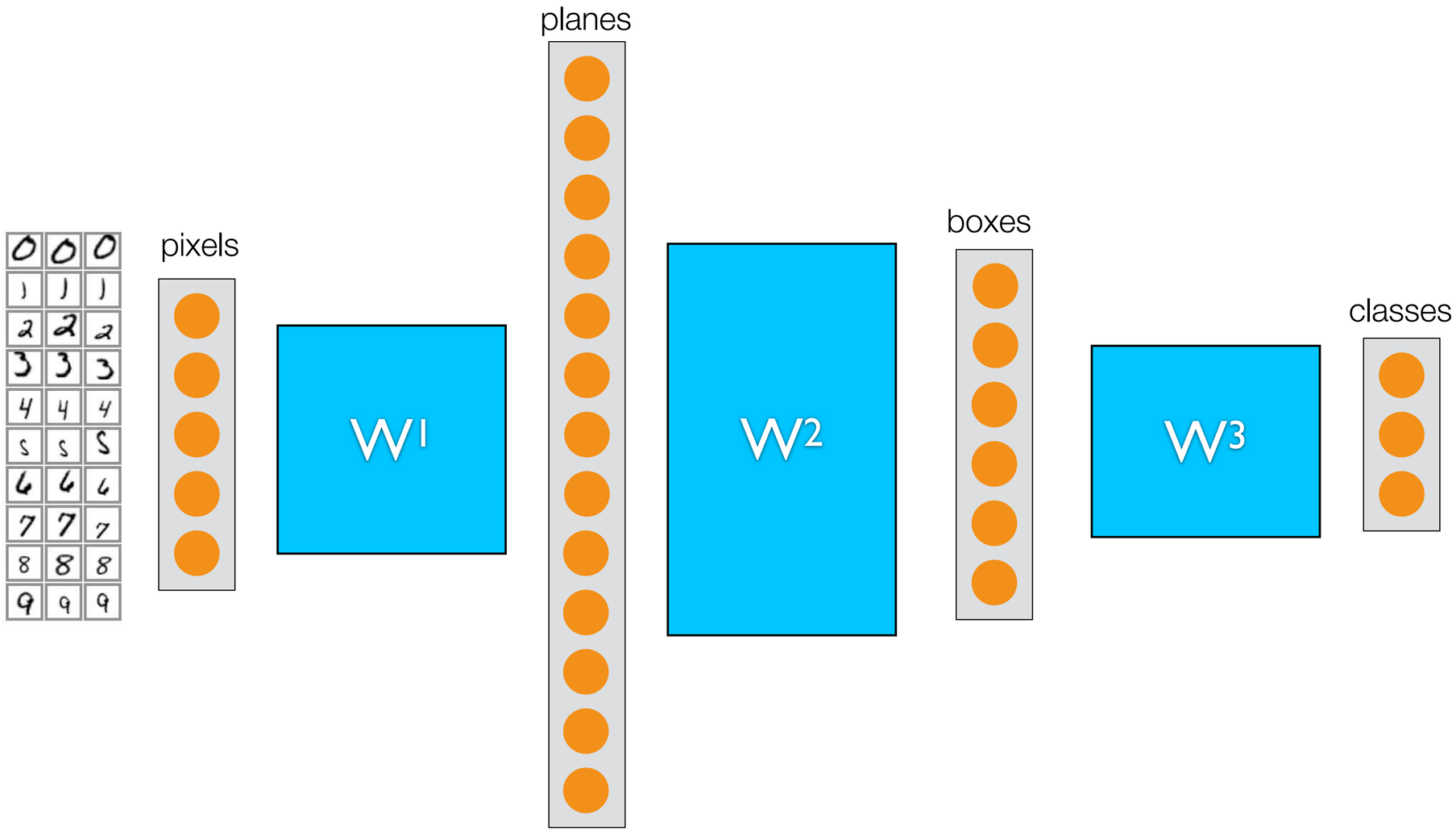

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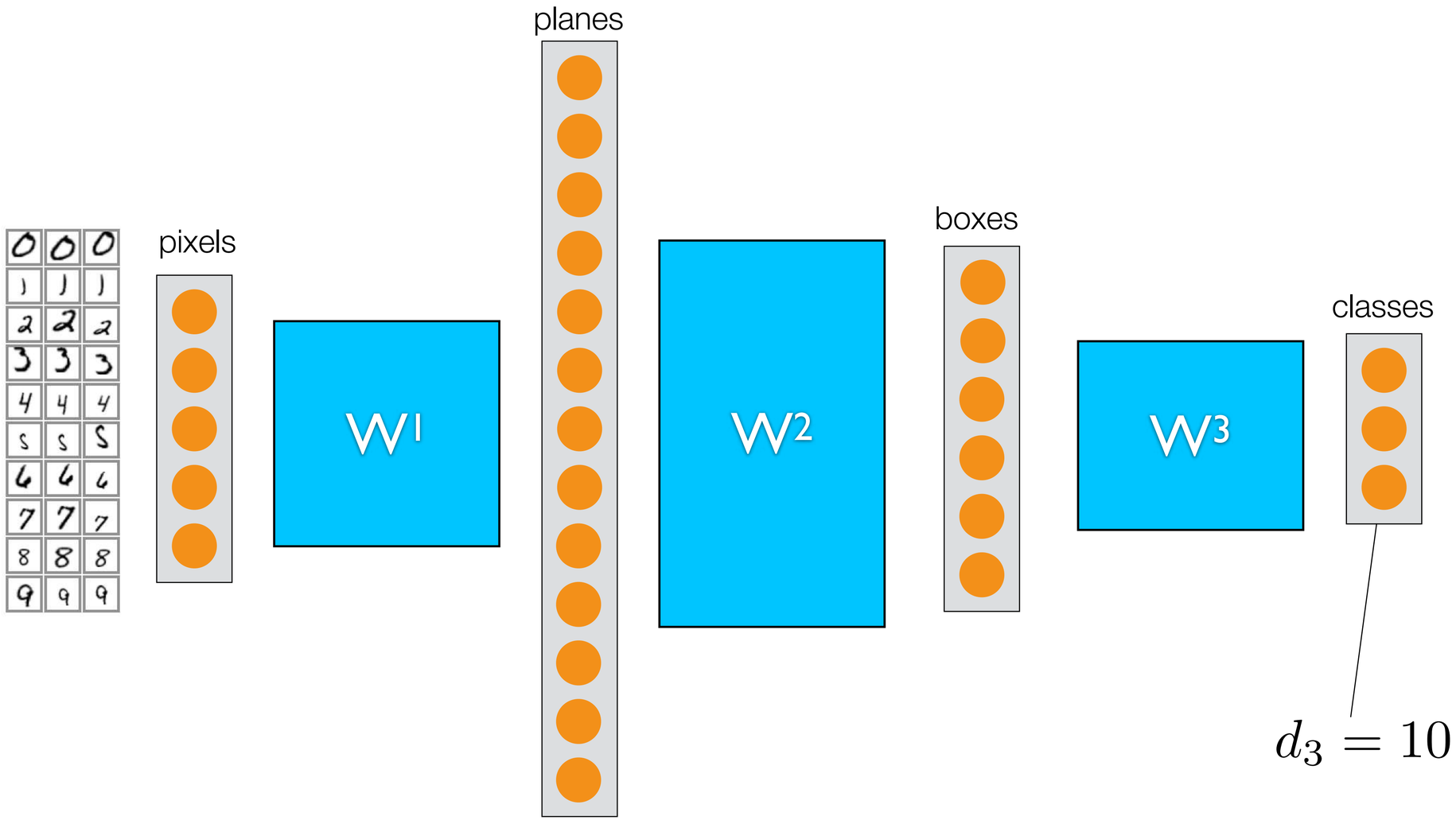

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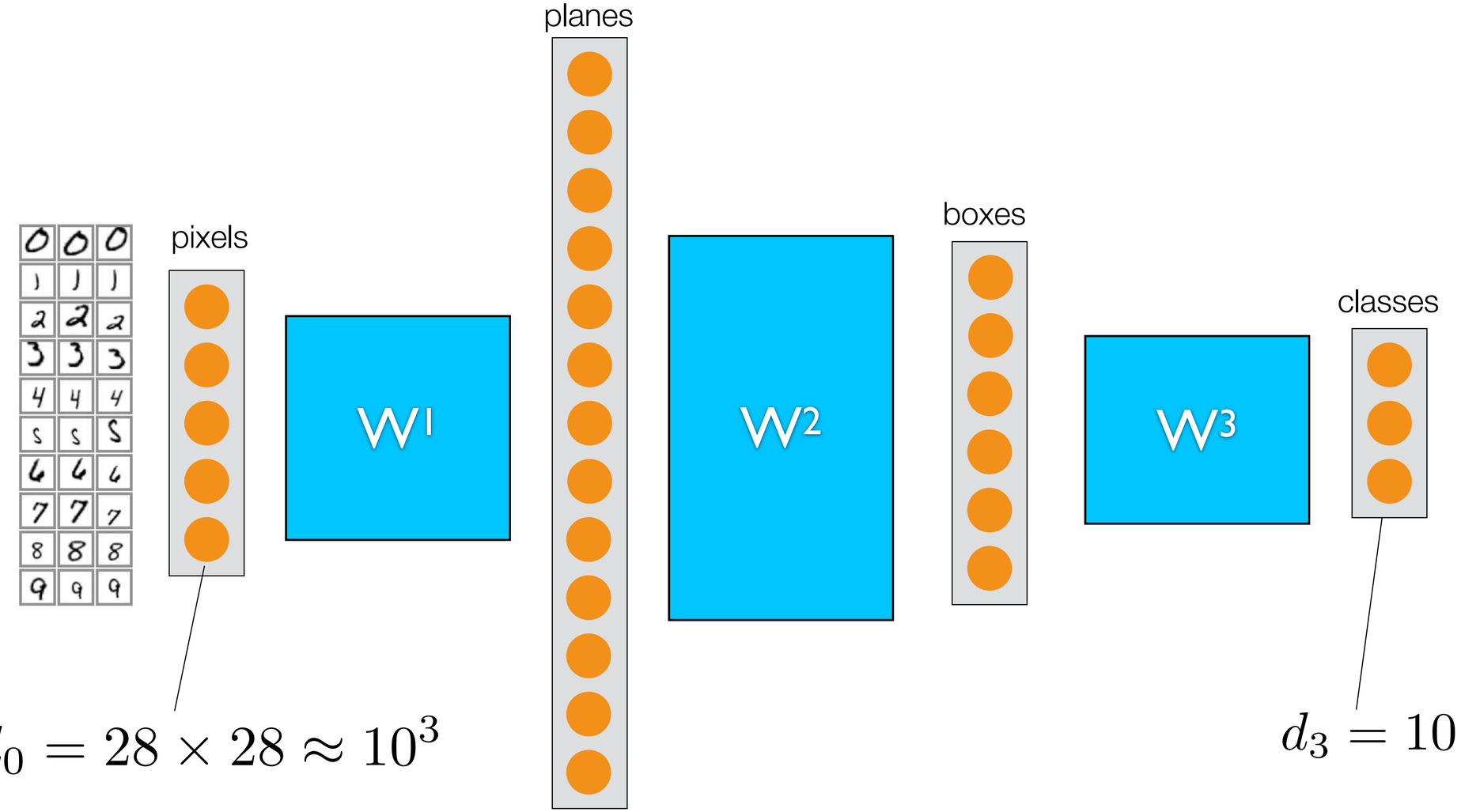
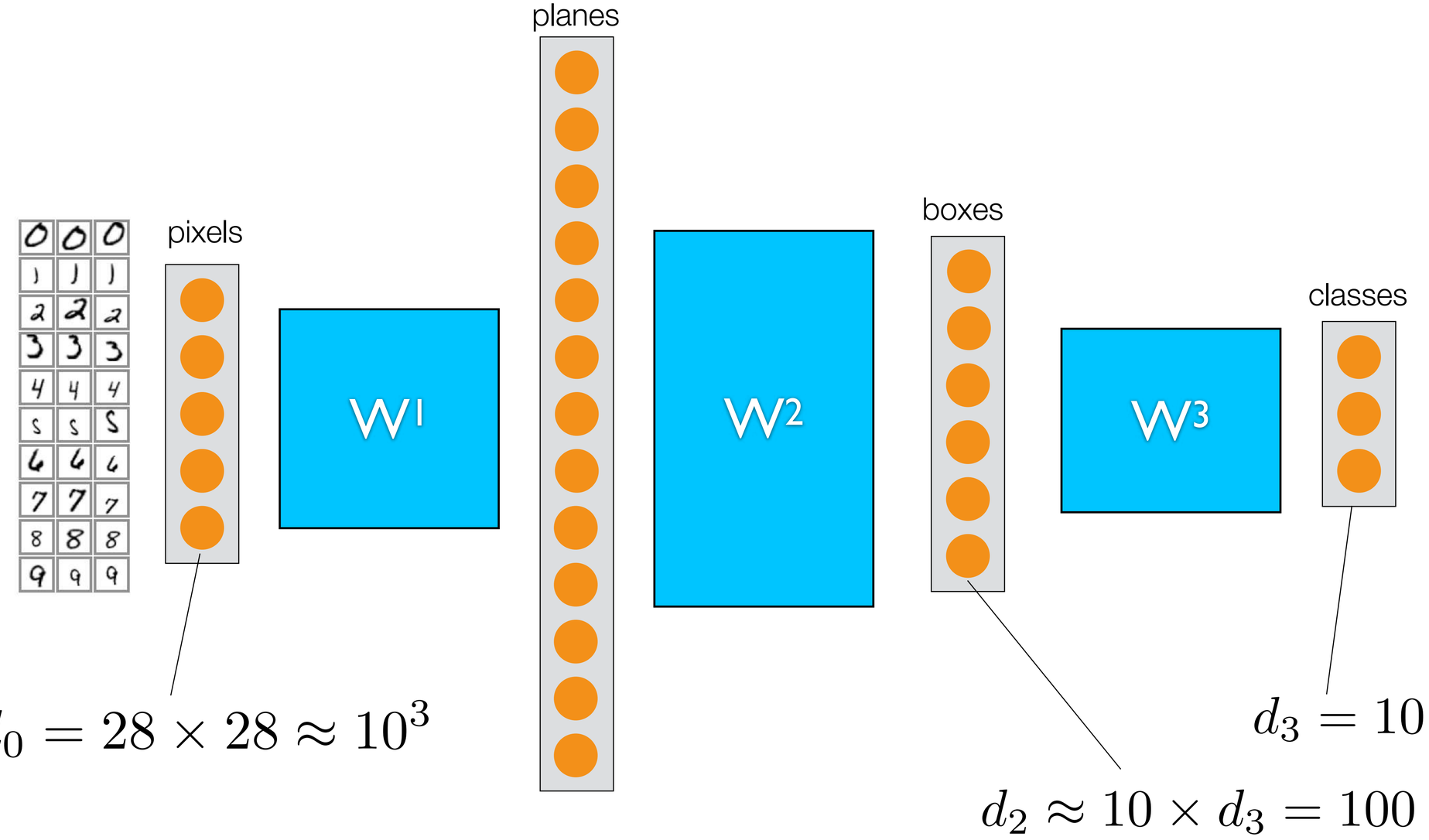


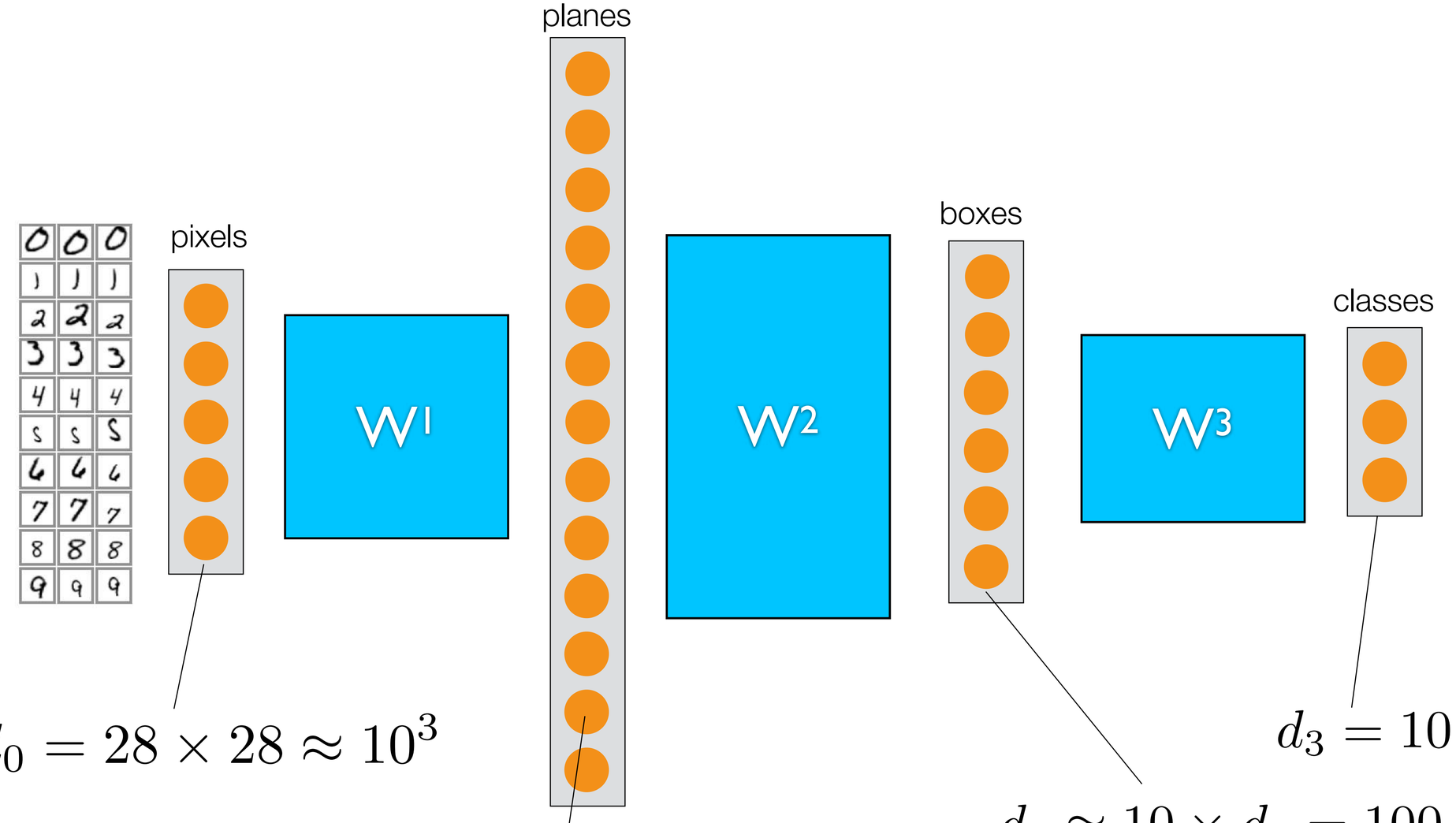
$$d_0 = 28 \times 28 \approx 10^3$$

$$d_3 = 10$$

Classifying MNIST: how many parameters?



Classifying MNIST: how many parameters?



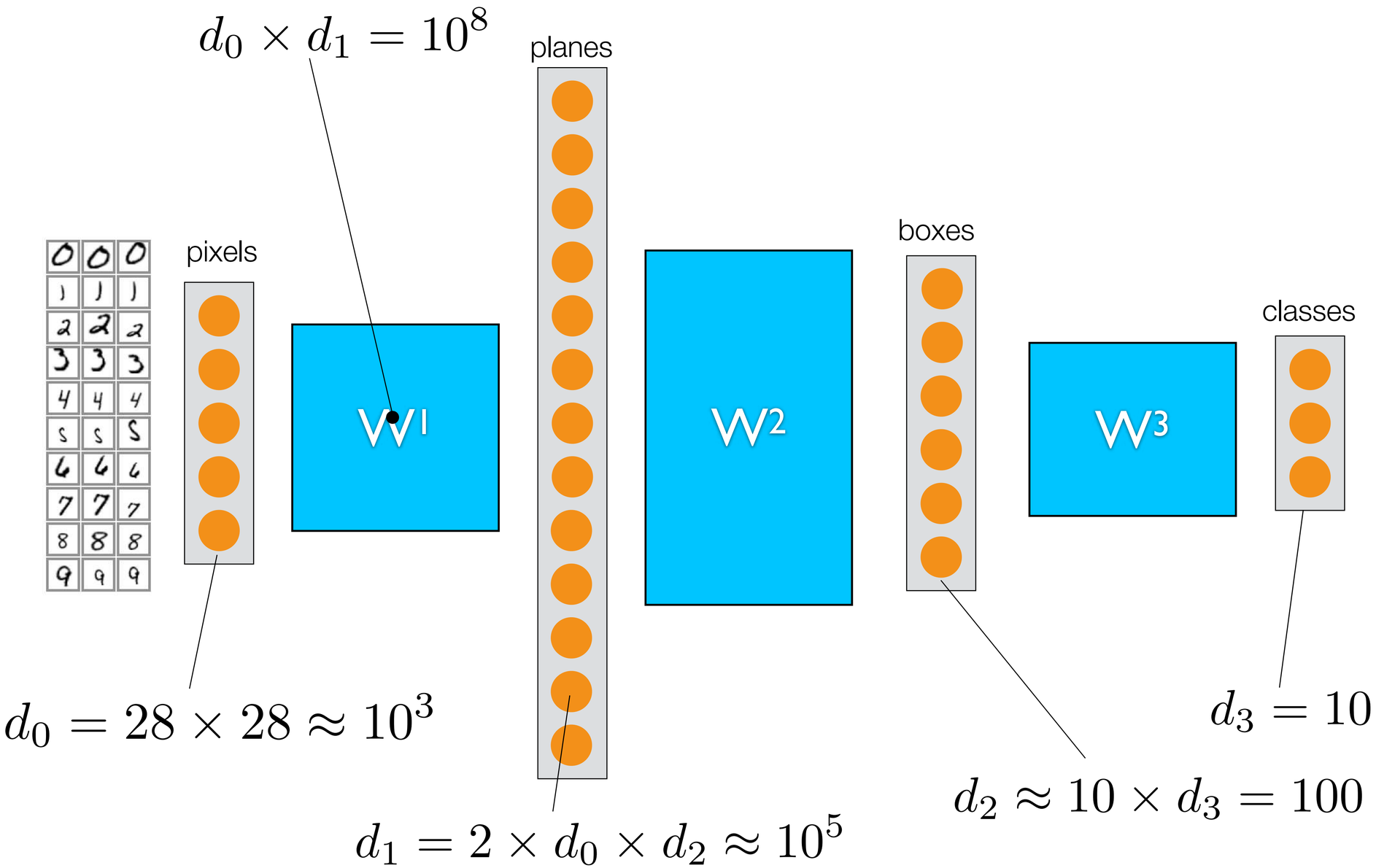
$$d_0 = 28 \times 28 \approx 10^3$$

$$d_1 = 2 \times d_0 \times d_2 \approx 10^5$$

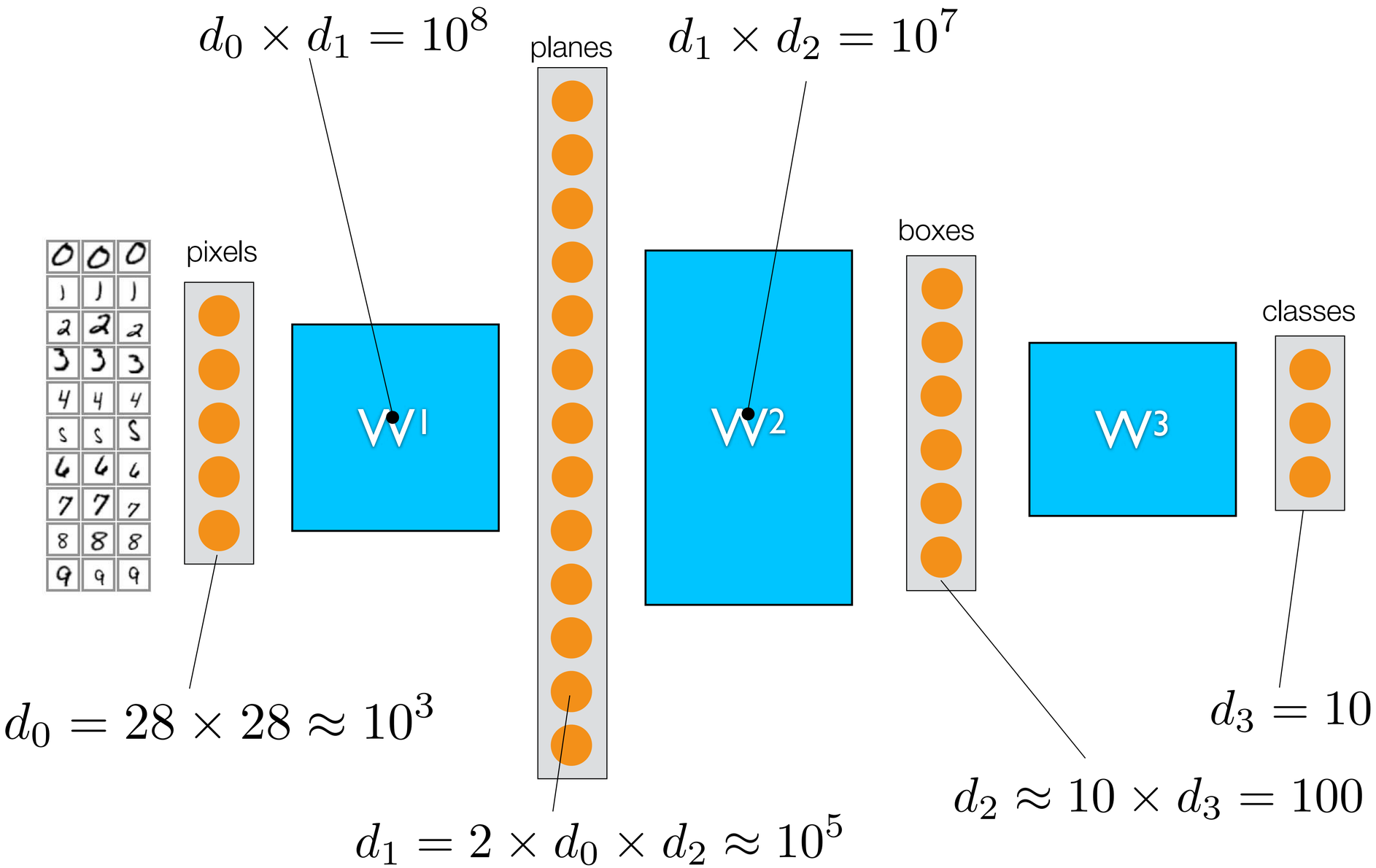
$$d_2 \approx 10 \times d_3 = 100$$

$$d_3 = 10$$

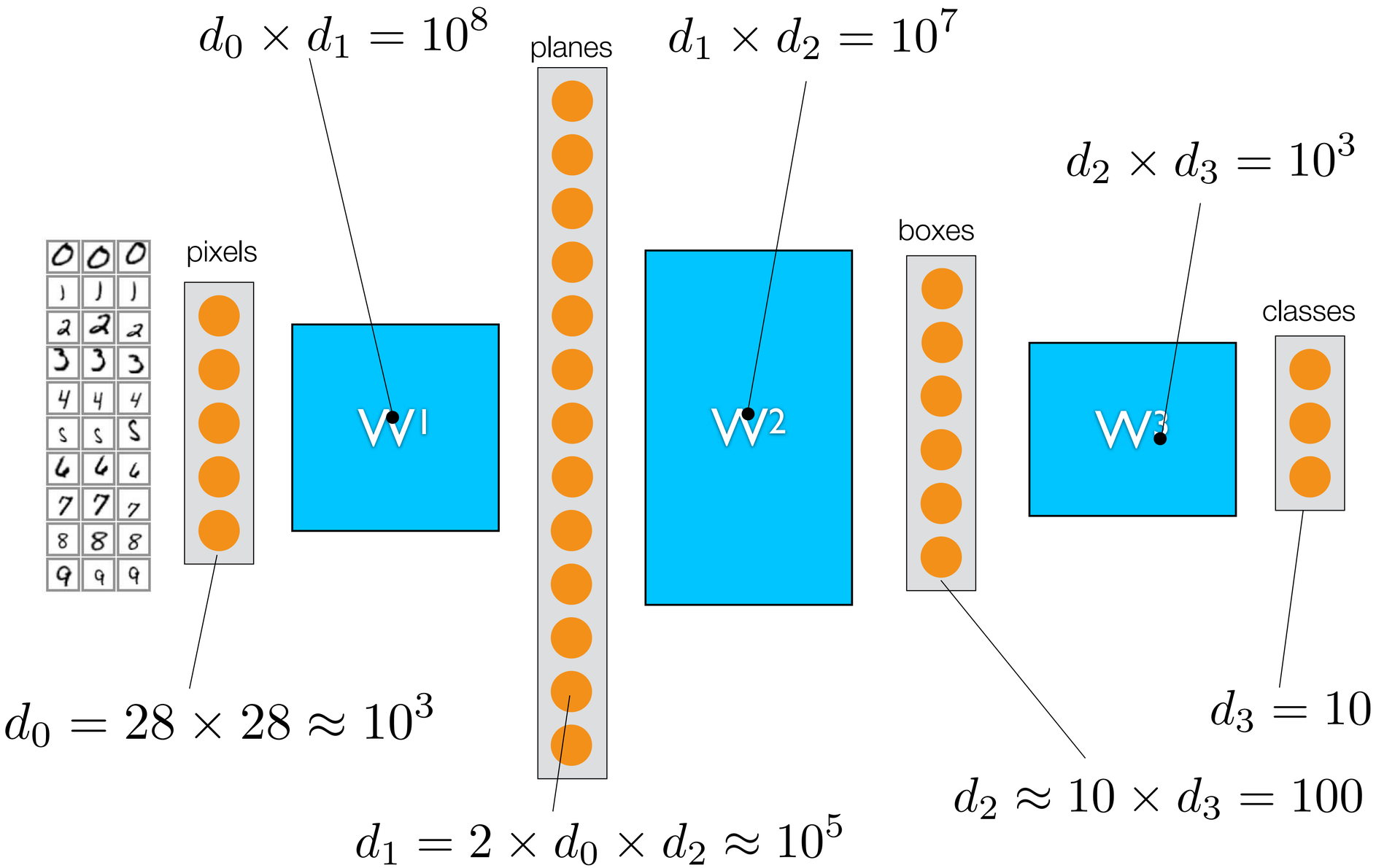
Classifying MNIST: how many parameters?



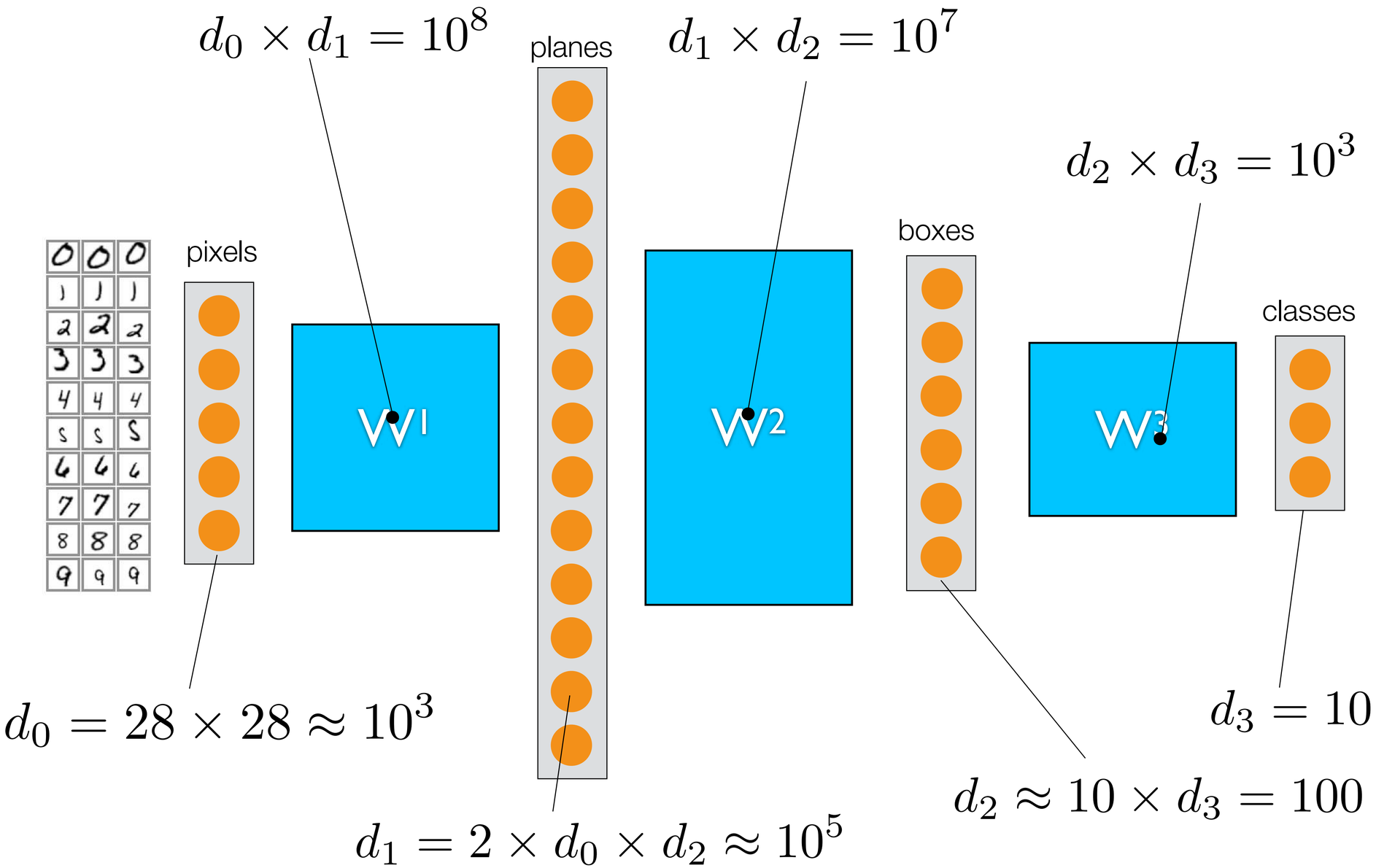
Classifying MNIST: how many parameters?

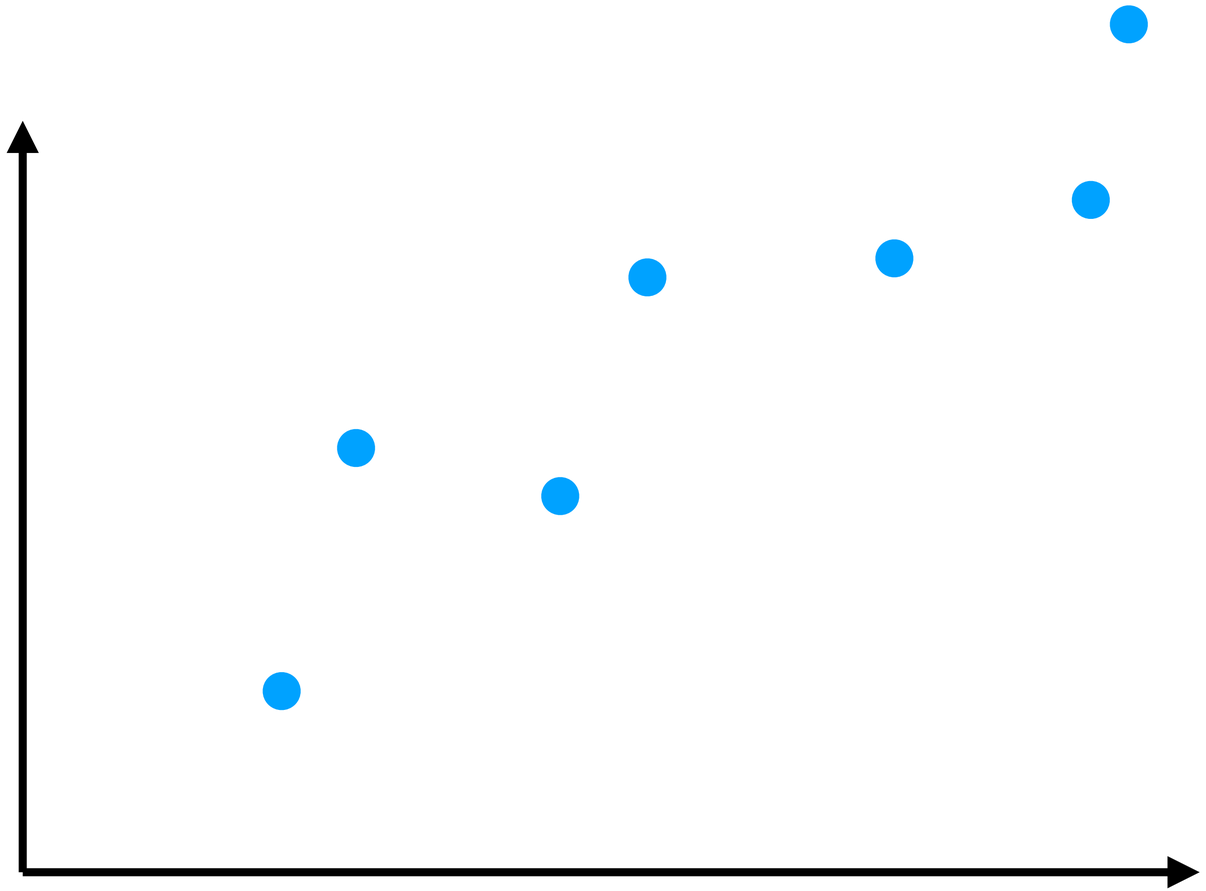


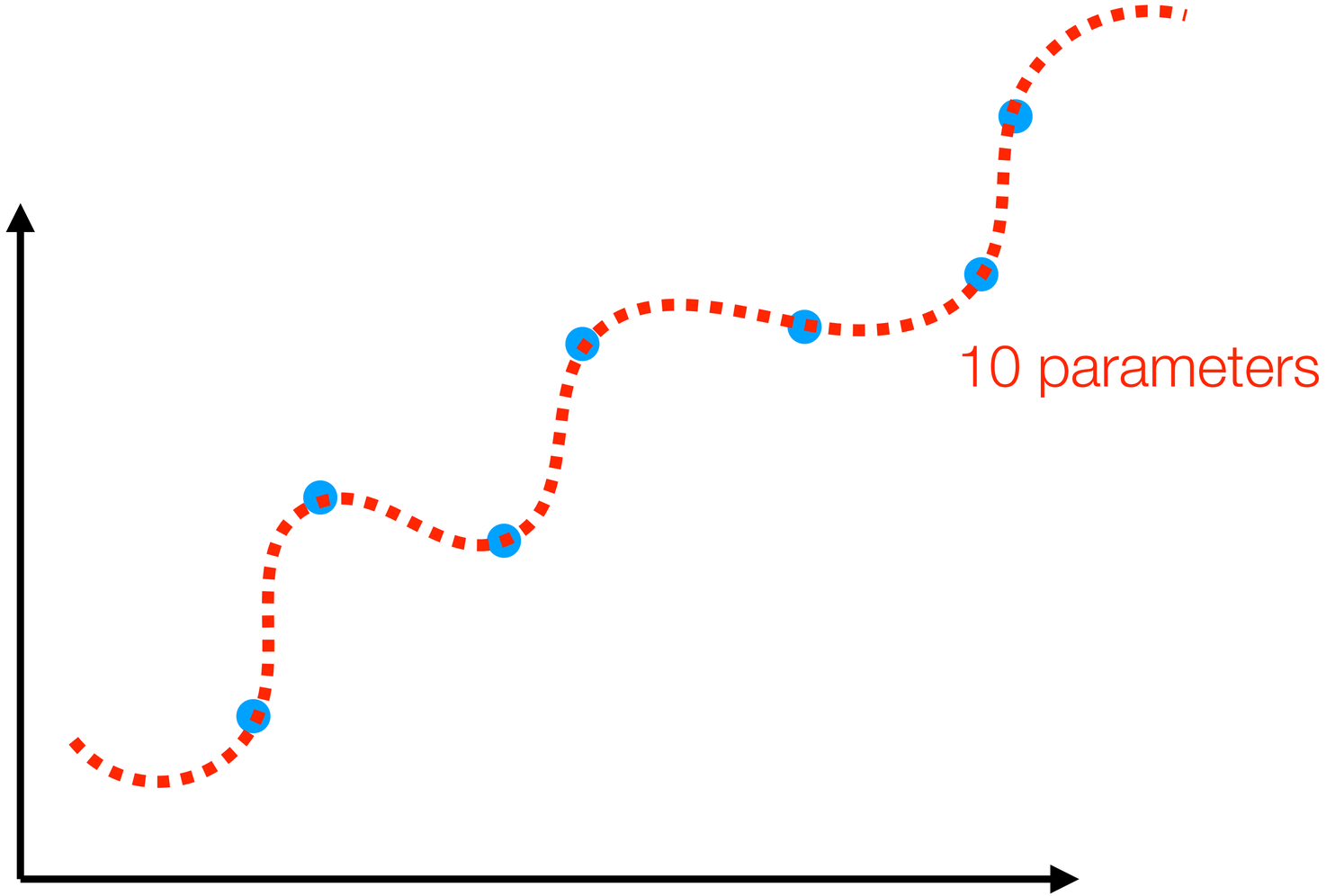
Classifying MNIST: how many parameters?

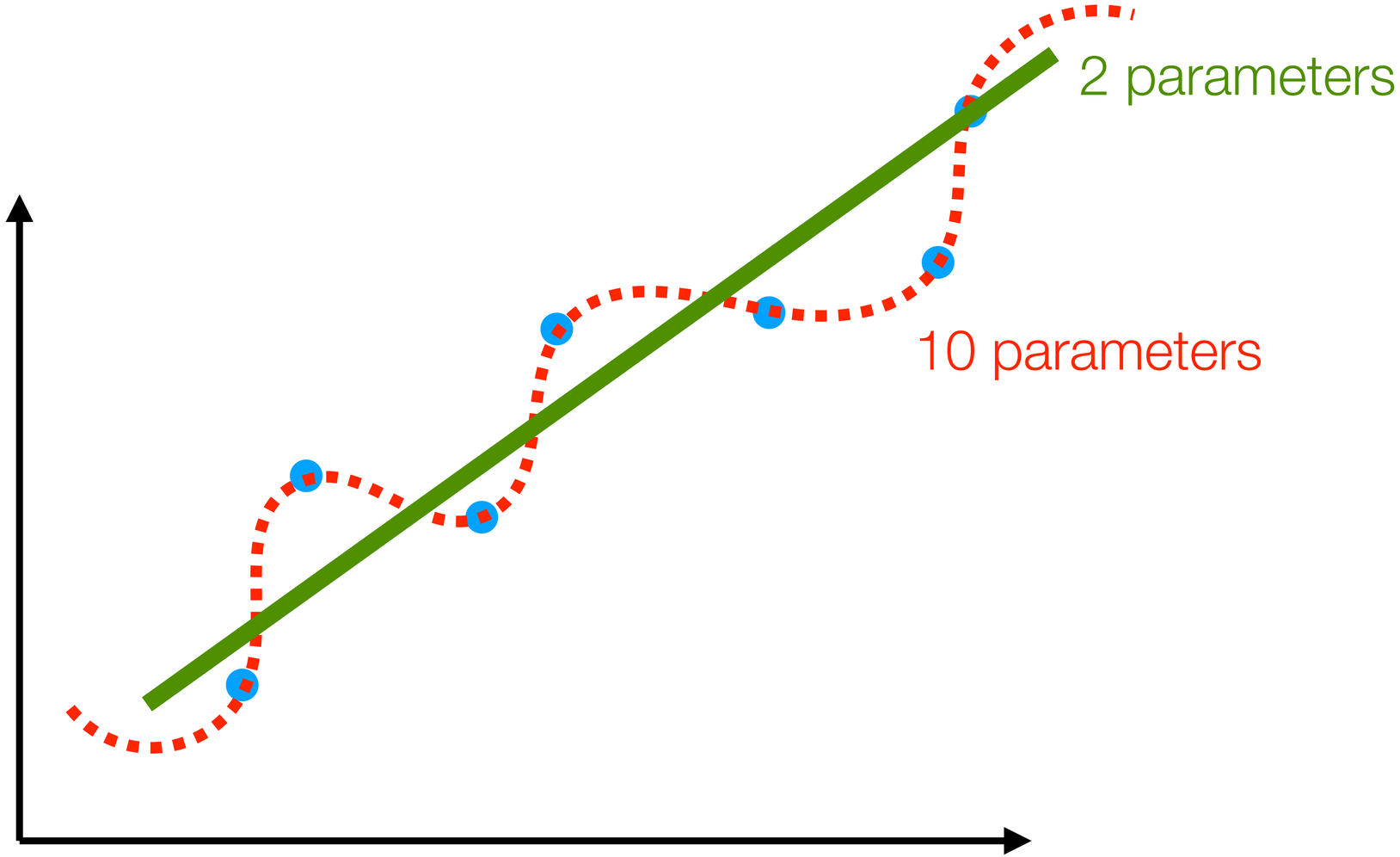


Classifying MNIST: how many parameters?

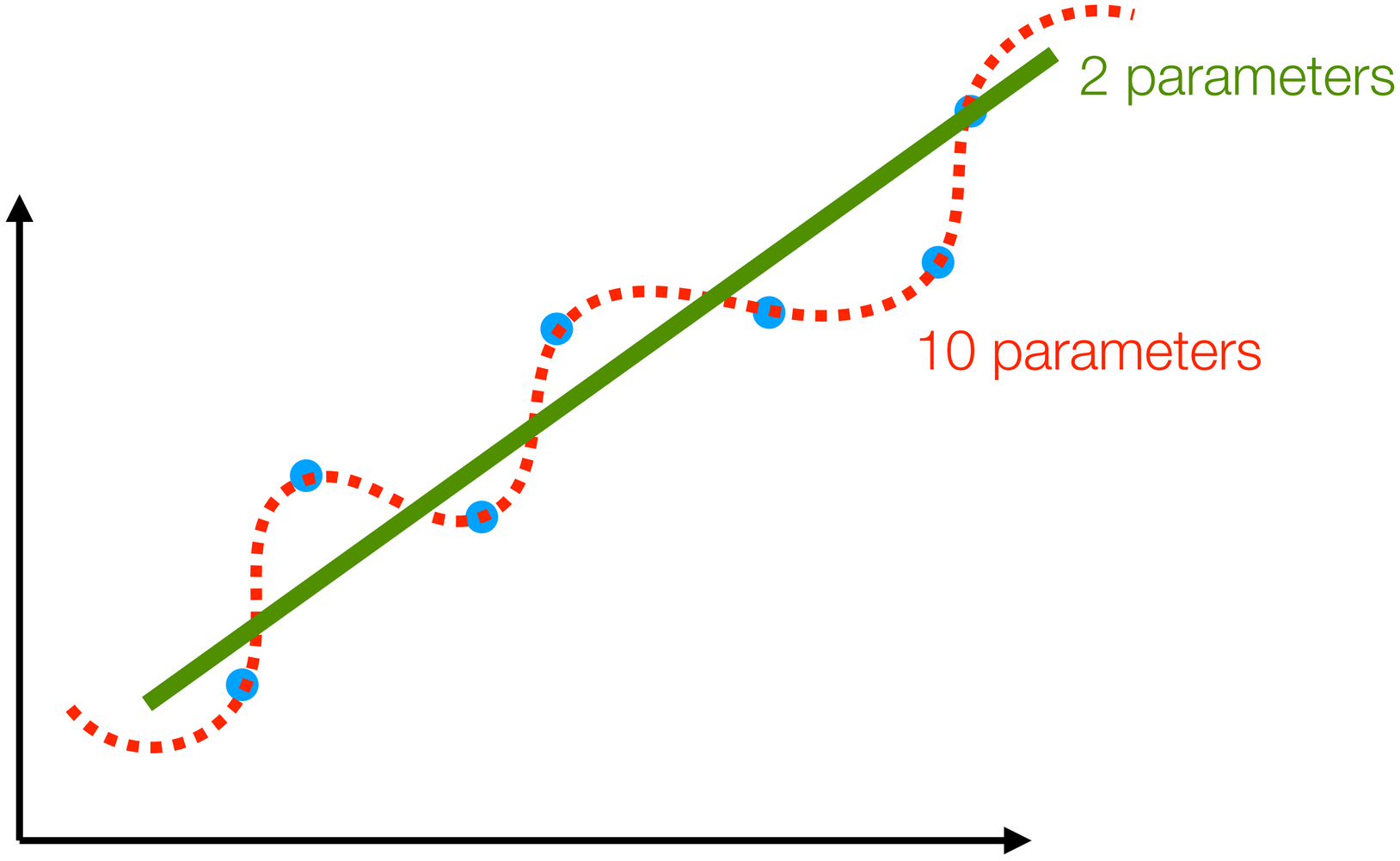





10 parameters

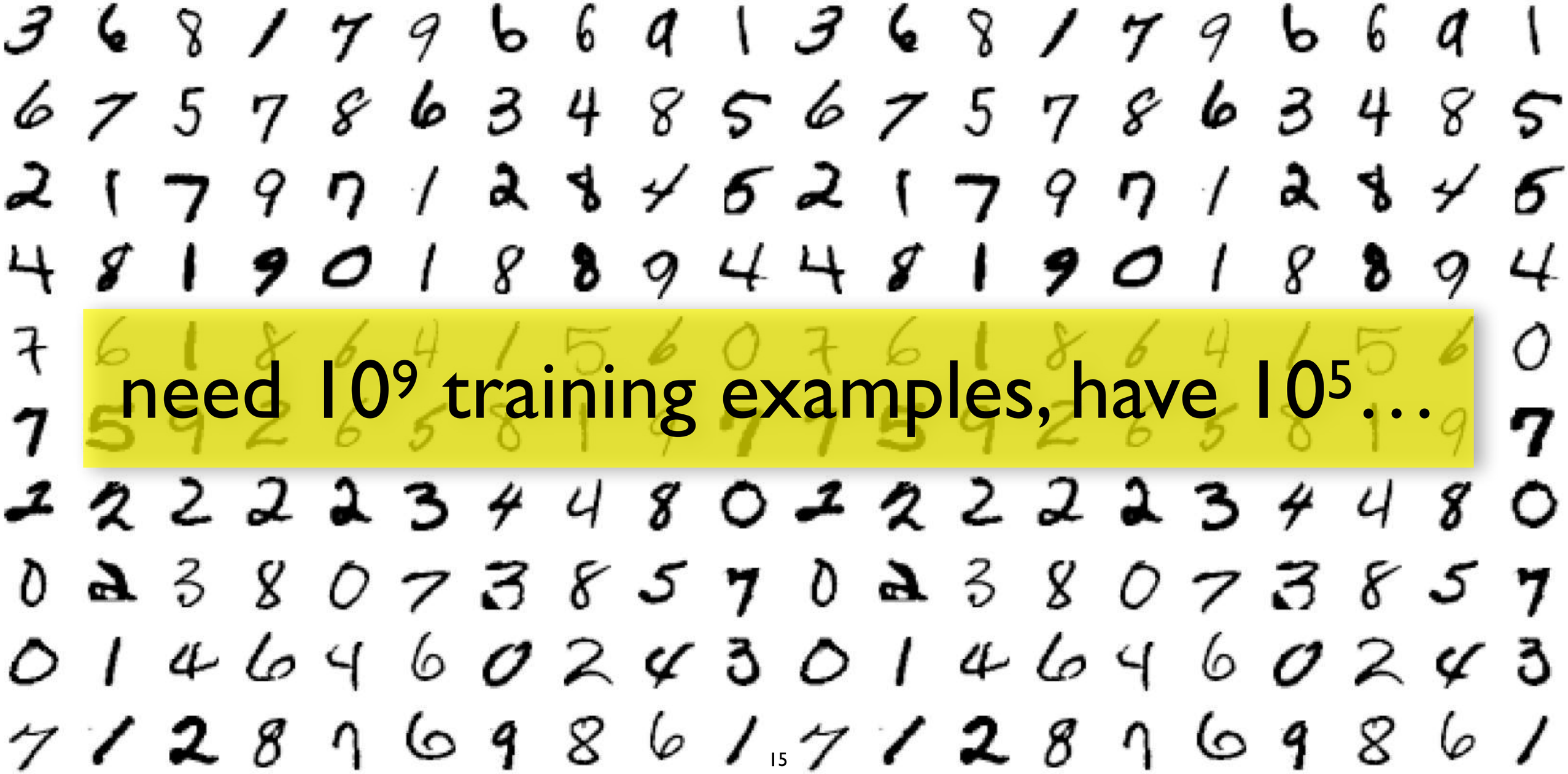

2 parameters
10 parameters

10^8 weights -> 10^9 training examples



MNIST dataset - 60K images

MNIST dataset - 60K images



need 10^9 training examples, have 10^5…

Training lots of weigths with few examples

Training lots of weigths with few examples

GENERAL ML IDEAS

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition
  - Sparsening (few nonz. weigths)

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition
  - Sparsening (few nonz. weigths)
- Data augmentation

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition
  - Sparsening (few nonz. weigths)
- Data augmentation

VISION-SPECIFIC IDEAS

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition
  - Sparsening (few nonz. weigths)
- Data augmentation

VISION-SPECIFIC IDEAS

- Local weights

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition
  - Sparsening (few nonz. weigths)
- Data augmentation

VISION-SPECIFIC IDEAS

- Local weights
- Weight sharing / conv. net

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition
  - Sparsening (few nonz. weigths)
- Data augmentation

VISION-SPECIFIC IDEAS

- Local weights
- Weight sharing / conv. net
- Subsampling

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition
  - Sparsening (few nonz. weigths)
- Data augmentation

VISION-SPECIFIC IDEAS

- Local weights
- Weight sharing / conv. net
- Subsampling
- Unsupervised-trained features

Training lots of weigths with few examples

GENERAL ML IDEAS

- Regularization
  - using prior on W
  - initial condition
  - Sparsening (few nonz. weigths)
- Data augmentation

VISION-SPECIFIC IDEAS

- Local weights ✔
- Weight sharing / conv. net
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Training lots of weigths with few examples

GENERAL ML IDEAS

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  - initial condition
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- Data augmentation

VISION-SPECIFIC IDEAS

- Local weights ✓
- Weight sharing / conv. net ✓
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  - Sparsening (few nonz. weigths)
- Data augmentation

VISION-SPECIFIC IDEAS

- Local weights ✓
- Weight sharing / conv. net ✓
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Training lots of weigths with few examples

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  - initial condition
  - Sparsening (few nonz. weigths)
- Data augmentation ✓

VISION-SPECIFIC IDEAS

- Local weights ✓
- Weight sharing / conv. net ✓
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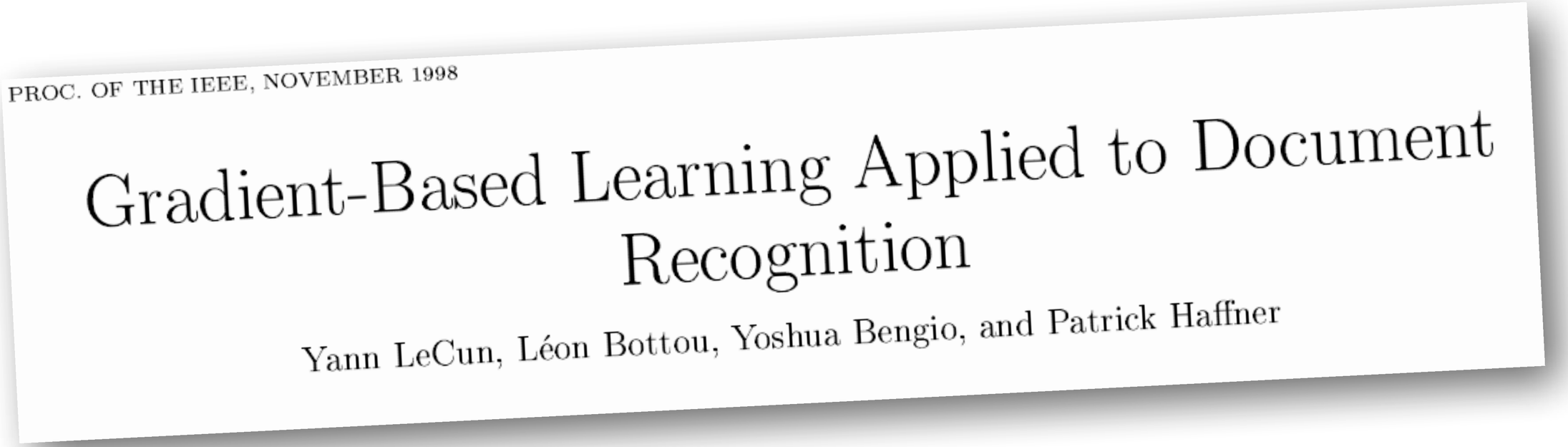


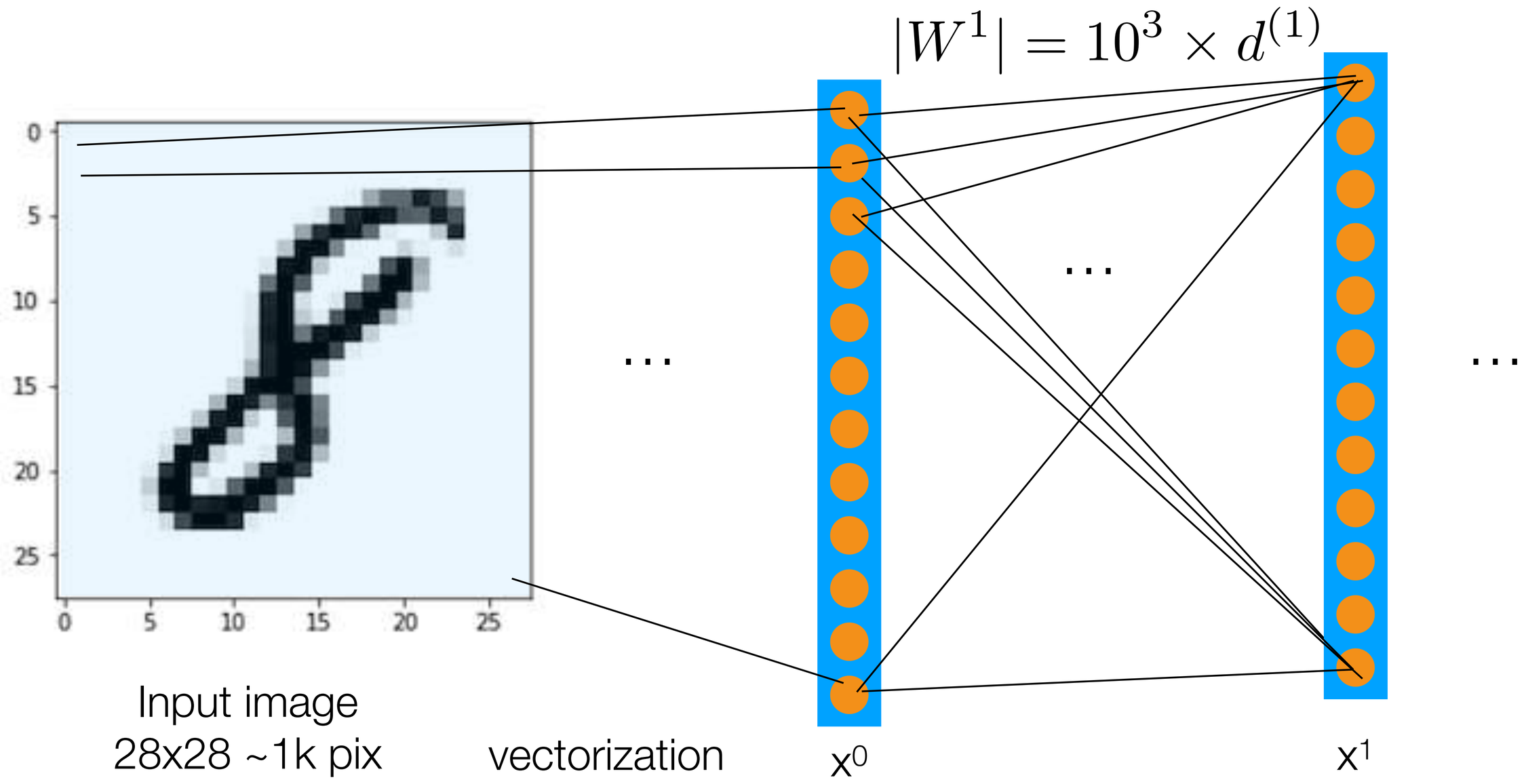
PROC. OF THE IEEE, NOVEMBER 1998

Gradient-Based Learning Applied to Document Recognition

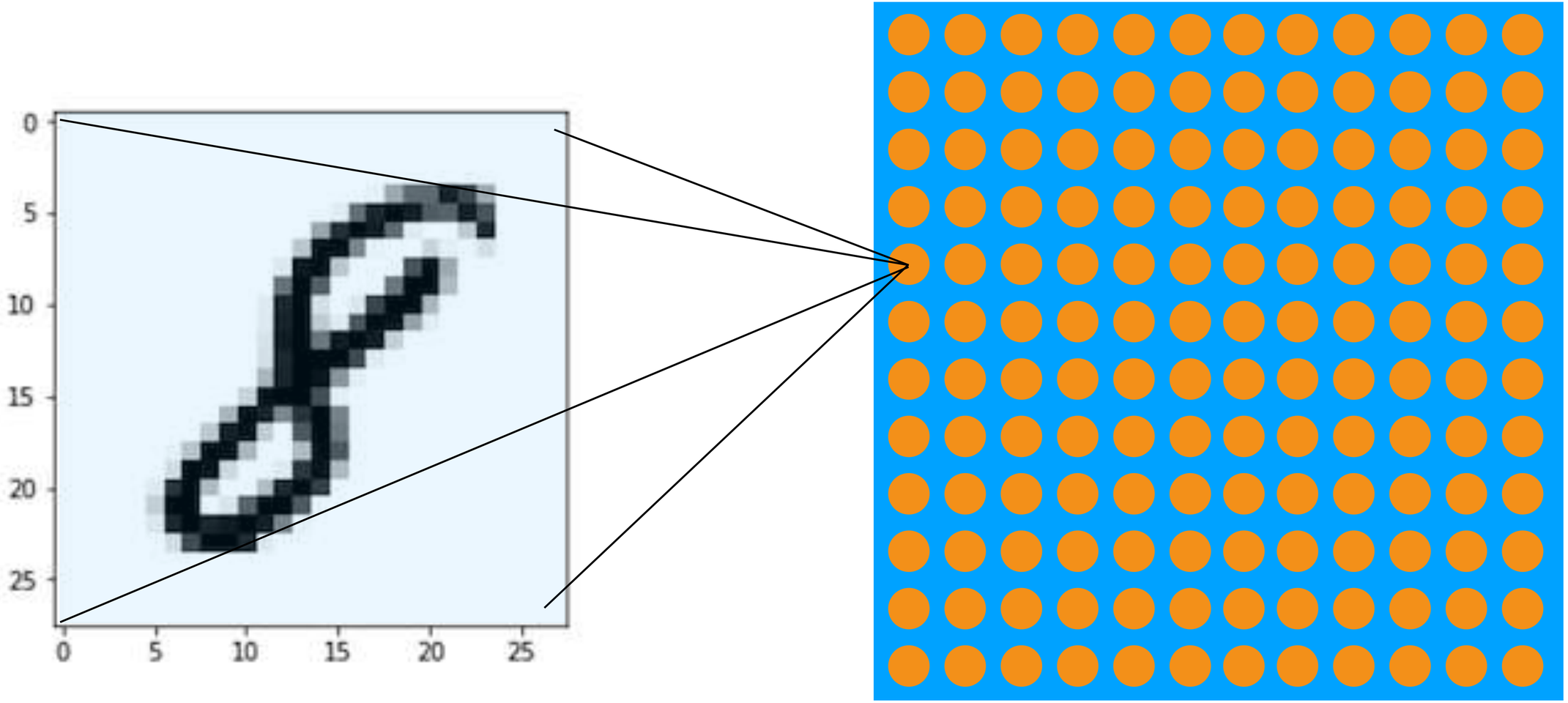
Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner

[Read pages 1-18 and relevant appendices]

Fully-connected deep neural net (FCDNN)



Idea 0: 2D maps

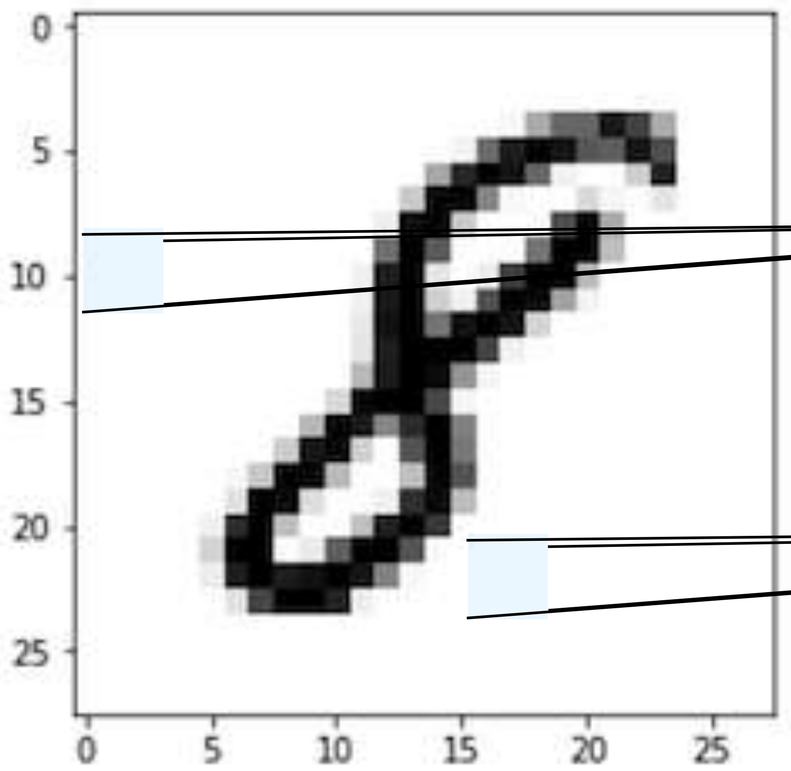


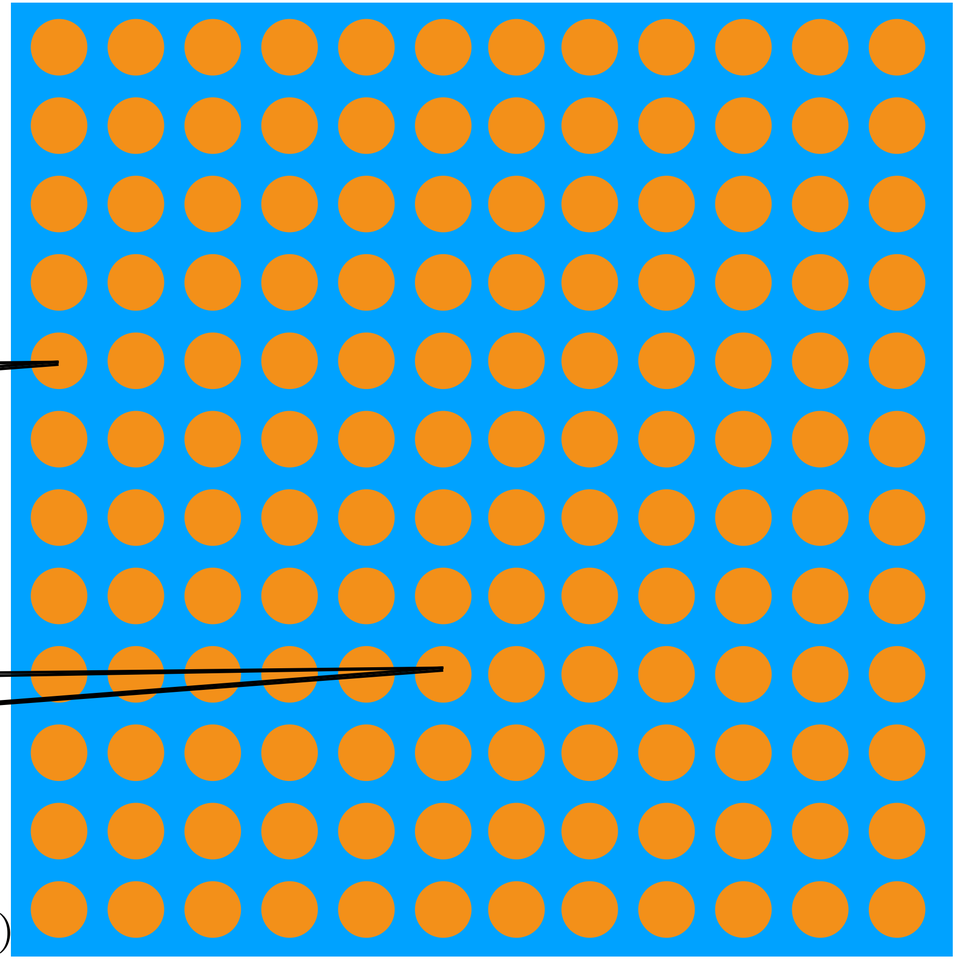
x^0

$$|W^1| = 10^3 \times d^{(1)}$$

x^1

Idea 1: Local weights





$$|W^1| = 3 \times 3 \times d^{(1)}$$

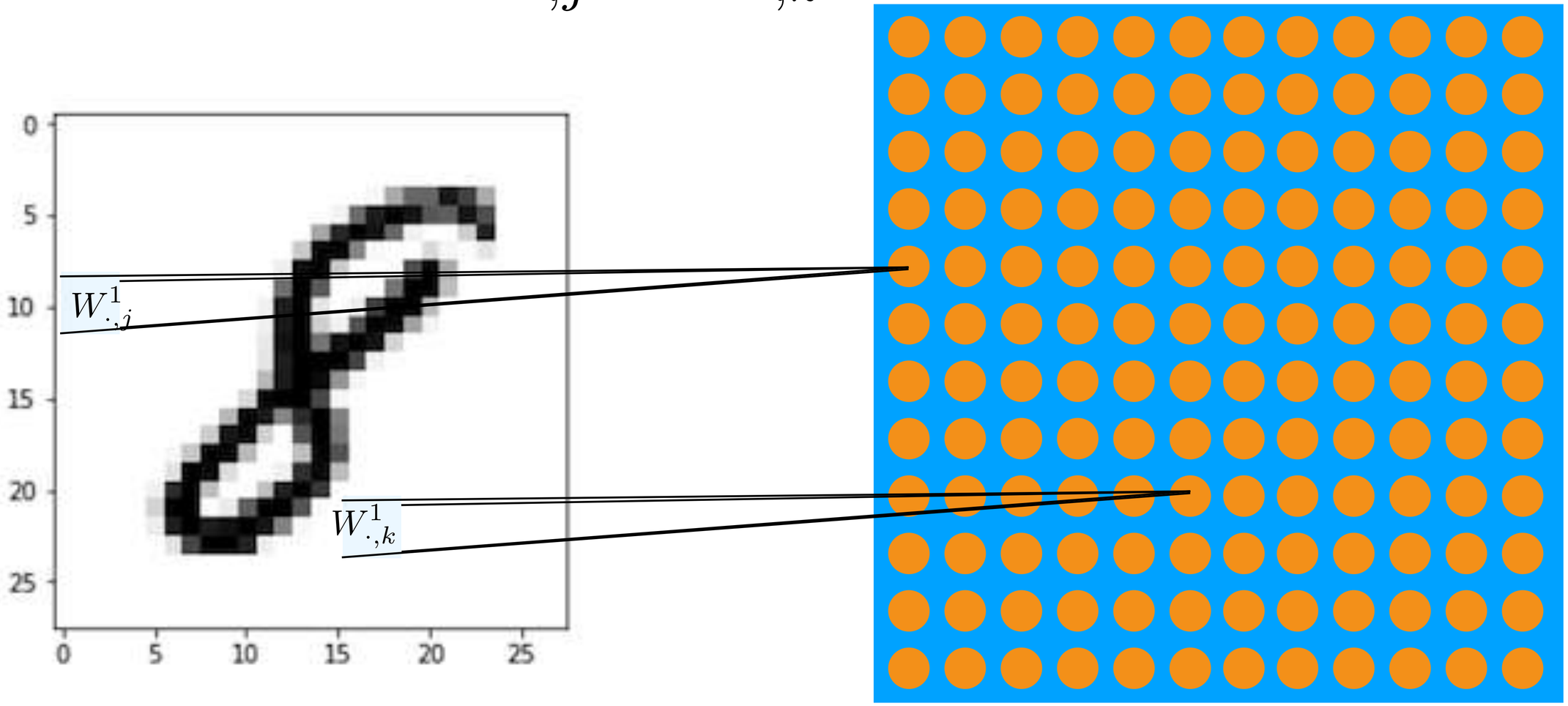
$$\approx 10^1_{20} \times d^{(1)}$$

x^1

x^2

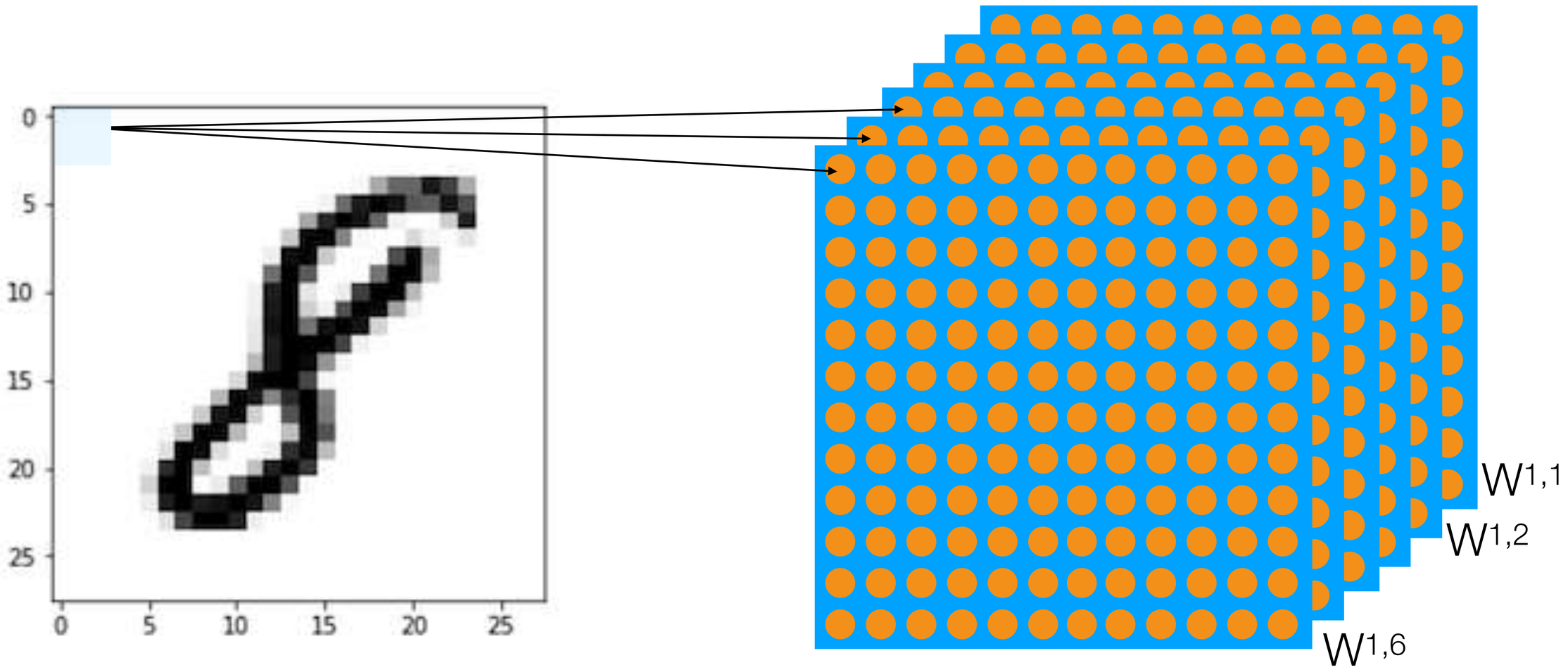
Idea 2: Convolutional weights

$$W^1_{\cdot,j} = W^1_{\cdot,k}$$



x^1 $|W^1| = 3 \times 3 \approx 10^1$ x^2

Idea 3: Channels

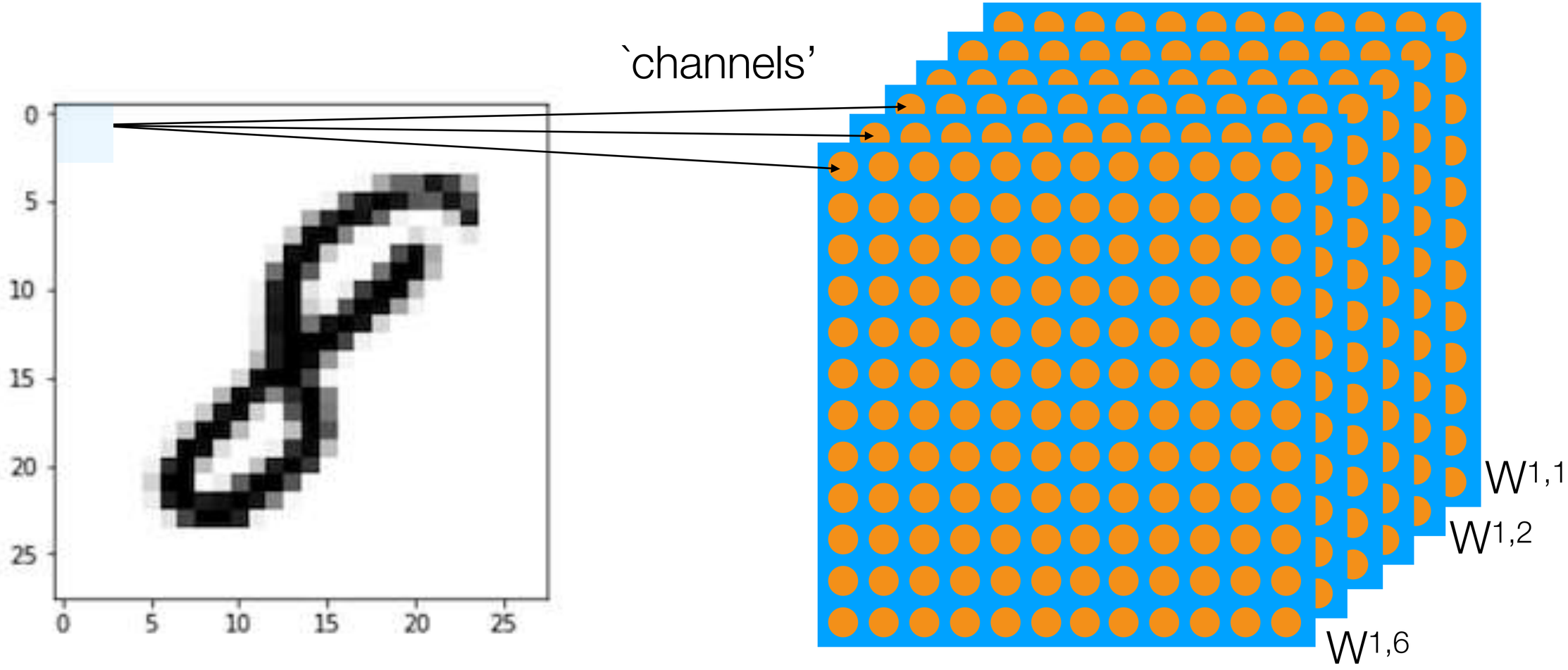


x^1

$$|W^1| = 10^1_{22} \times C$$

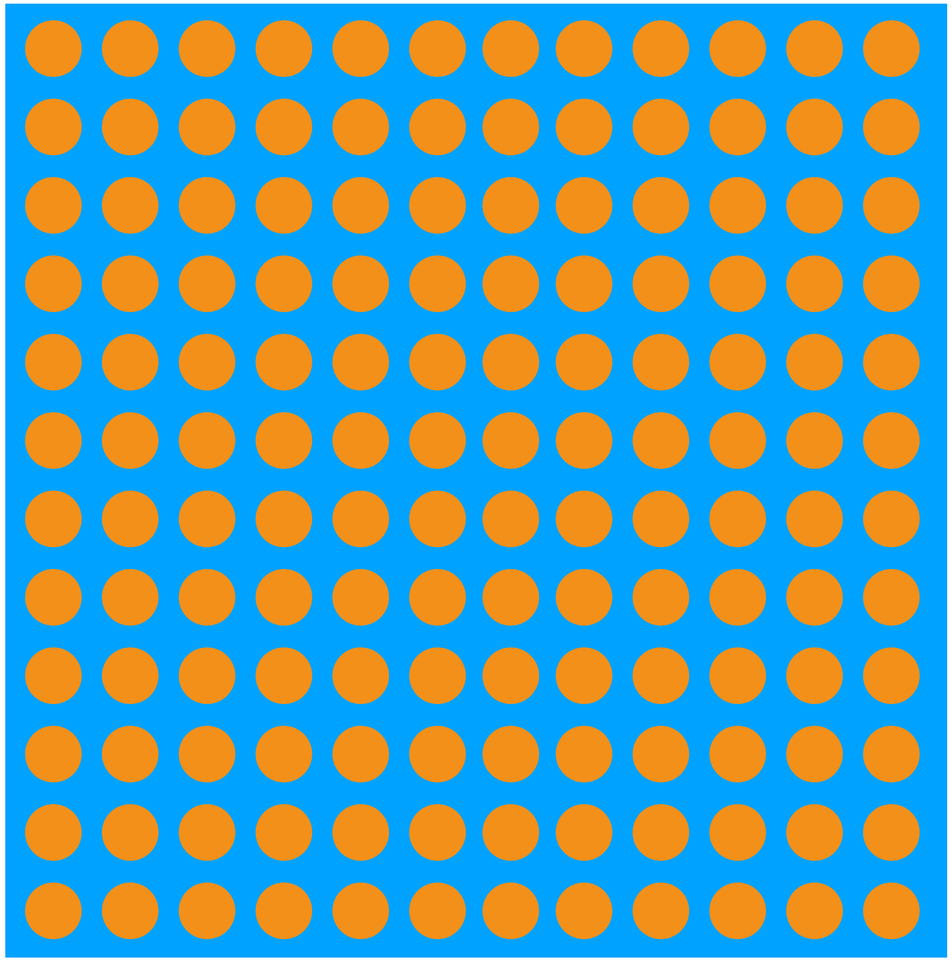
x^2

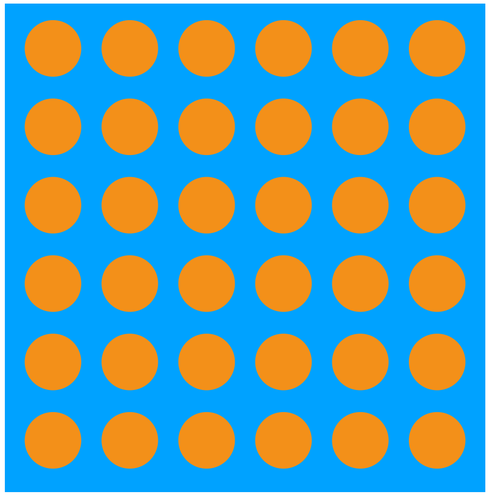
Idea 3: Channels



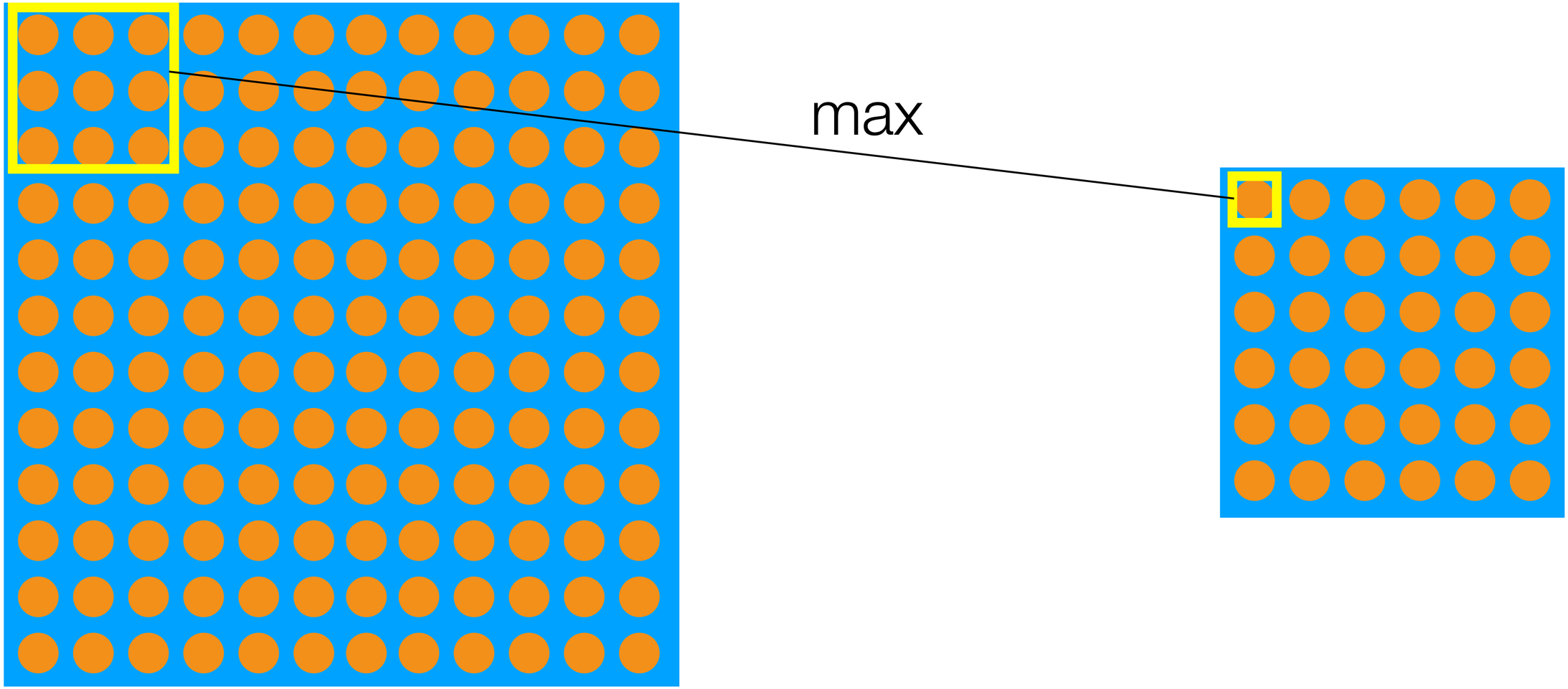
$$|W^1| = 10^1_{22} \times C$$

Idea 4: Subsampling

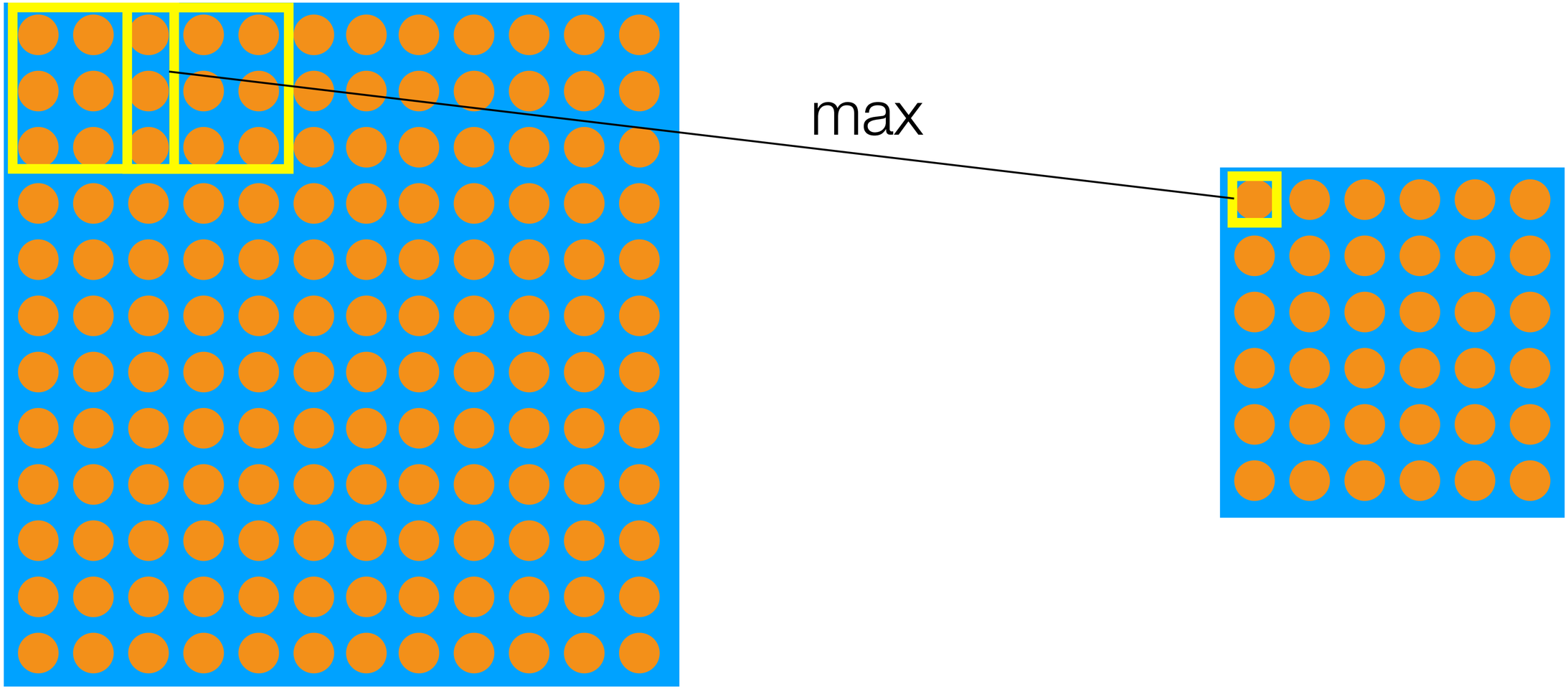




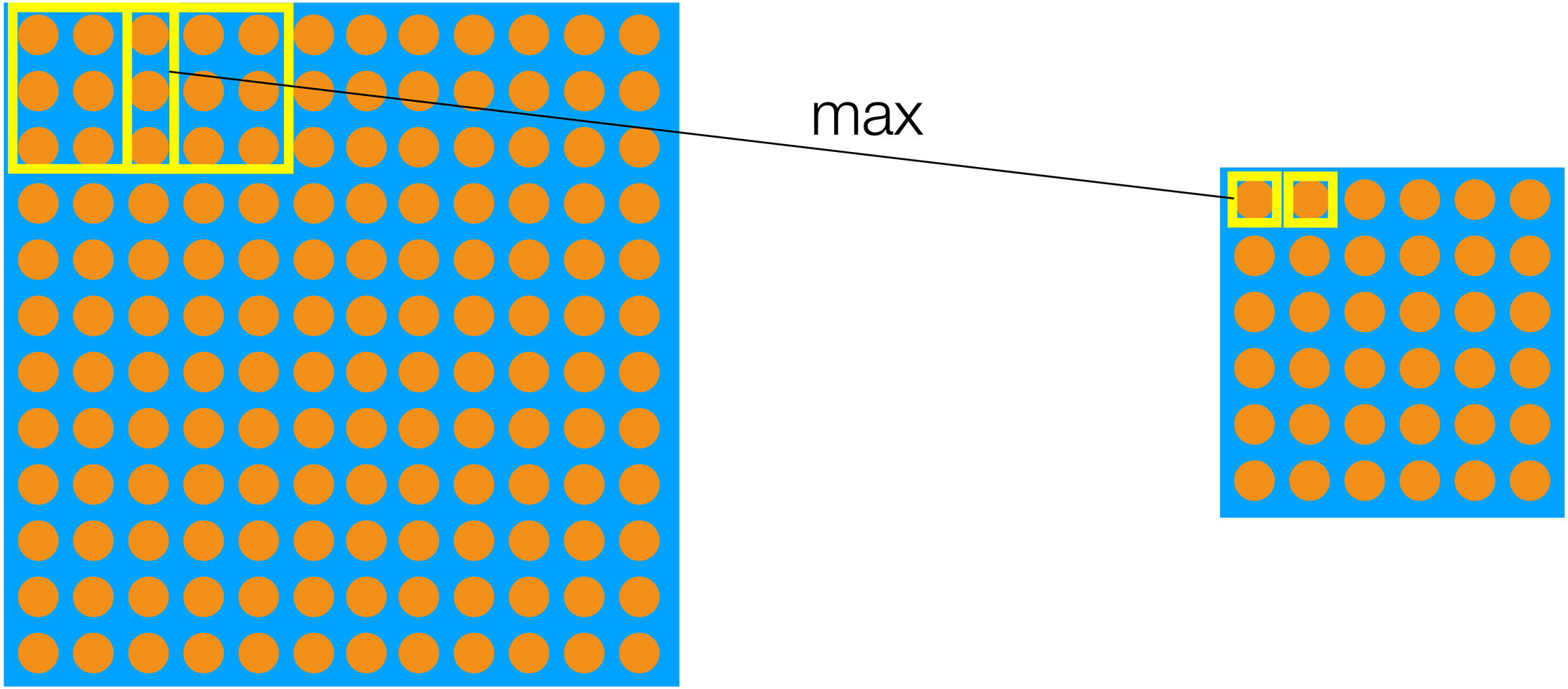
Idea 4: Subsampling



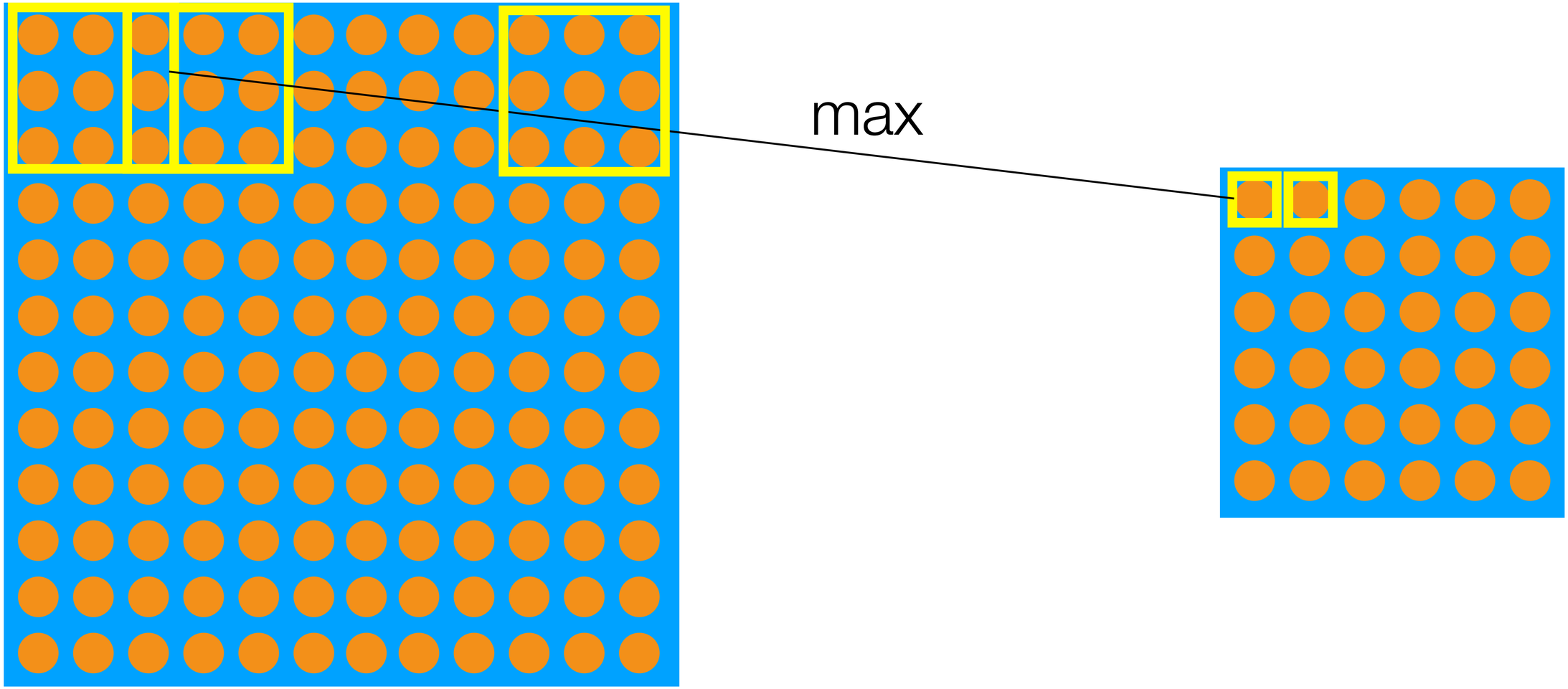
Idea 4: Subsampling



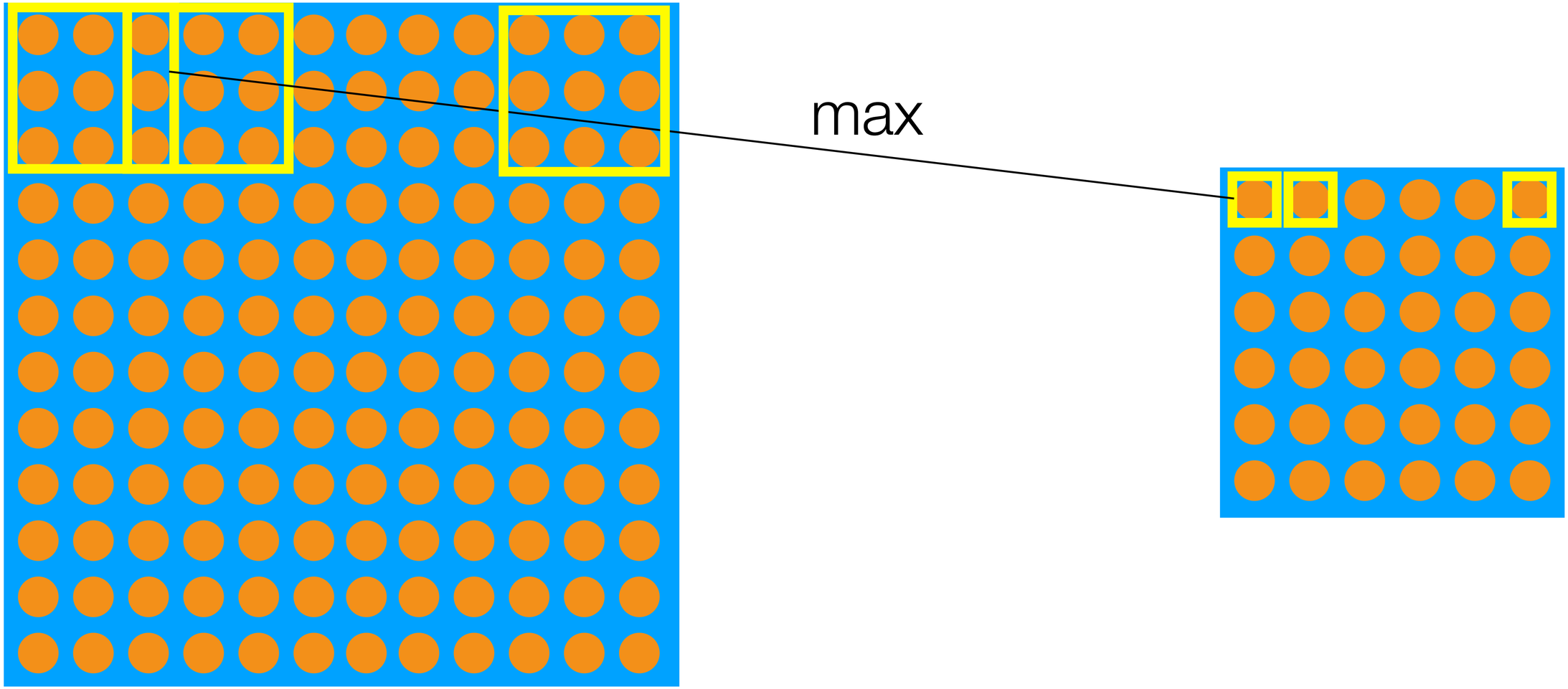
Idea 4: Subsampling



Idea 4: Subsampling



Idea 4: Subsampling



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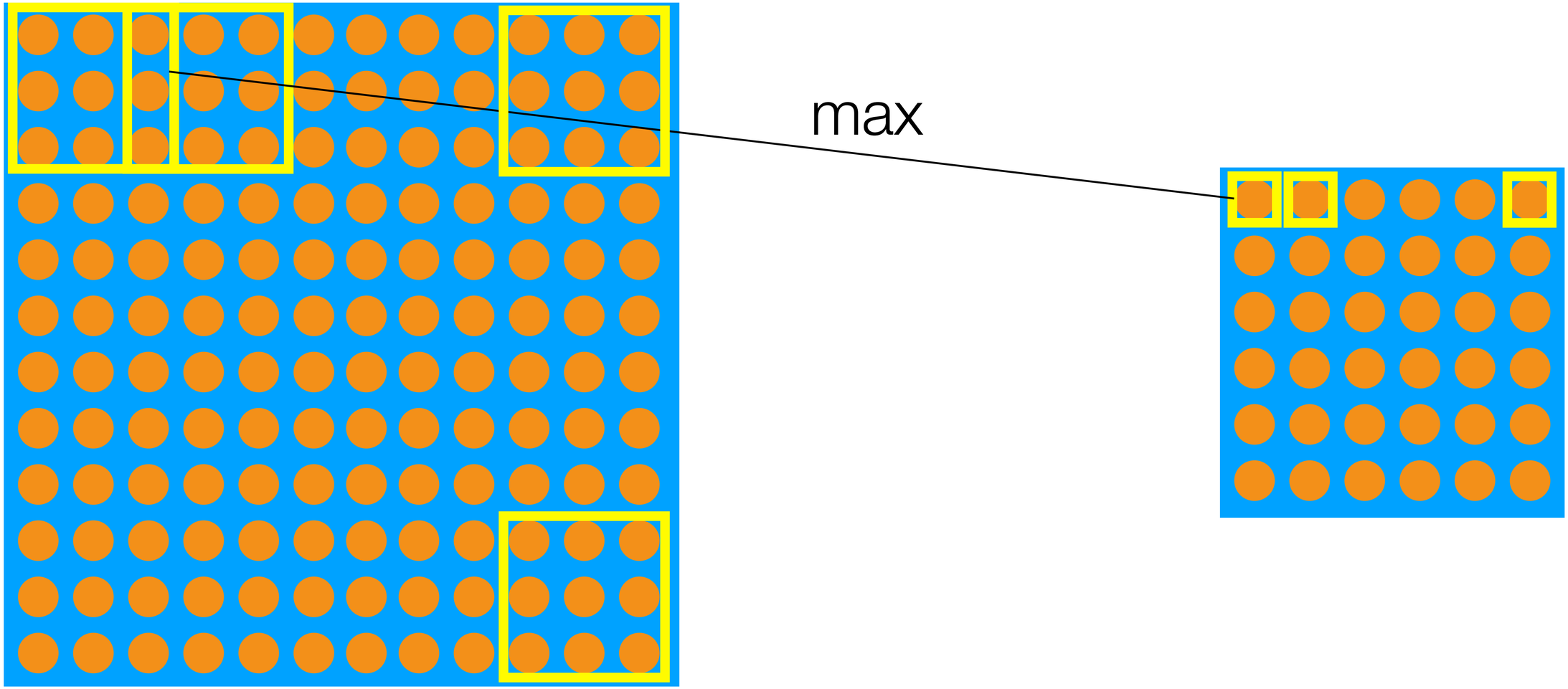
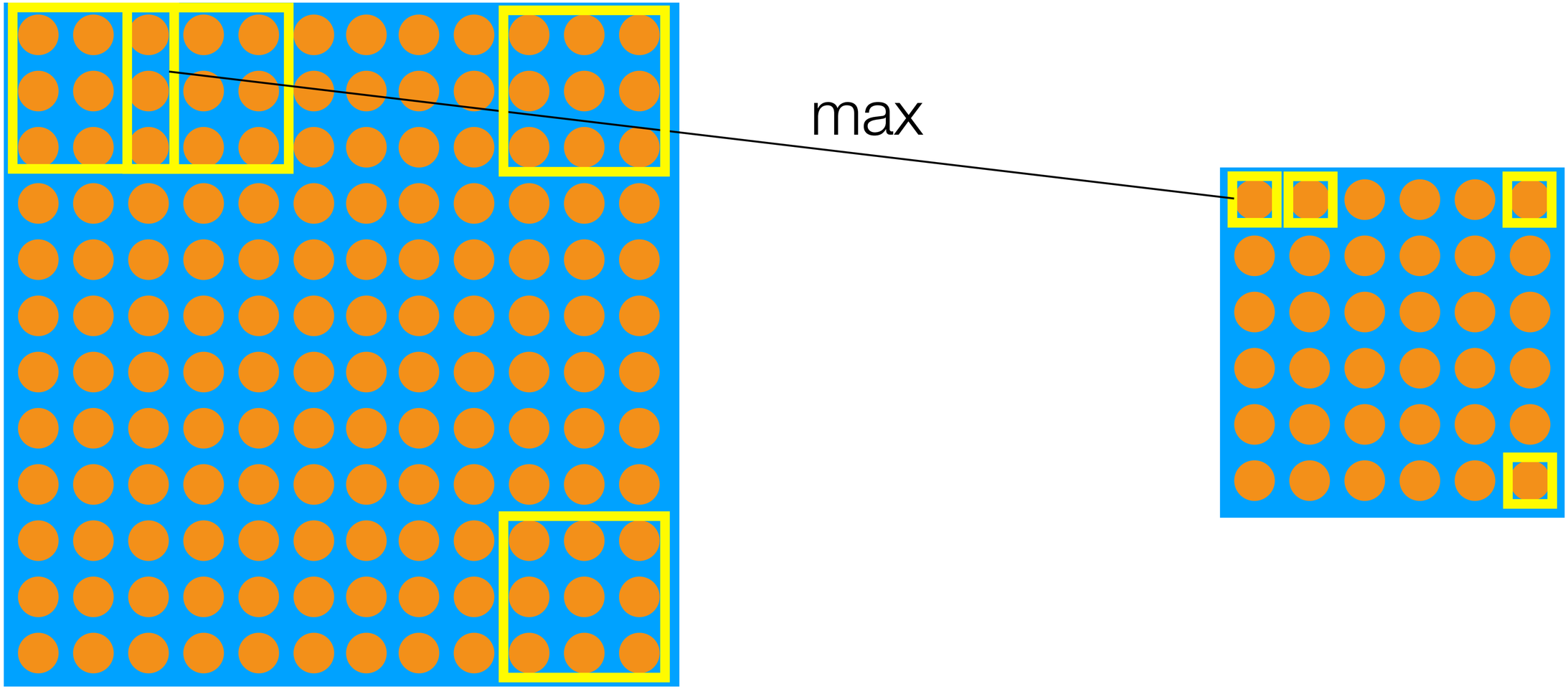
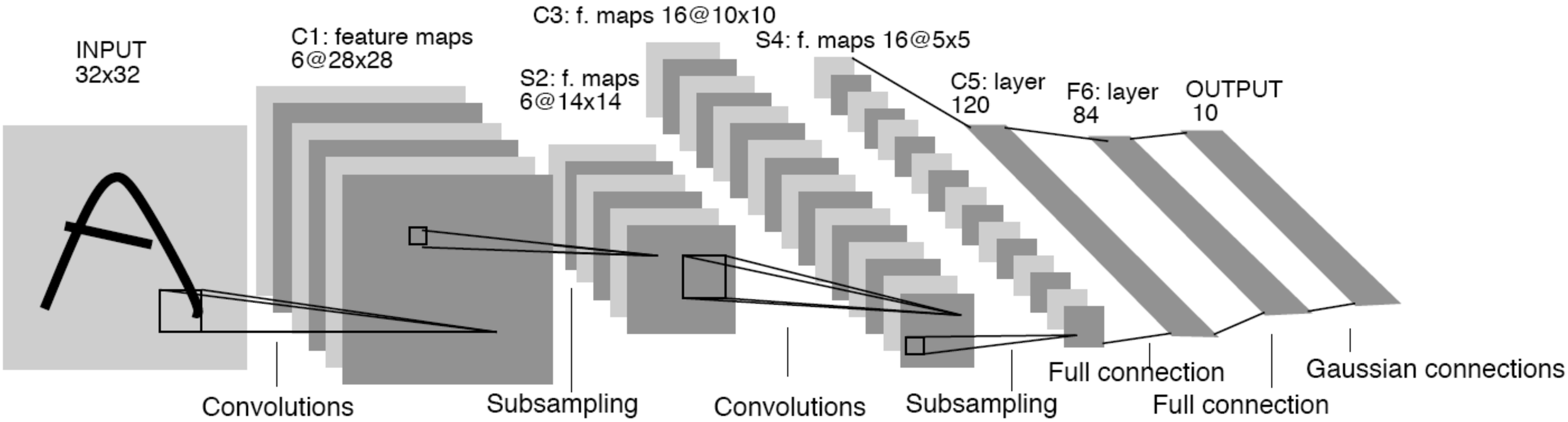


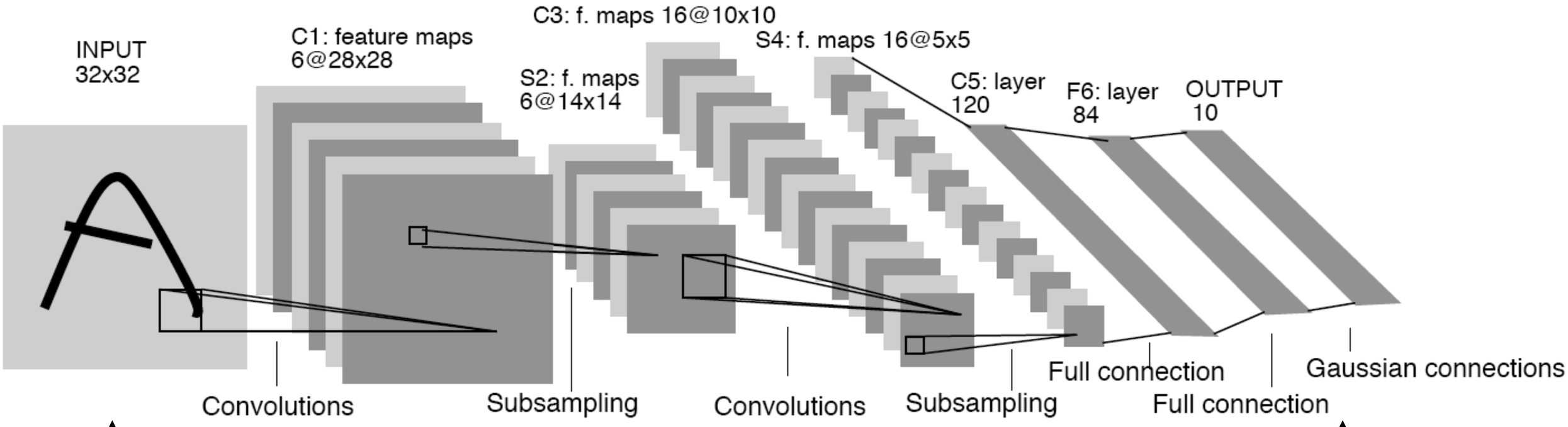
Idea 4: Subsampling



LeNet


INPUT
32x32
C1: feature maps
6@28x28
S2: f. maps
6@14x14
C3: f. maps 16@10x10
S4: f. maps 16@5x5
C5: layer
120
F6: layer
84
OUTPUT
10
Convolutions
Subsampling
Convolutions
Subsampling
Full connection
Full connection
Gaussian connections

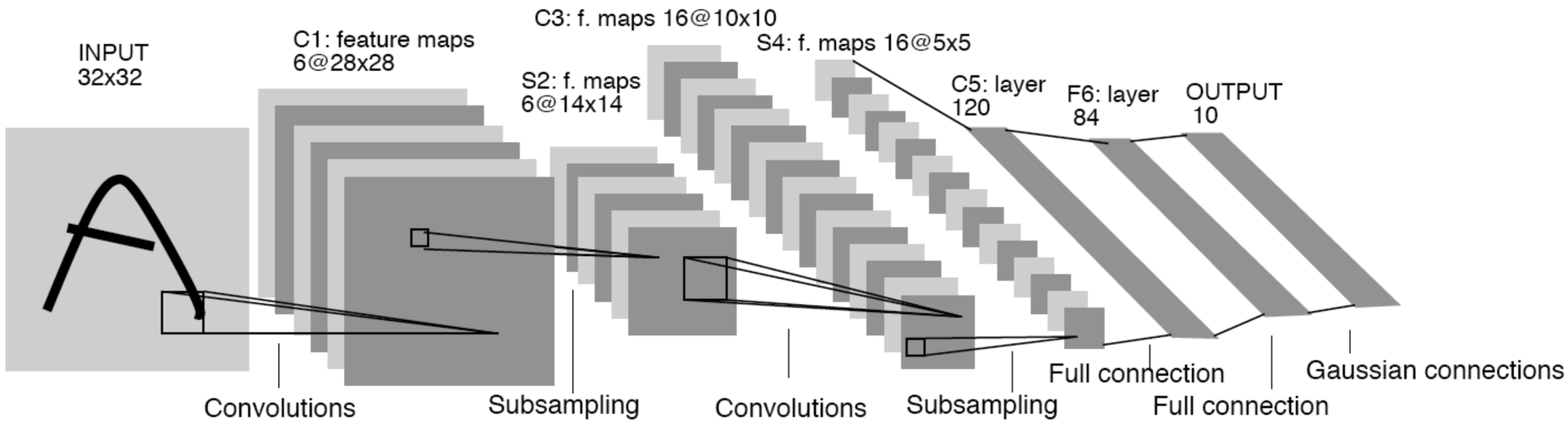
LeNet



28 x 28 pixels
1 channel

1 x 1 pixels
10 channels

LeNet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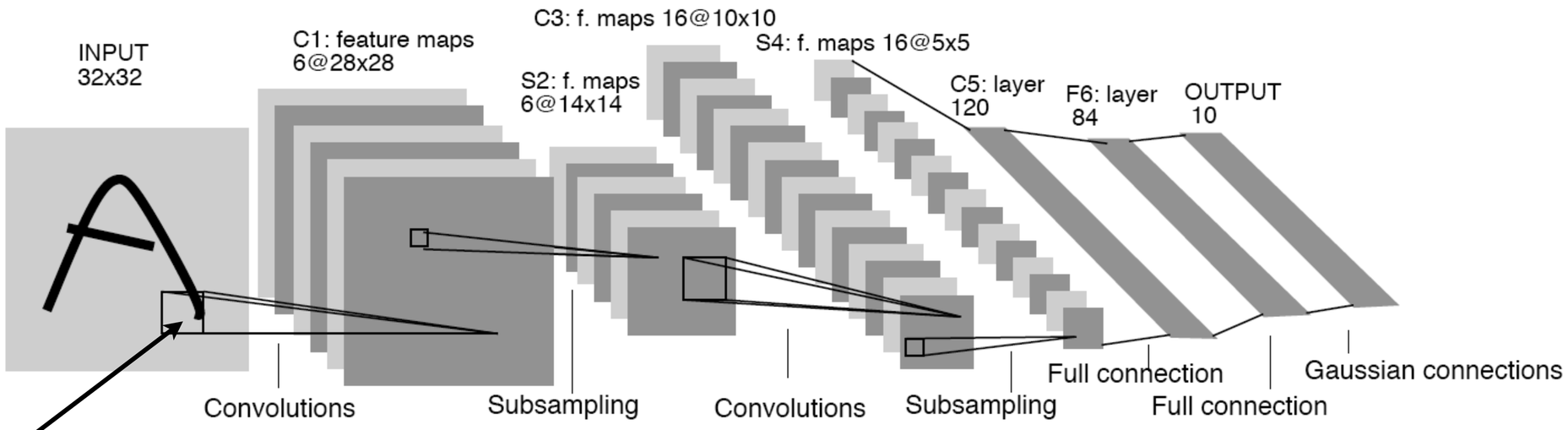
LeNet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Subsampling
Convolutions
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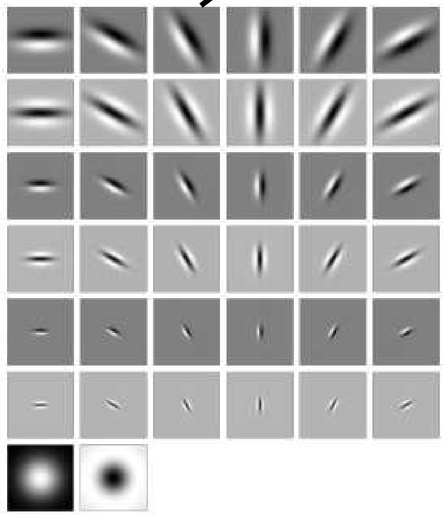
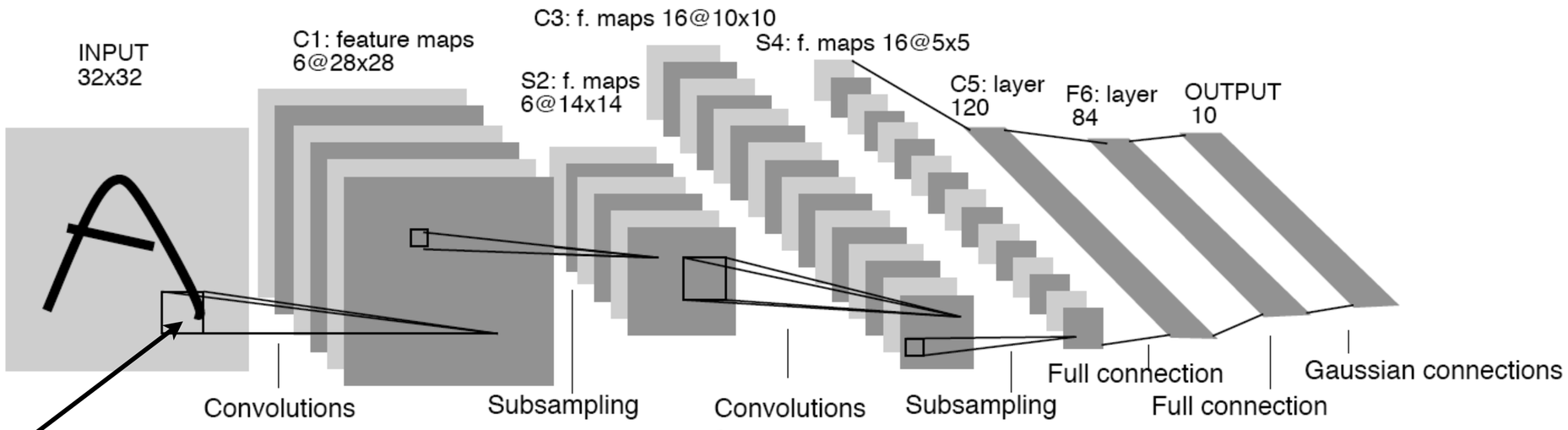
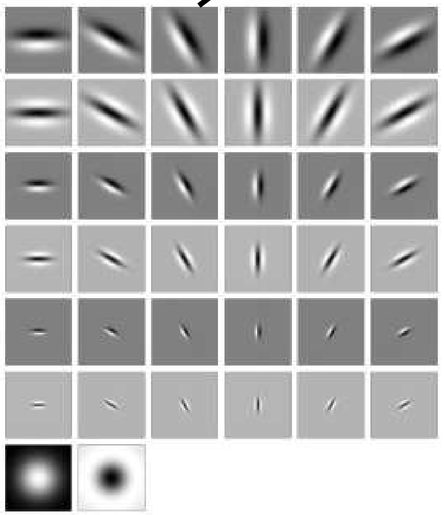
Full connection
Full connection
Gaussian connections



LeNet



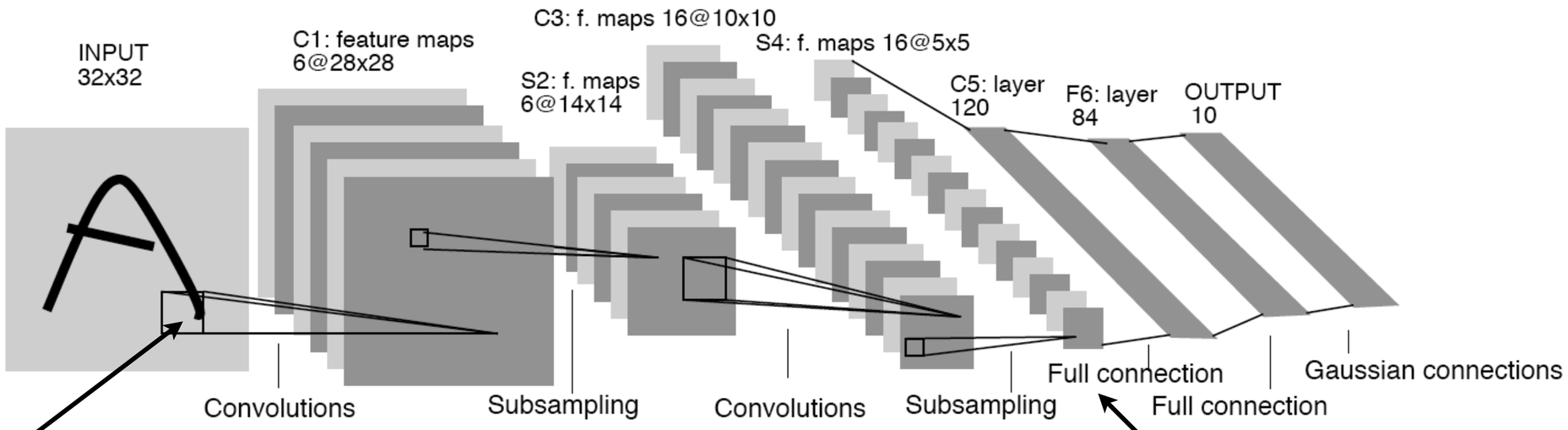


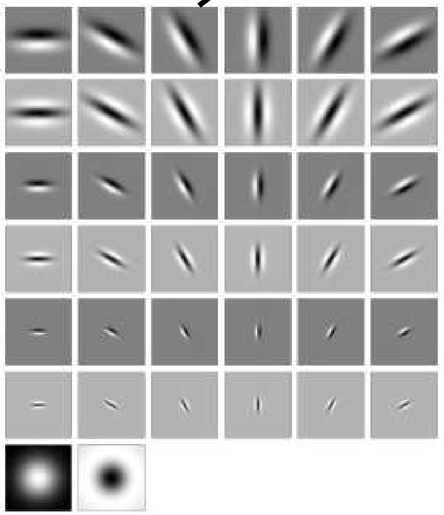
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	X				X	X	X			X	X	X	X		X	X
1	X	X				X	X	X			X	X	X	X		X
2	X	X	X				X	X	X			X		X	X	X
3		X	X	X			X	X	X	X			X		X	X
4			X	X	X			X	X	X	X		X	X		X
5				X	X	X			X	X	X	X		X	X	X

TABLE I

EACH COLUMN INDICATES WHICH FEATURE MAP IN S2 ARE COMBINED BY THE UNITS IN A PARTICULAR FEATURE MAP OF C3.

LeNet





	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	X				X	X	X			X	X	X	X		X	X
1	X	X				X	X	X			X	X	X	X		X
2	X	X	X				X	X	X			X		X	X	X
3		X	X	X			X	X	X	X			X		X	X
4			X	X	X			X	X	X	X		X	X		X
5				X	X	X			X	X	X	X		X	X	X

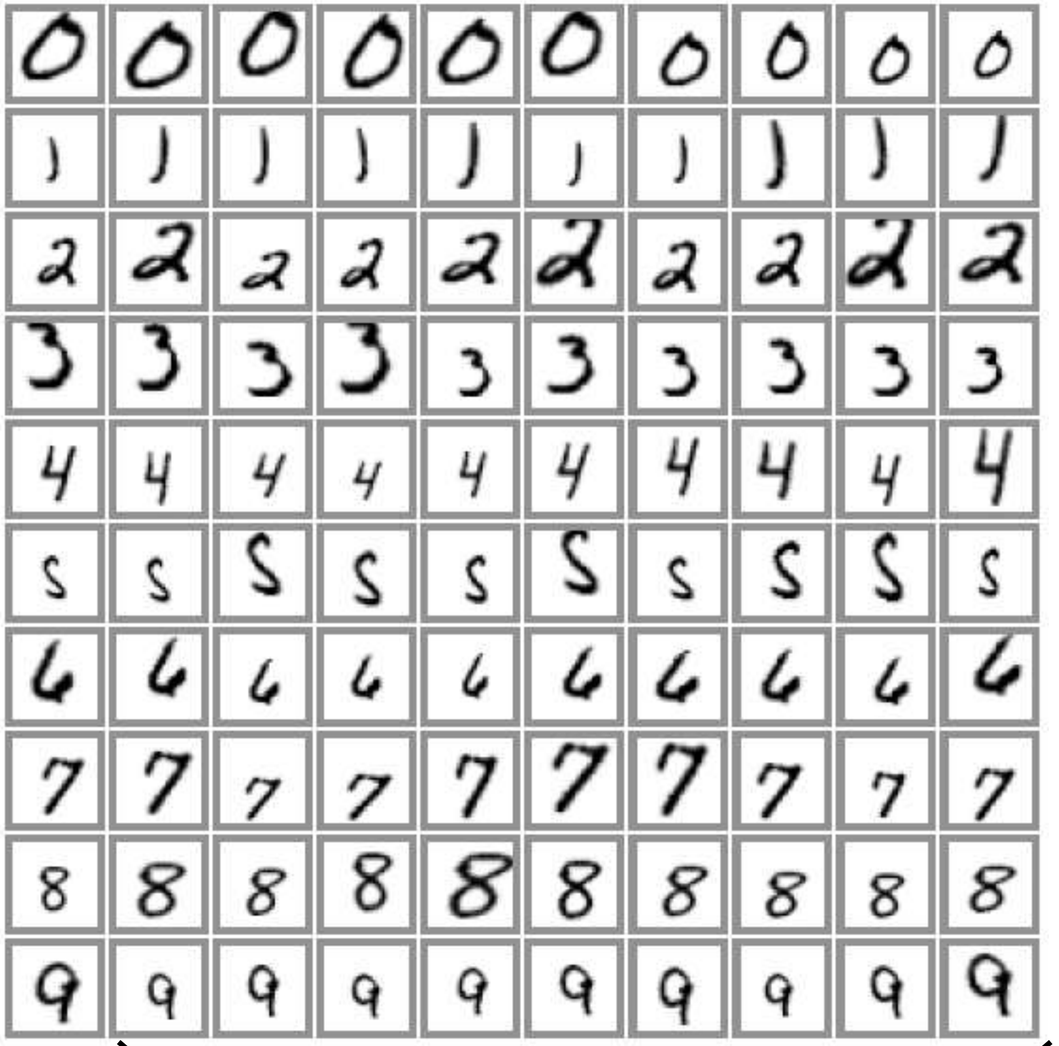
TABLE I

EACH COLUMN INDICATES WHICH FEATURE MAP IN S2 ARE COMBINED BY THE UNITS IN A PARTICULAR FEATURE MAP OF C3.

Layer C5 is a convolutional layer with 120 feature maps. Each unit is connected to a 5x5 neighborhood on all 16 of S4's feature maps. Here, because the size of S4 is also 5x5, the size of C5's feature maps is 1x1: this amounts to a full connection between S4 and C5. C5 is labeled

NIST training set

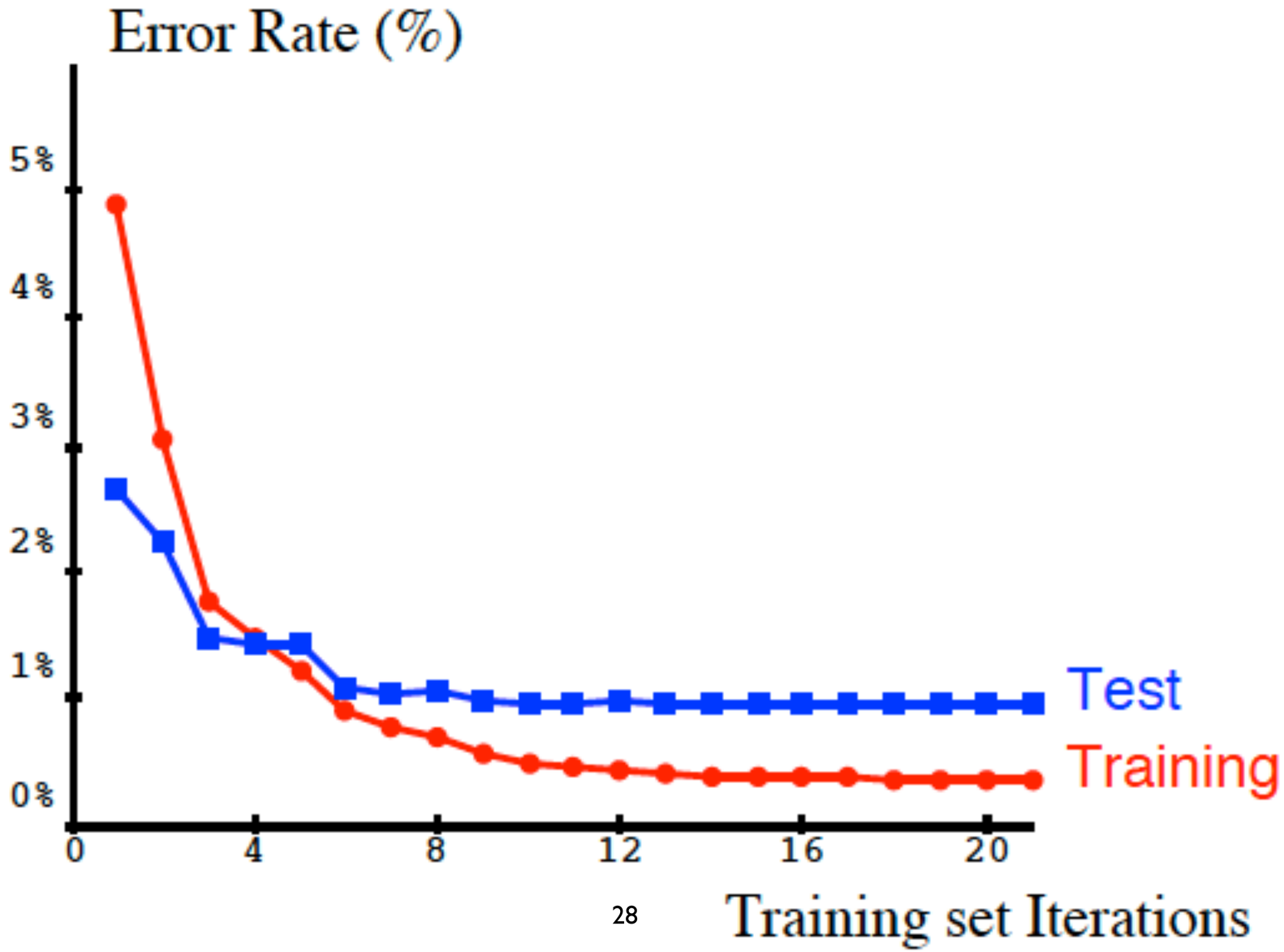
Idea 5: Data augmentation



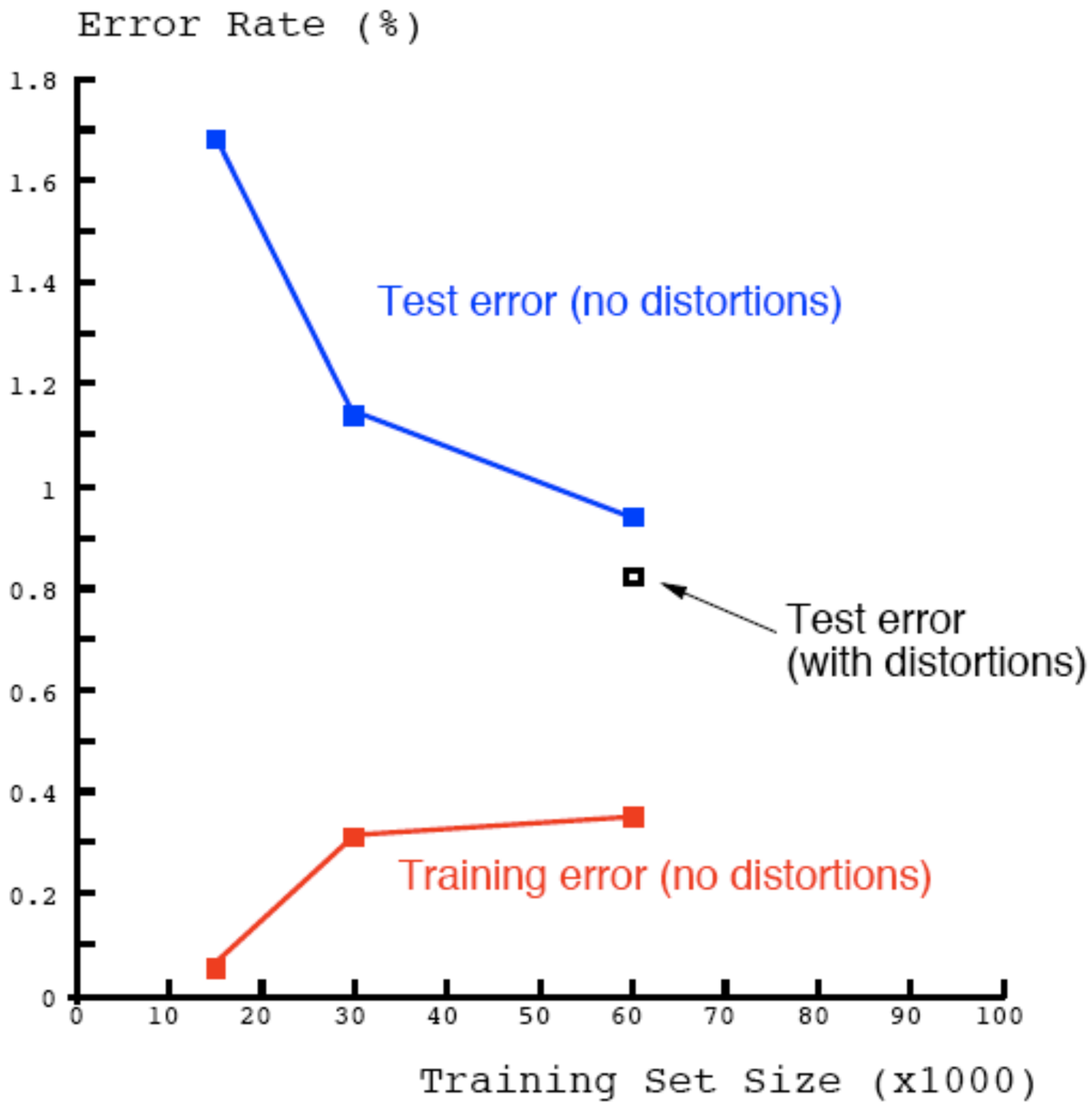
original example

synthetic distortions

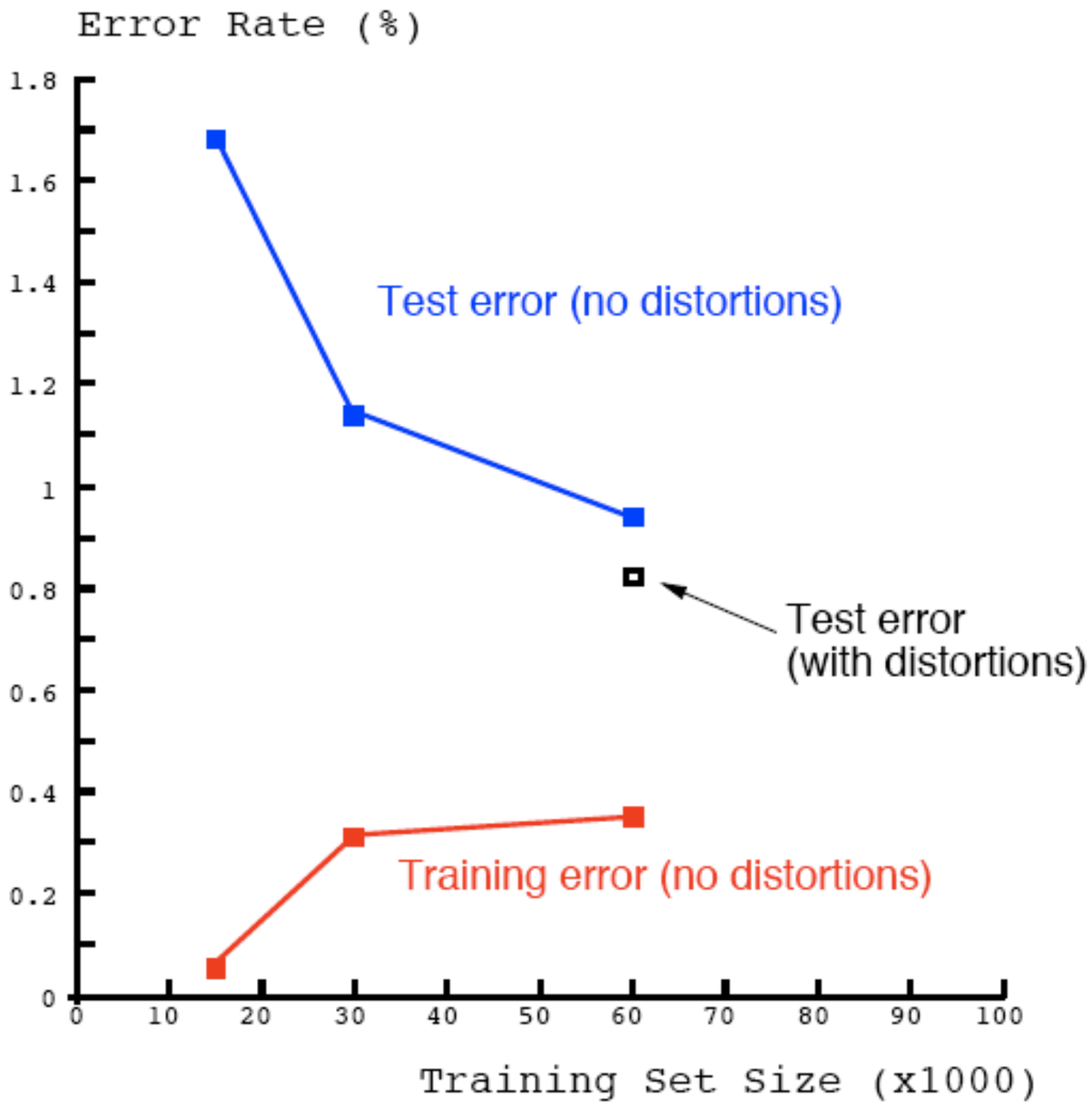
Training by error backpropagation



Errors on test set

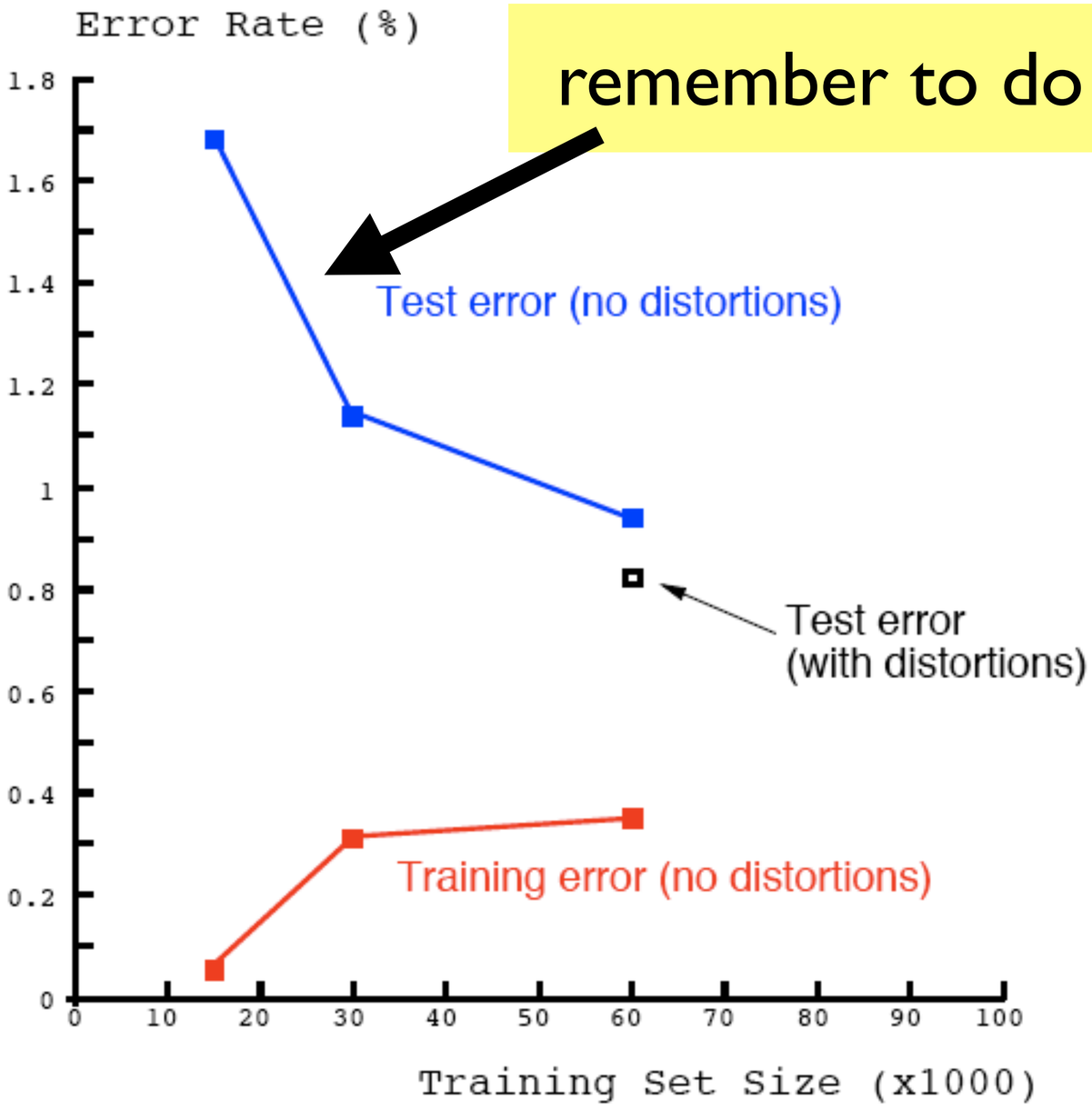


Errors on test set



Errors on test set

remember to do this in your projects



Fast forward to 2009

ImageNet: A Large-Scale Hierarchical Image Database

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li and Li Fei-Fei
Dept. of Computer Science, Princeton University, USA
{jiadeng, wdong, rsocher, jial, li, feifeili}@cs.princeton.edu

IEEE Computer Vision and Pattern Recognition (CVPR), 2009

mammal → placental → carnivore → canine → dog → working dog → husky

vehicle → craft → watercraft → sailing vessel → sailboat → trimaran

ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky
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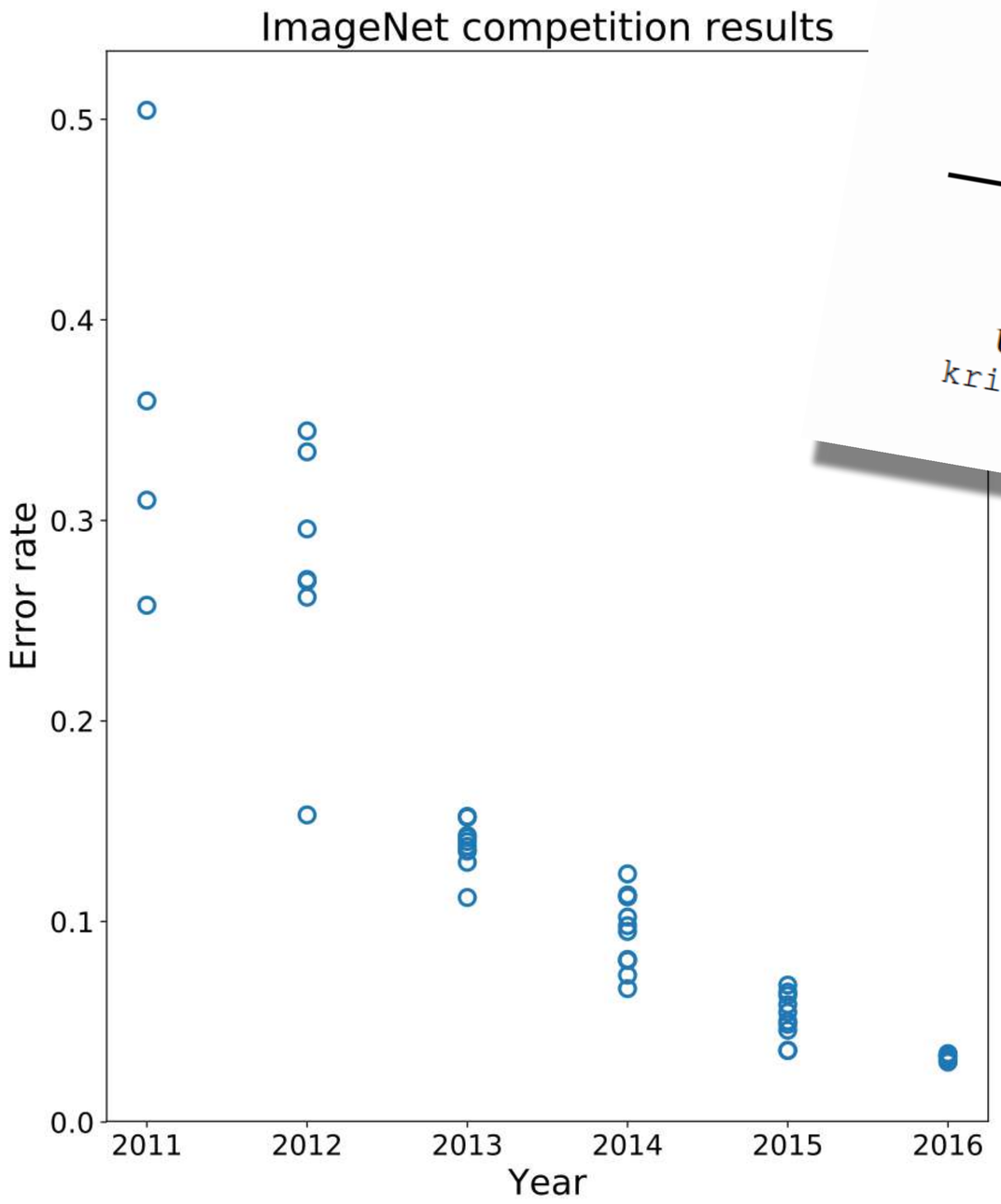
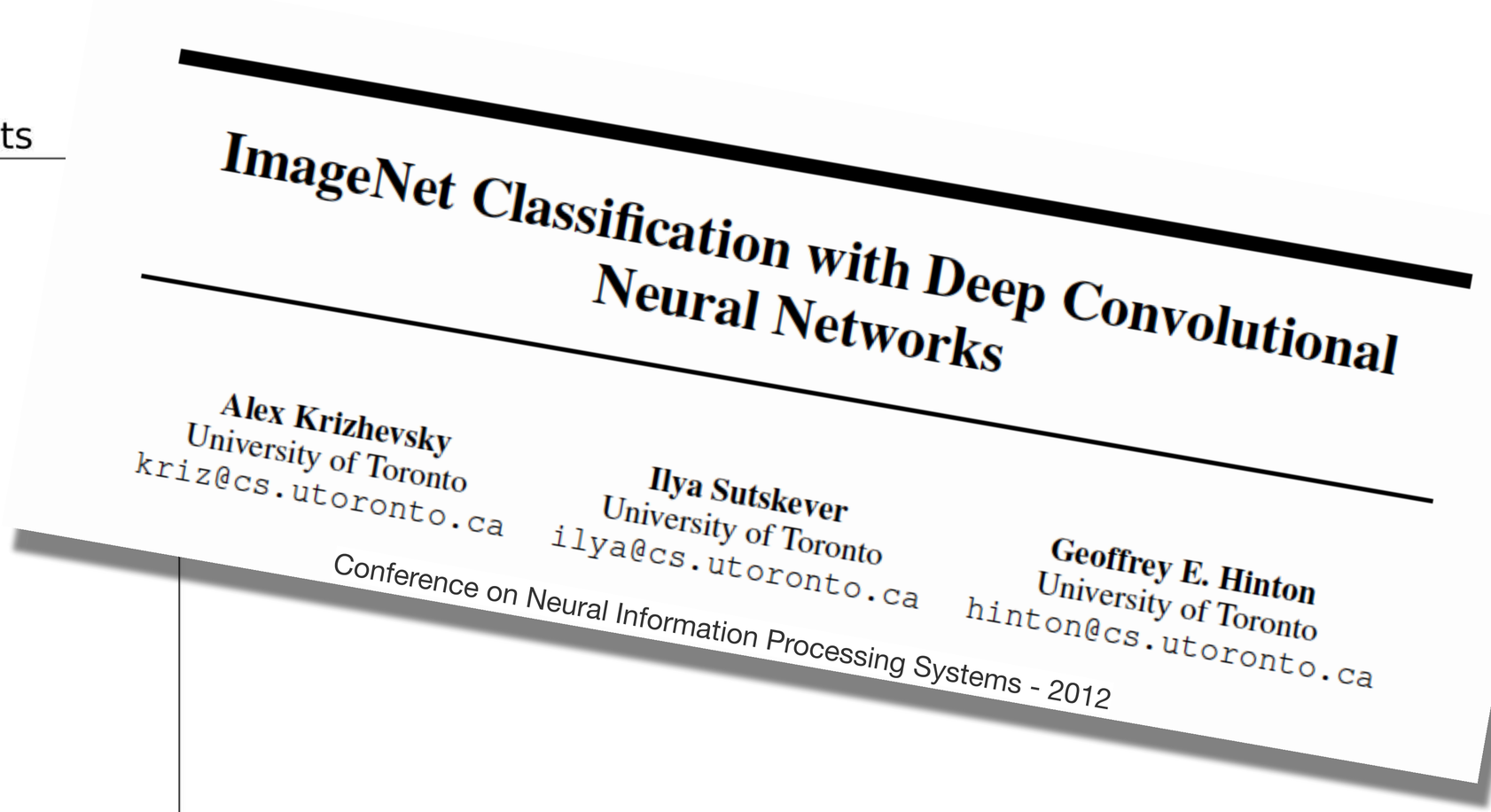
Ilya Sutskever
University of Toronto
ilya@cs.utoronto.ca

Geoffrey E. Hinton
University of Toronto
hinton@cs.utoronto.ca

Conference on Neural Information Processing Systems - 2012





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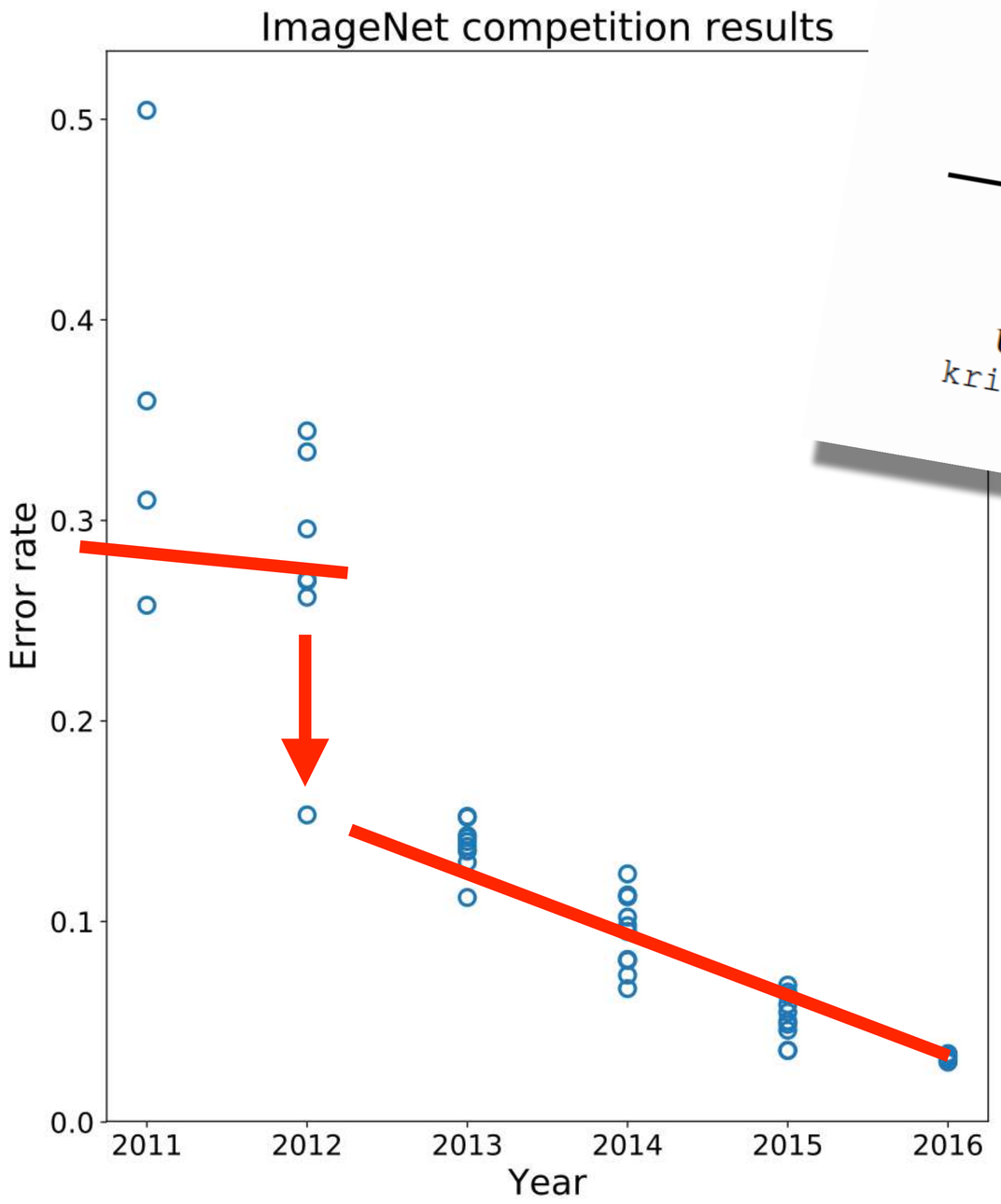

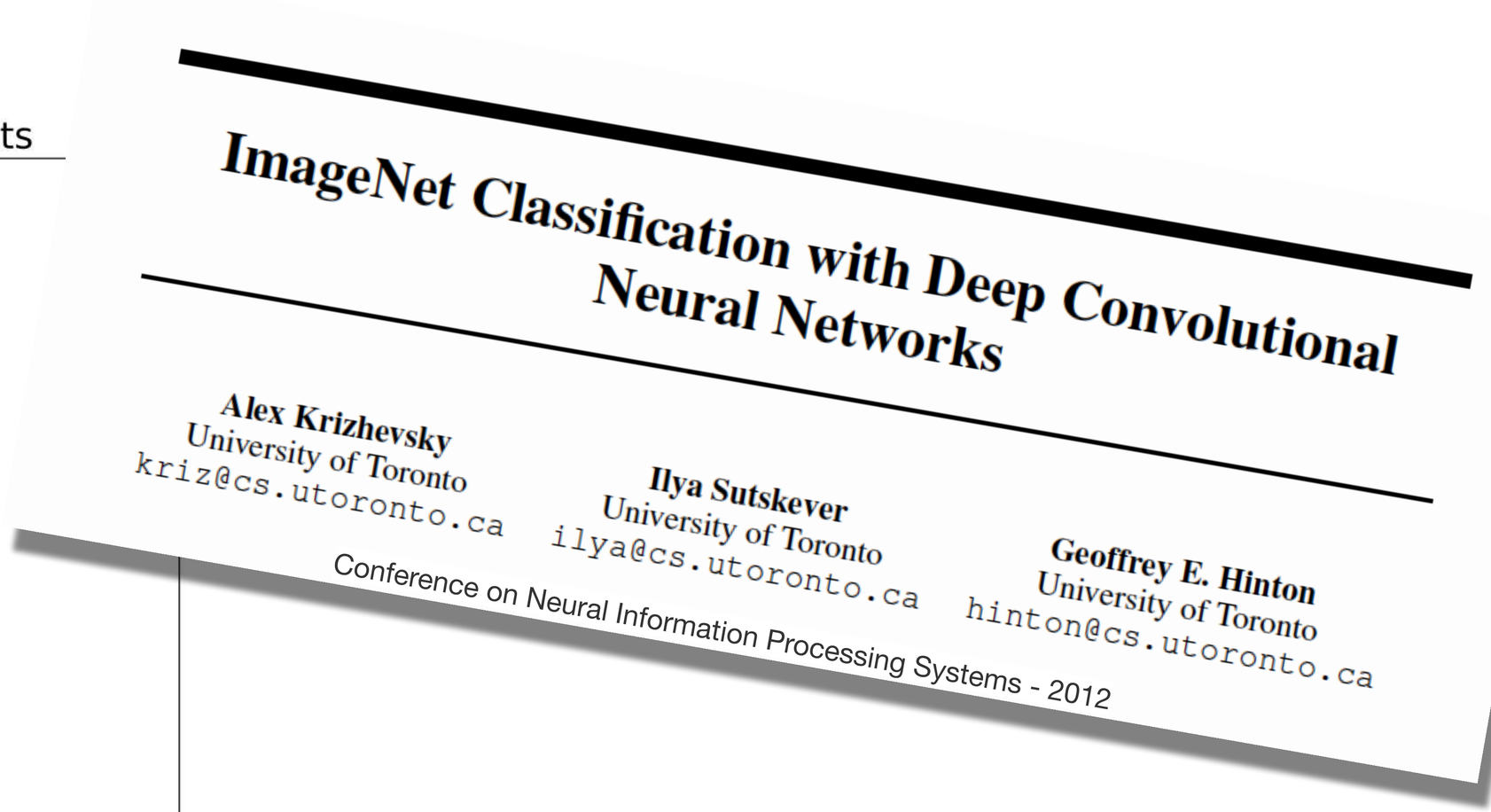

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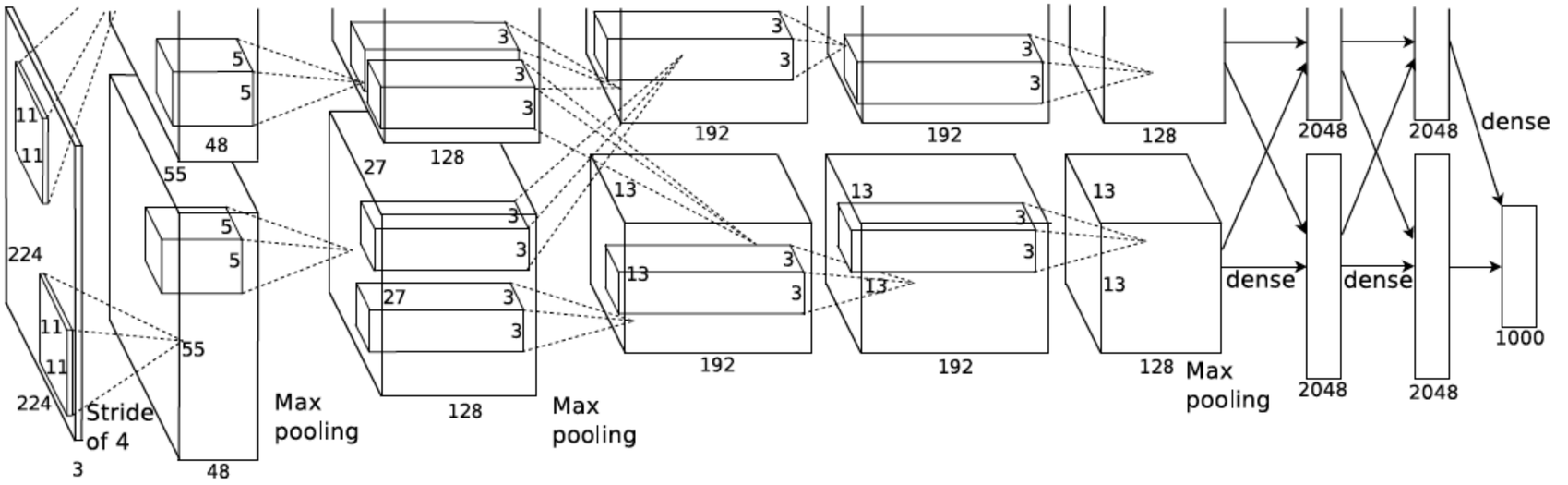
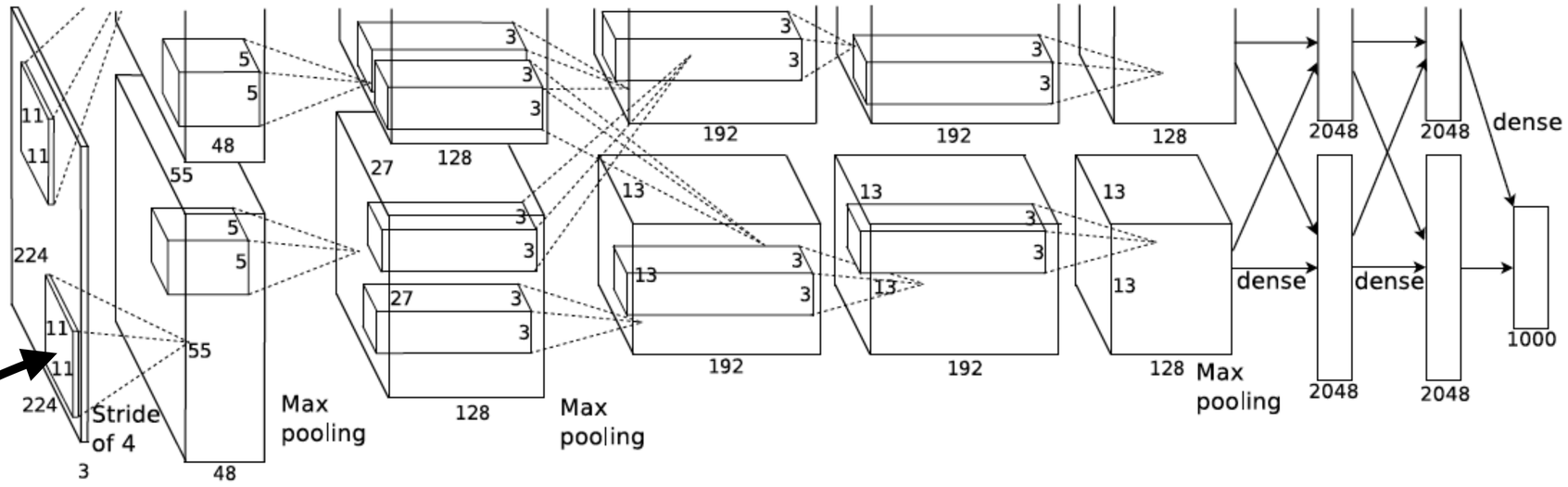
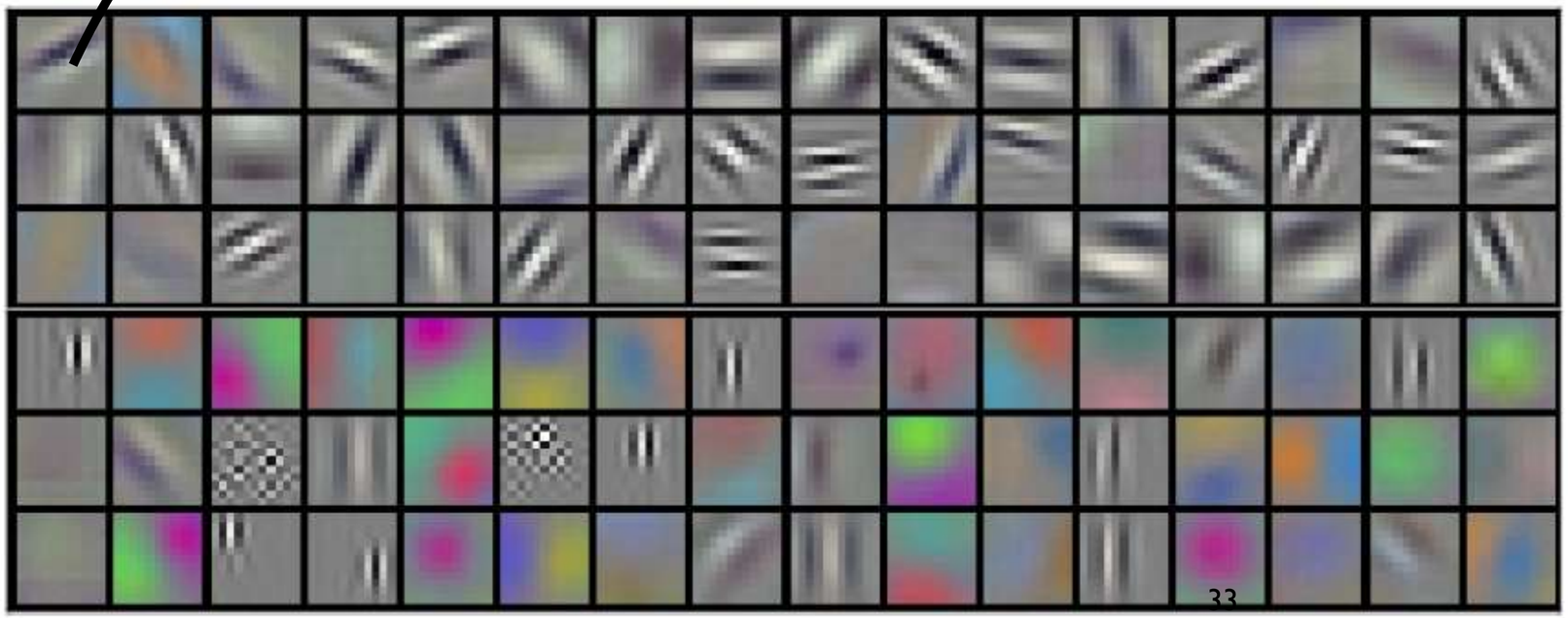
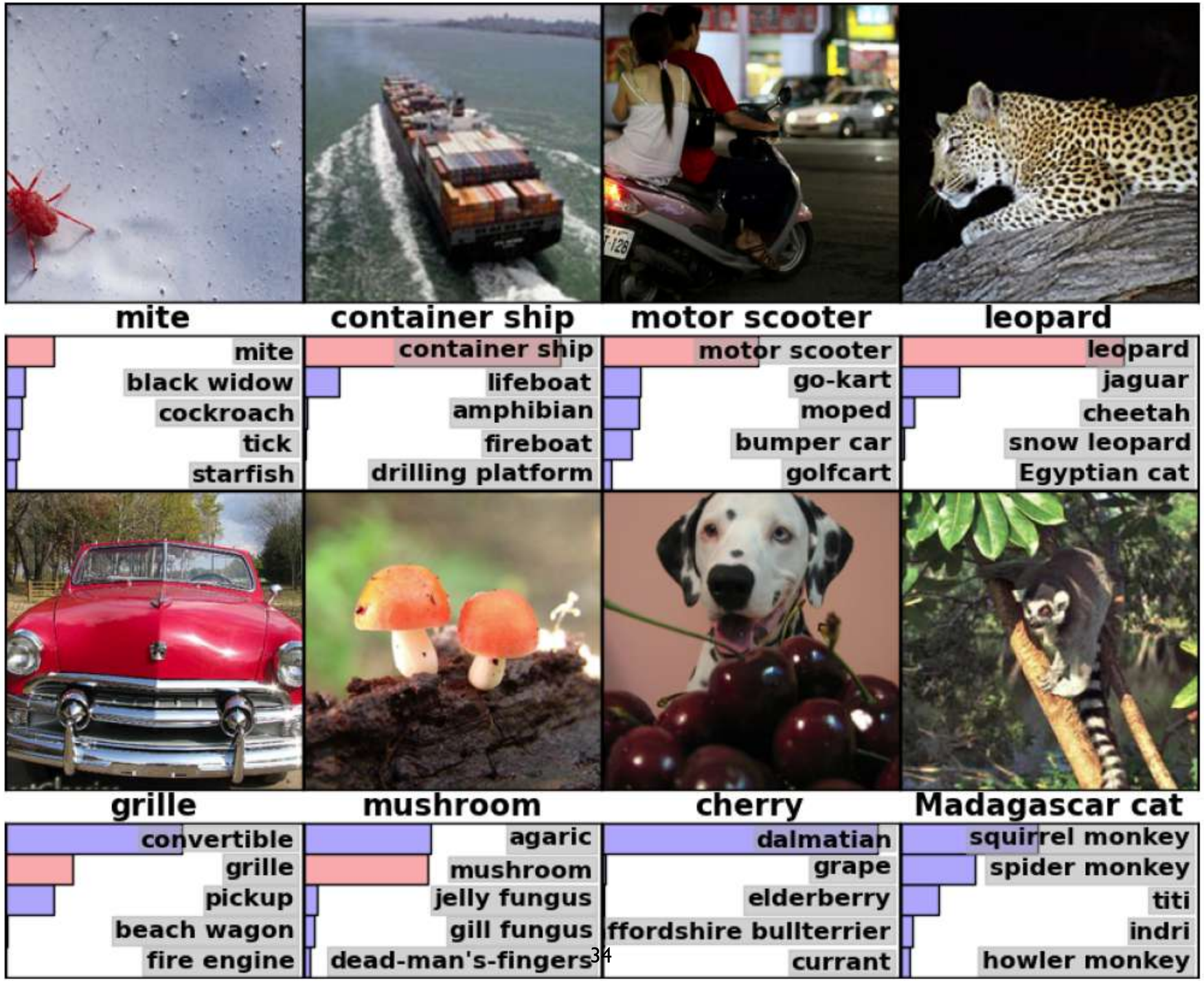
hinton@cs.utoronto.ca

Conference on Neural Information Processing Systems - 2012

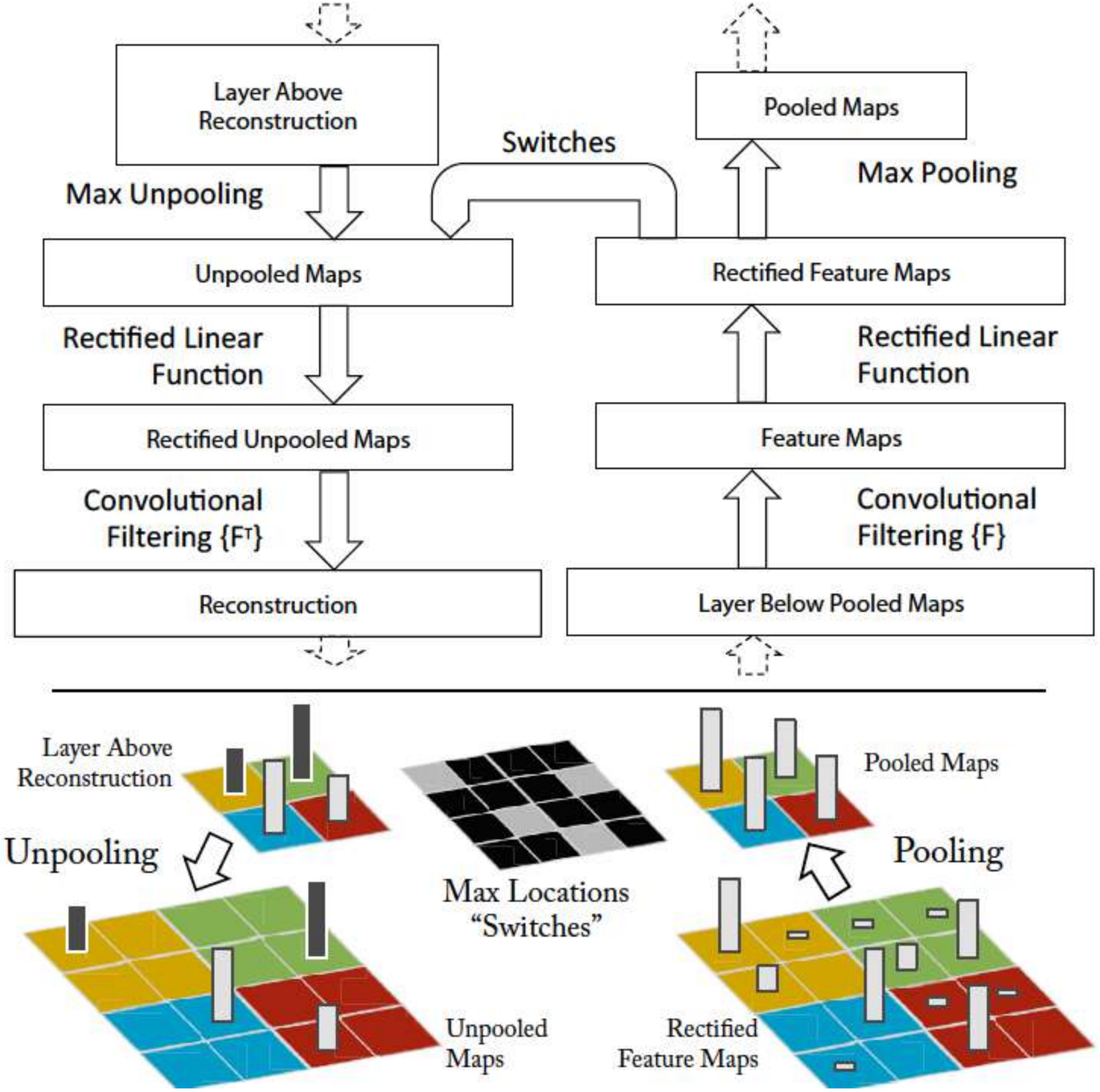

224
224
3
11
11
11
11
Stride
of 4
55
55
48
48
5
5
5
5
Max
pooling
27
27
128
128
3
3
3
3
3
3
3
3
Max
pooling
13
13
13
13
192
192
3
3
3
3
3
3
192
192
13
13
13
13
128
128
Max
pooling
dense
dense
dense
2048
2048
2048
2048
1000


224
224
3
11
11
11
11
Stride
of 4
55
55
48
48
5
5
5
5
27
27
128
128
3
3
Max
pooling
Max
pooling
13
13
192
192
192
192
128
128
13
13
Max
pooling
dense
dense
dense
2048
2048
2048
2048
1000




mite
mite
black widow
cockroach
tick
starfish
container ship
container ship
lifeboat
amphibian
fireboat
drilling platform
motor scooter
motor scooter
go-kart
moped
bumper car
golfcart
leopard
leopard
jaguar
cheetah
snow leopard
Egyptian cat
grille
convertible
grille
pickup
beach wagon
fire engine
mushroom
agaric
mushroom
jelly fungus
gill fungus
dead-man's-fingers
cherry
dalmatian
grape
elderberry
ffordshire bullterrier
currant
Madagascar cat
squirrel monkey
spider monkey
titi
indri
howler monkey

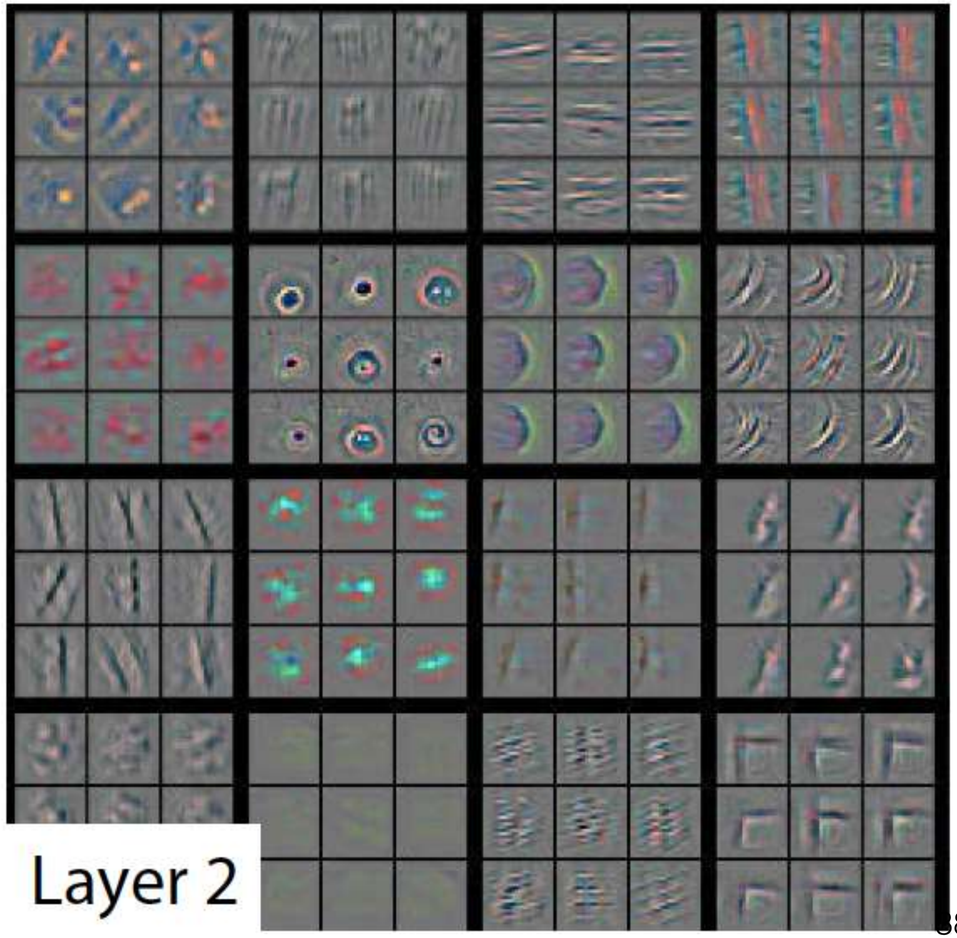
What do the units code for?



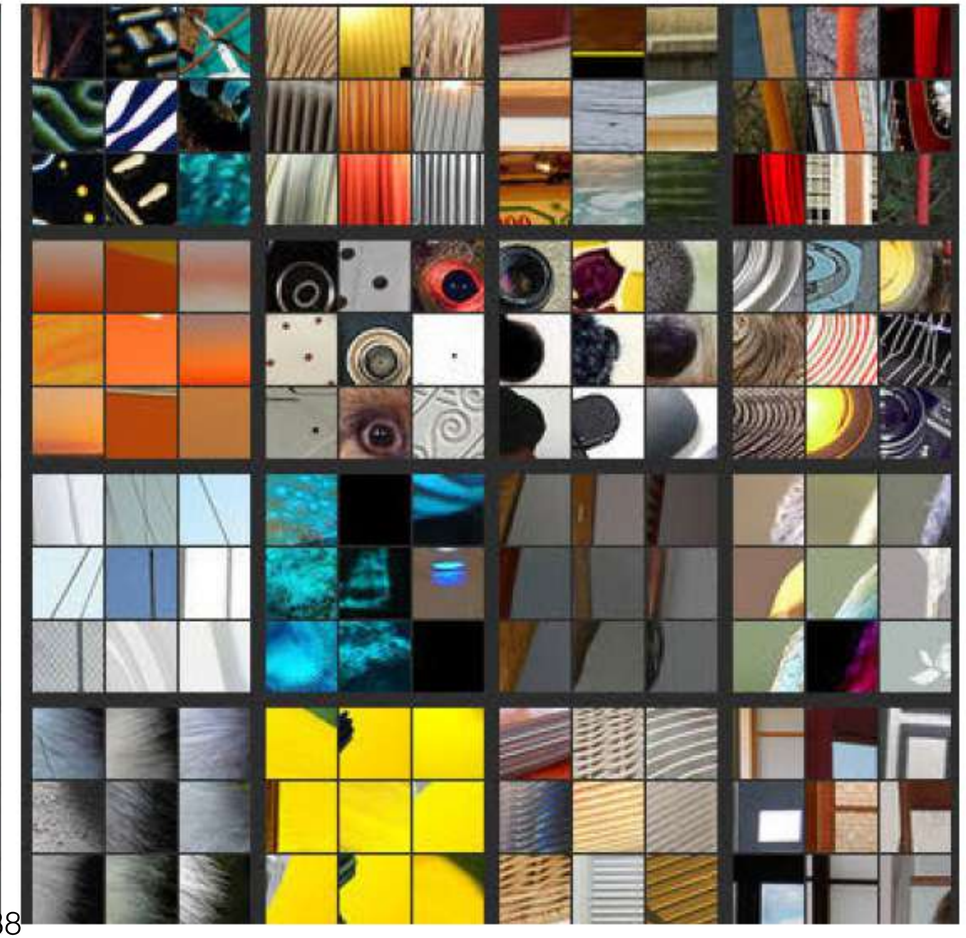
[Zeiler and Fegus, '14]

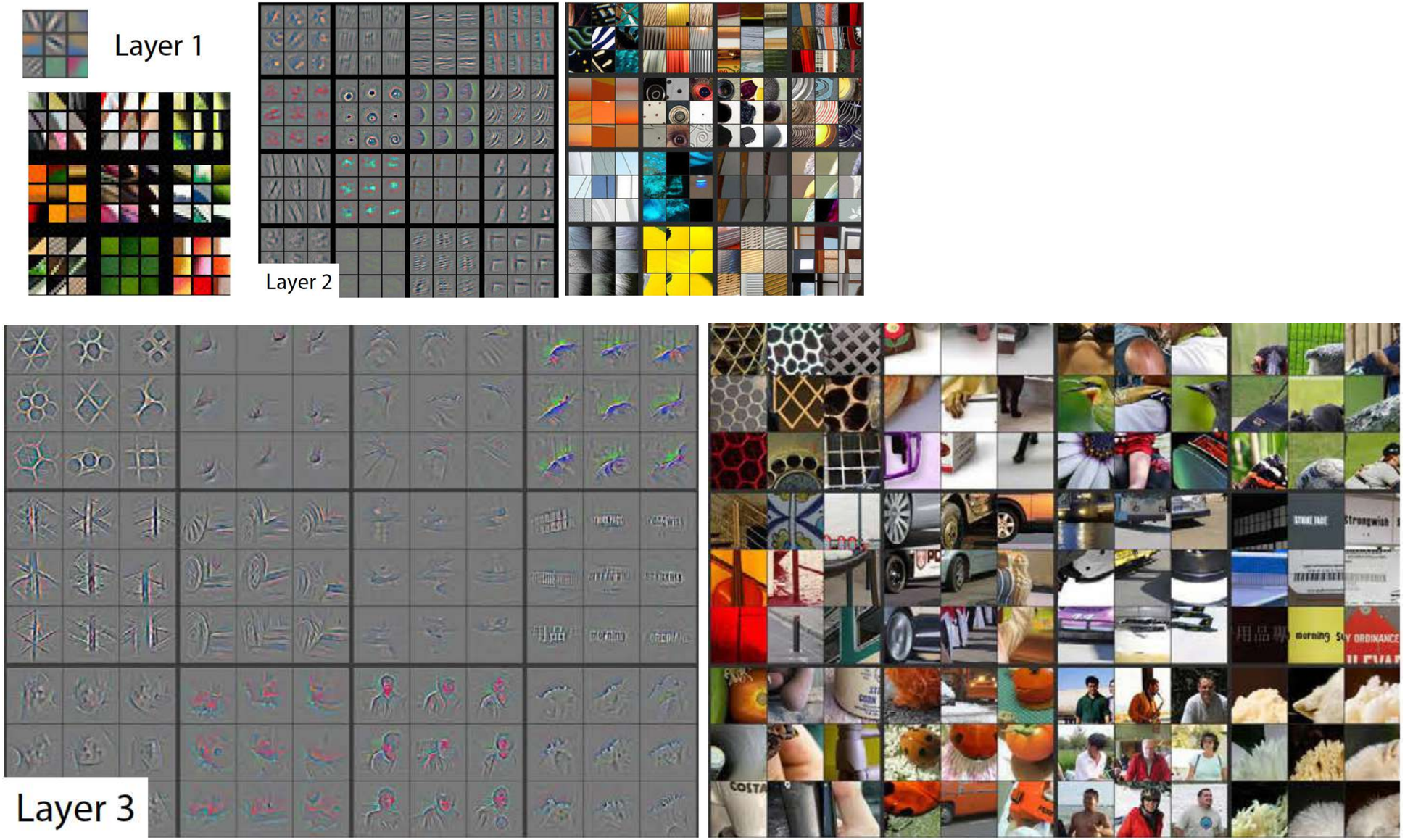
Layer 1

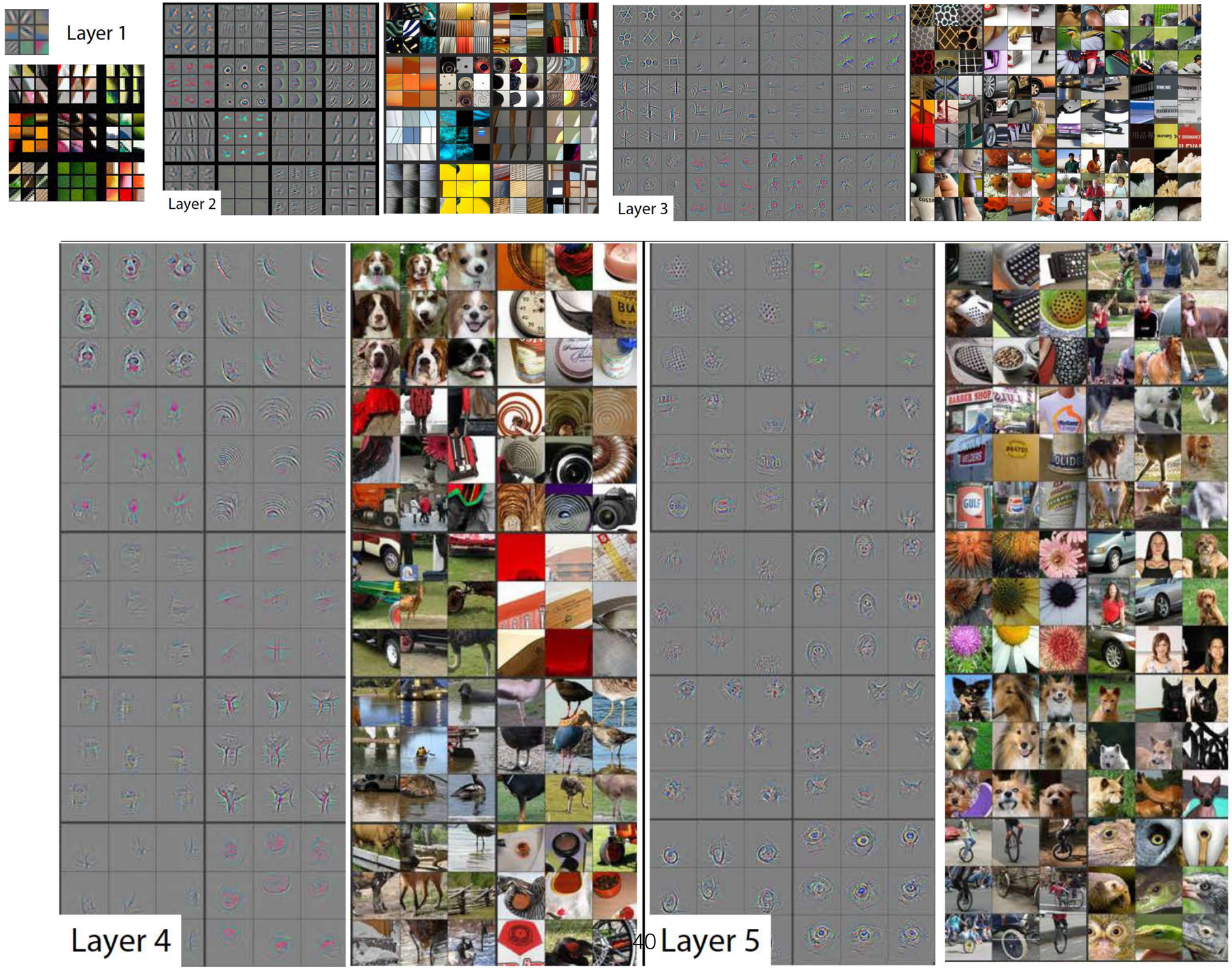
Layer 1



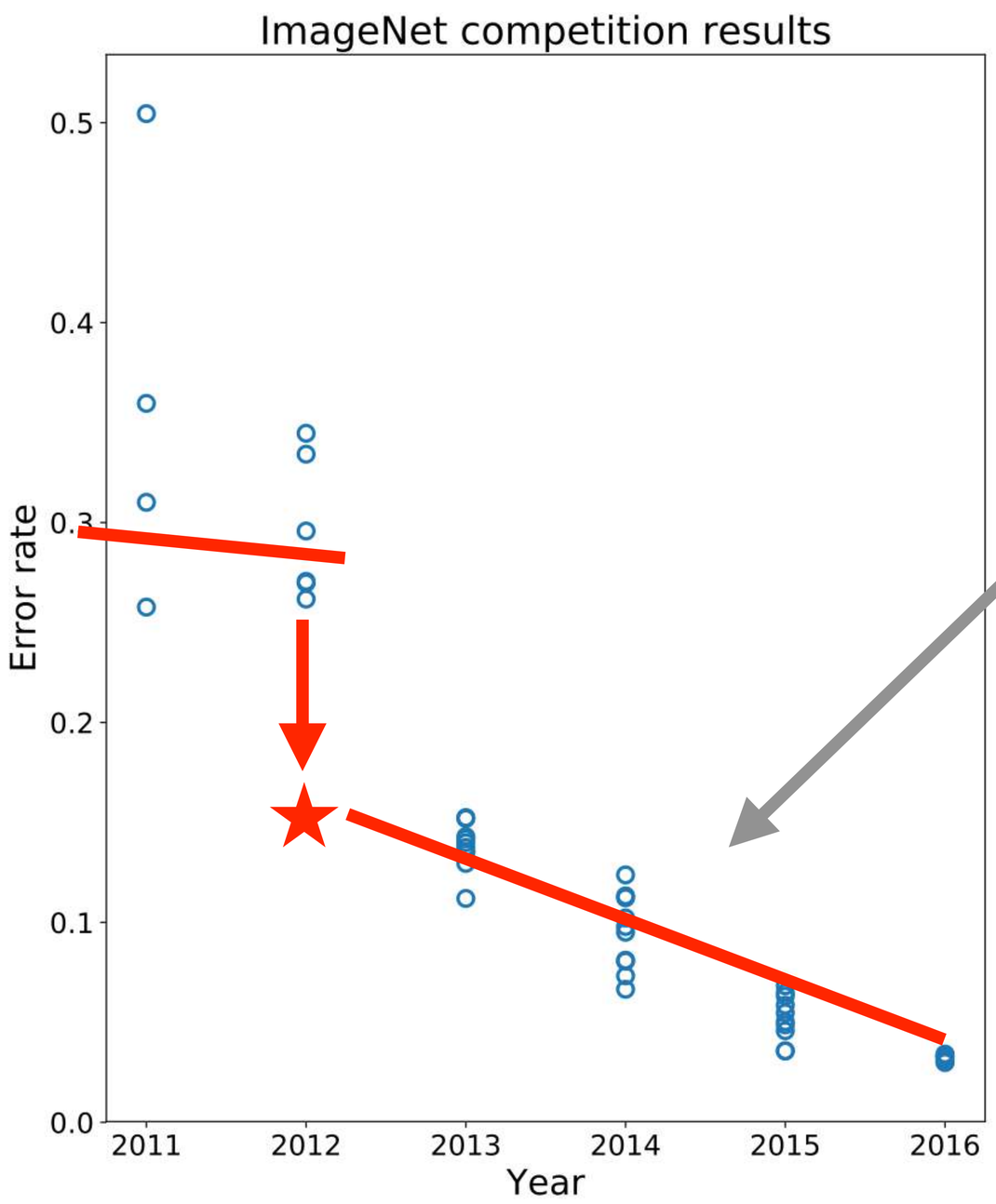
Layer 2

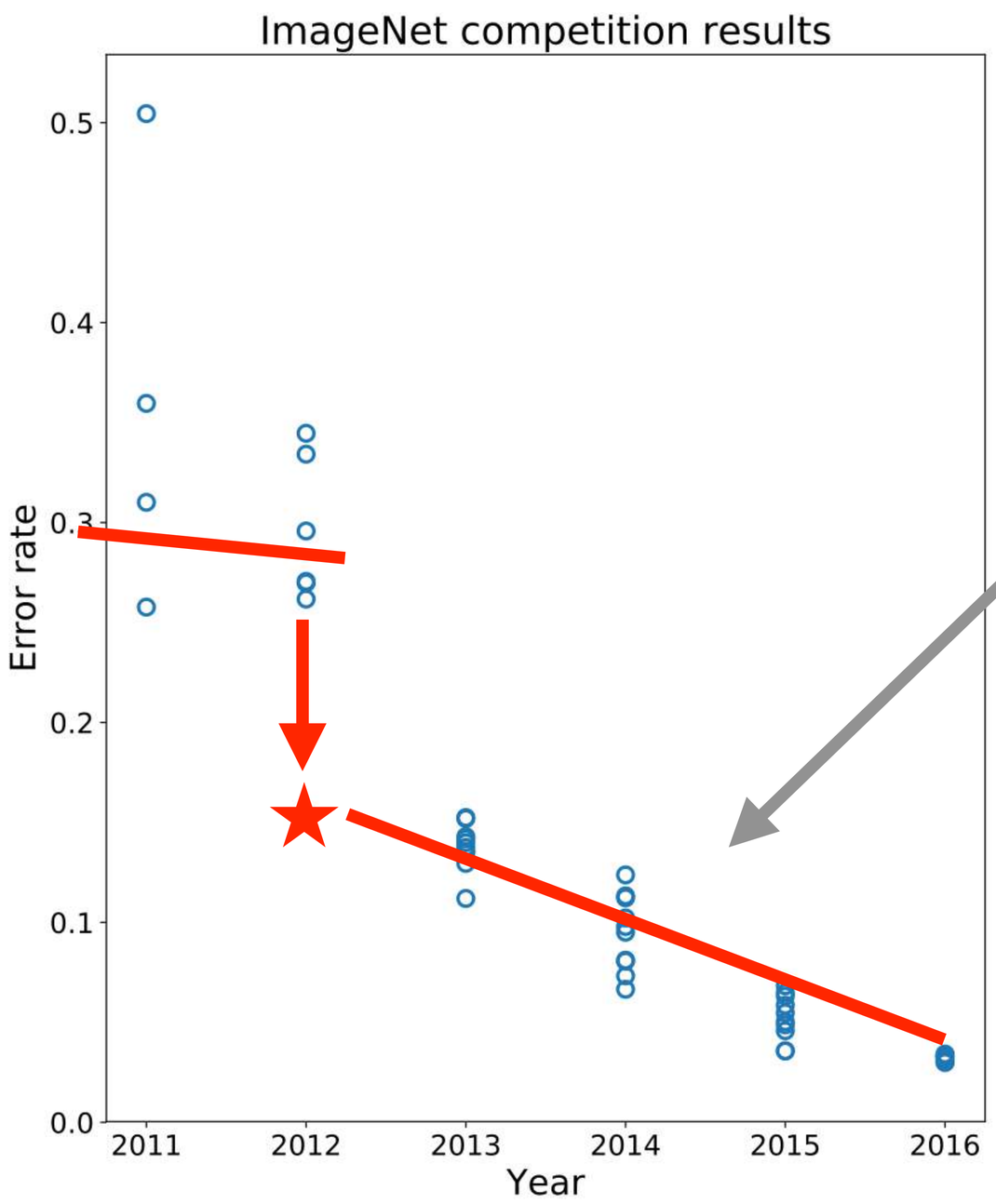



Layer 1
Layer 2
Layer 3


Layer 1
Layer 2
Layer 3
Layer 4
Layer 5

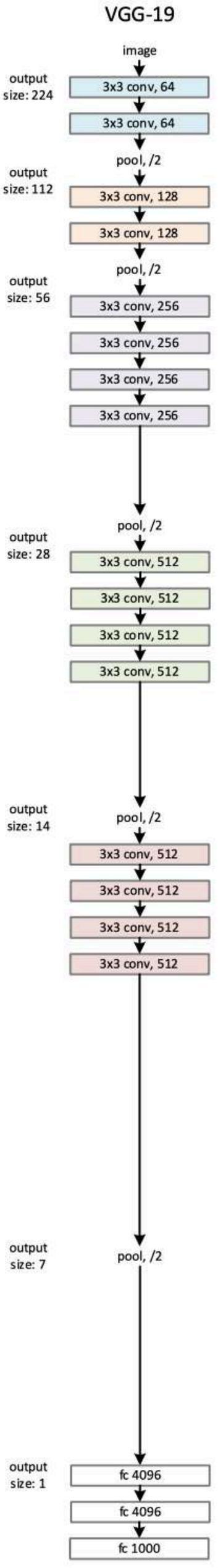
Next ideas?

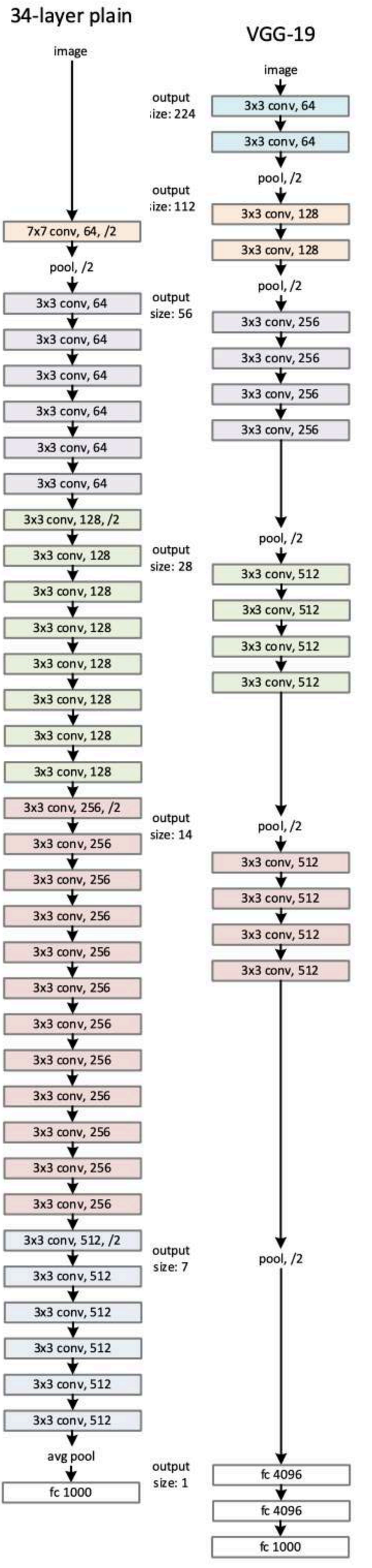

ImageNet competition results
Error rate
0.0
0.1
0.2
0.3
0.4
0.5
2011
2012
2013
2014
2015
2016
Year

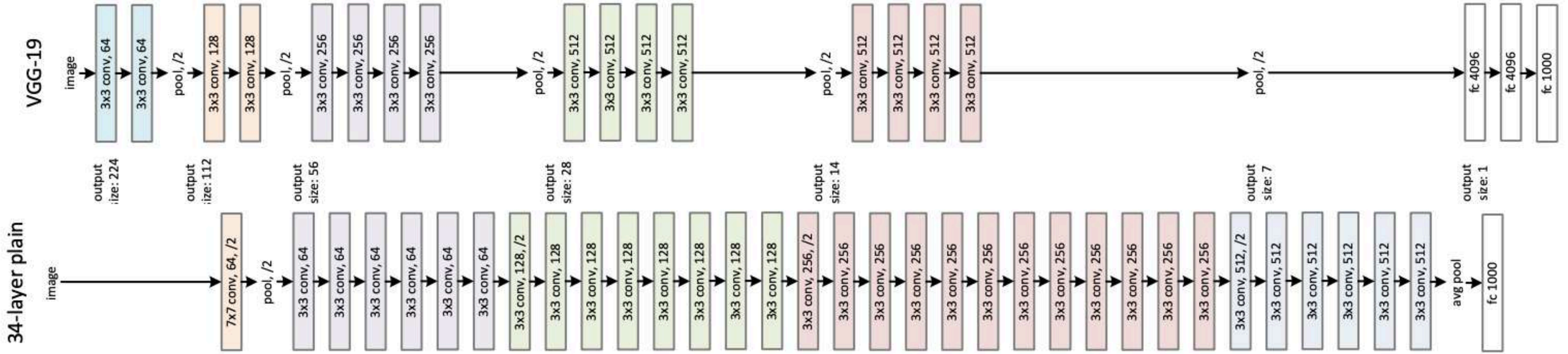


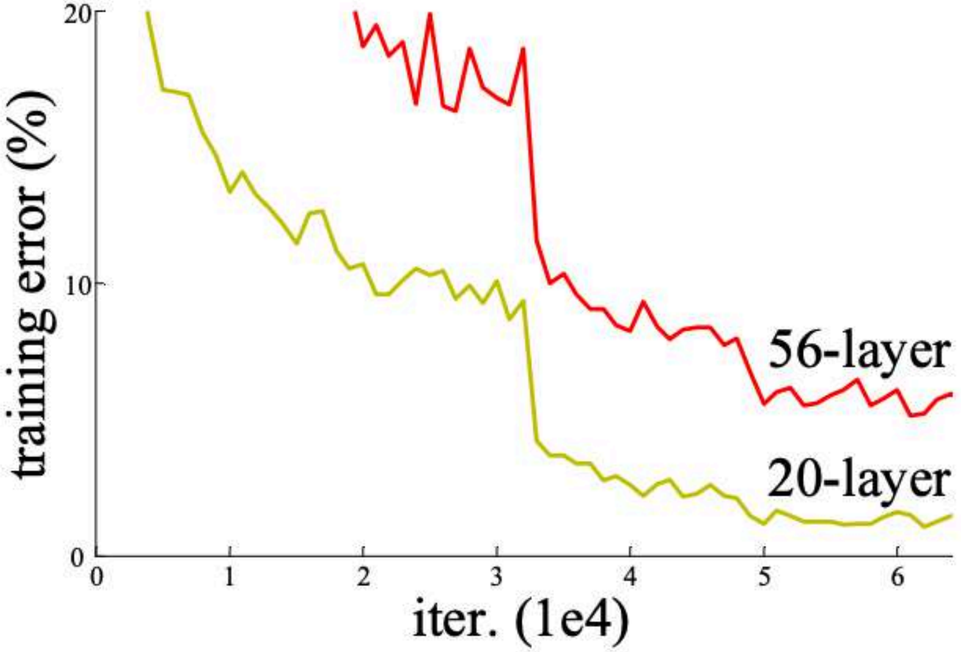
Next ideas?

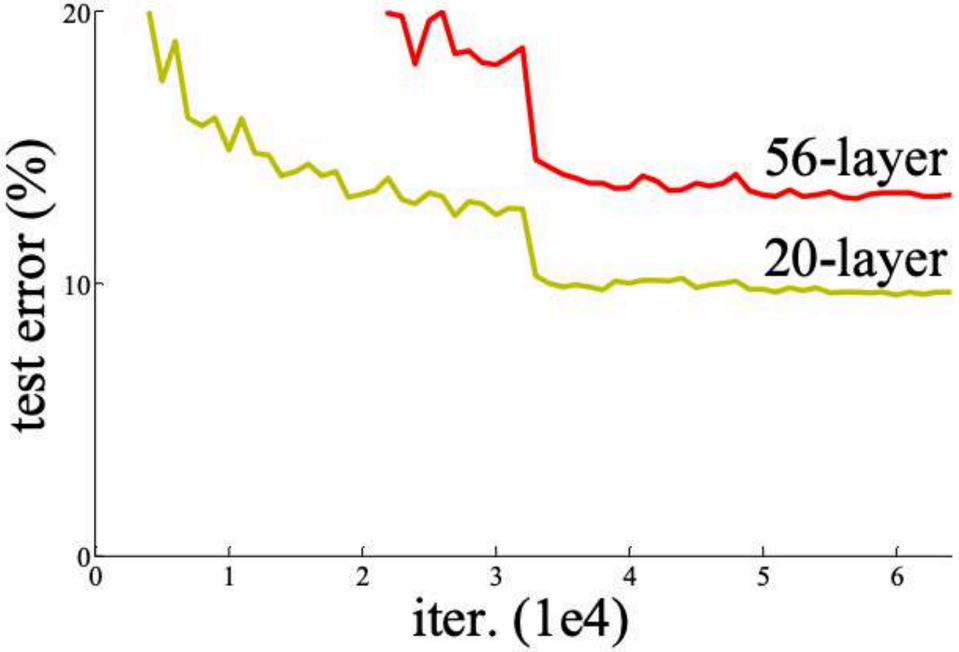
More parameters?
More layers?


VGG-19
image
output size: 224
3x3 conv, 64
3x3 conv, 64
pool, /2
output size: 112
3x3 conv, 128
3x3 conv, 128
pool, /2
output size: 56
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
pool, /2
output size: 28
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
pool, /2
output size: 14
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
pool, /2
output size: 7
fc 4096
output size: 1
fc 4096
fc 1000


34-layer plain
VGG-19
image
7x7 conv, 64, /2
pool, /2
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 256, /2
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 512, /2
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
avg pool
fc 1000
output size: 224
output size: 112
output size: 56
output size: 28
output size: 14
output size: 7
output size: 1
image
3x3 conv, 64
3x3 conv, 64
pool, /2
3x3 conv, 128
3x3 conv, 128
pool, /2
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
pool, /2
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
pool, /2
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
pool, /2
fc 4096
fc 4096
fc 1000

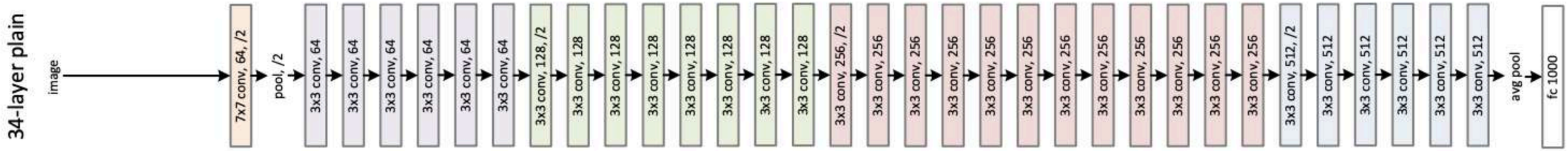

VGG-19
image
3x3 conv, 64
3x3 conv, 64
pool, /2
3x3 conv, 128
3x3 conv, 128
pool, /2
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
pool, /2
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
pool, /2
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
pool, /2
fc 4096
fc 4096
fc 1000
output size: 224
output size: 112
output size: 56
output size: 28
output size: 14
output size: 7
output size: 1
34-layer plain
image
7x7 conv, 64, /2
pool, /2
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128
3x3 conv, 256, /2
3x3 conv, 256
3x3 conv, 512, /2
3x3 conv, 512
avg pool
fc 1000


training error (%)
iter. (1e4)
56-layer
20-layer
0
10
20
0
1
2
3
4
5
6


test error (%)
iter. (1e4)
56-layer
20-layer
0
10
20
0
1
2
3
4
5
6

Wait a minute…

Wait a minute…



Wait a minute…

Wait a minute…

Re-learning the identity each time?

Proceedings of the IEEE conference on computer vision and pattern recognition

Deep Residual Learning for Image Recognition

Kaiming He Xiangyu Zhang Shaoqing Ren Jian Sun
Microsoft Research
{kahe, v-xiangz, v-shren, jiansun}@microsoft.com

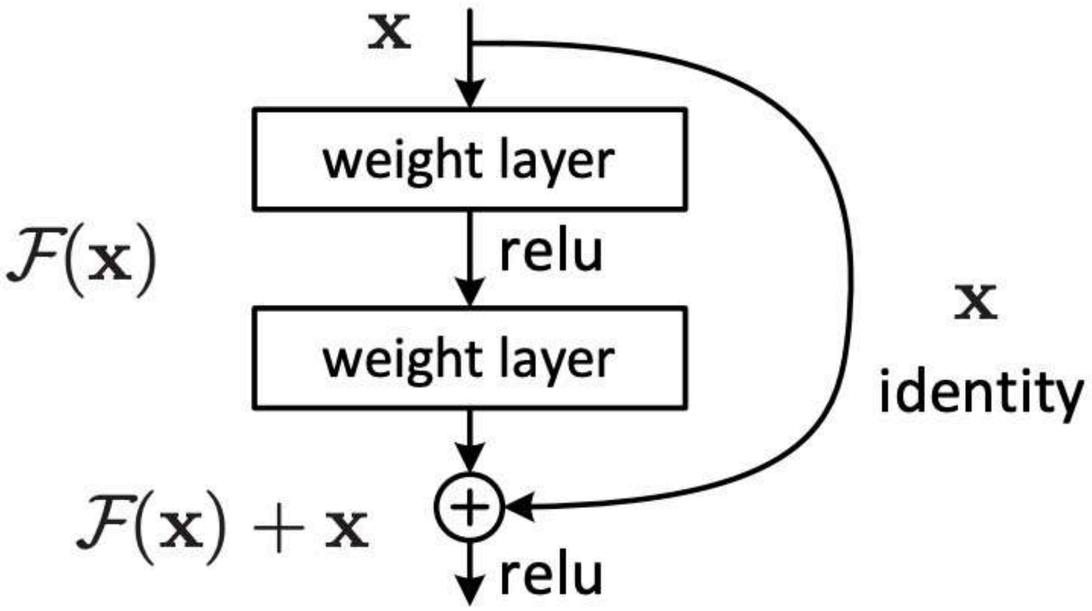
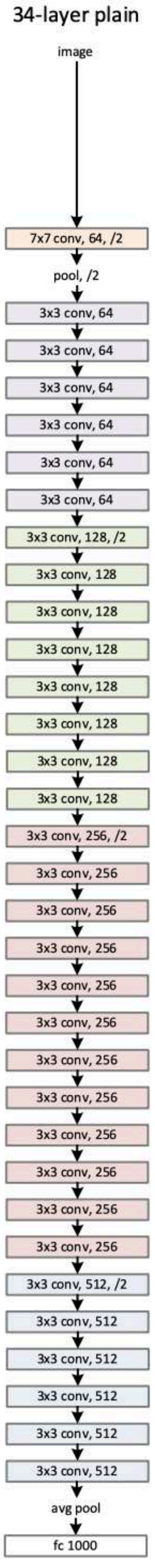
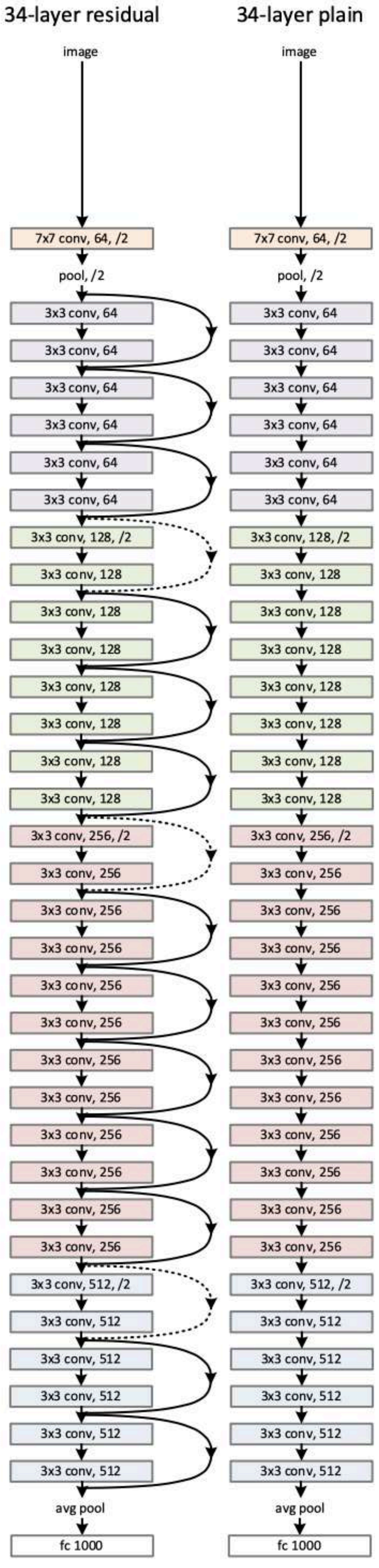
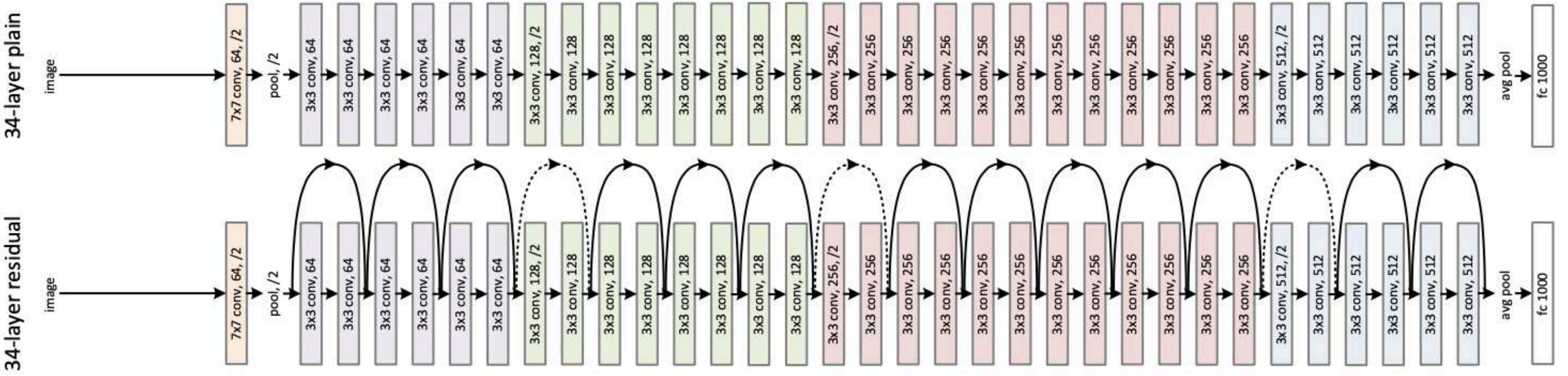
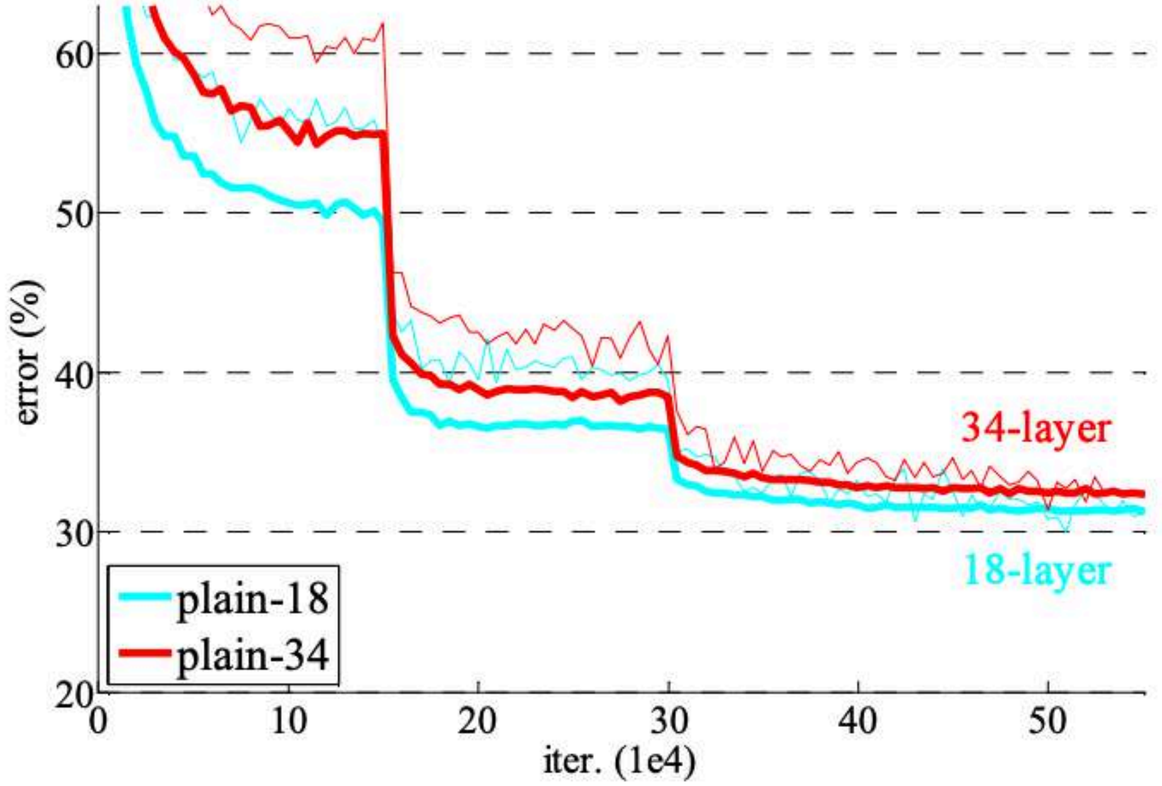


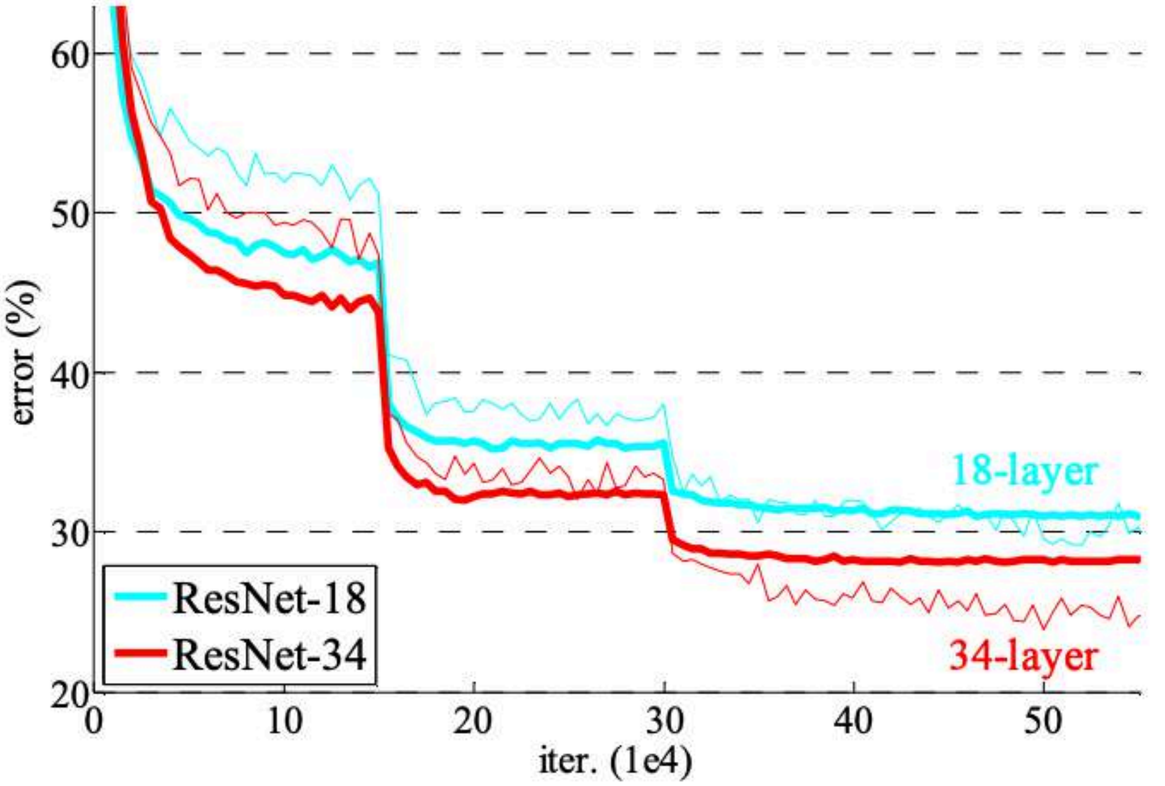
Figure 2. Residual learning: a building block.


34-layer plain
image
7x7 conv, 64, /2
pool, /2
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 128
3x3 conv, 256, /2
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 256
3x3 conv, 512, /2
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
3x3 conv, 512
avg pool
fc 1000


34-layer residual
34-layer plain
image
7x7 conv, 64, /2
pool, /2
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128
3x3 conv, 256, /2
3x3 conv, 256
3x3 conv, 512, /2
3x3 conv, 512
avg pool
fc 1000


34-layer plain
image
7x7 conv, 64, /2
pool, /2
3x3 conv, 64
3x3 conv, 128, /2
3x3 conv, 128
3x3 conv, 256, /2
3x3 conv, 256
3x3 conv, 512, /2
3x3 conv, 512
avg pool
fc 1000
34-layer residual


error (%)
iter. (1e4)
34-layer
18-layer
plain-18
plain-34


error (%)
iter. (1e4)
18-layer
34-layer
ResNet-18
ResNet-34

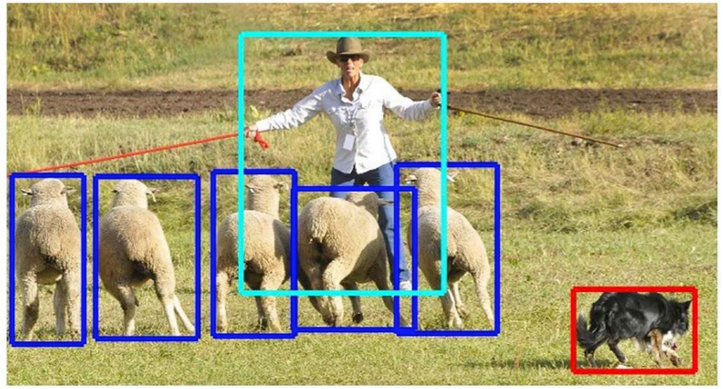
Semantic segmentation

Microsoft COCO: Common Objects in Context

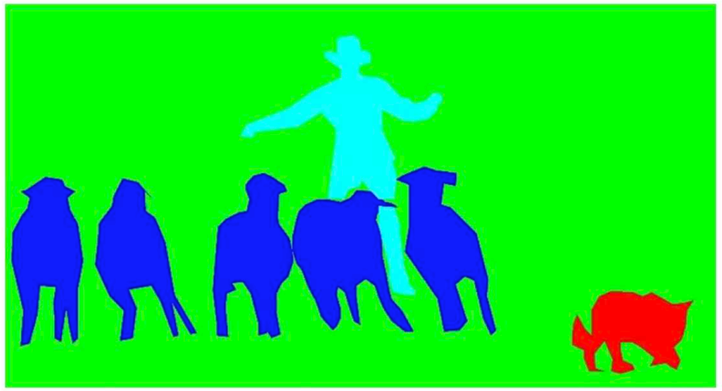
Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, Piotr Dollár

arXiv:1405.0312v3 [cs.CV] 21 Feb 2015

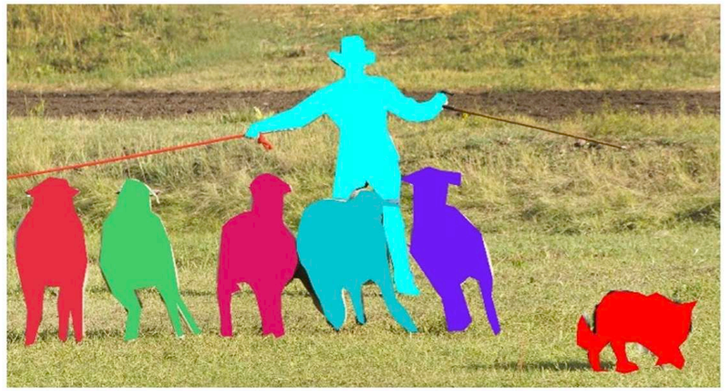
(a) Image classification



(b) Object localization



(c) Semantic segmentation



(d) This work

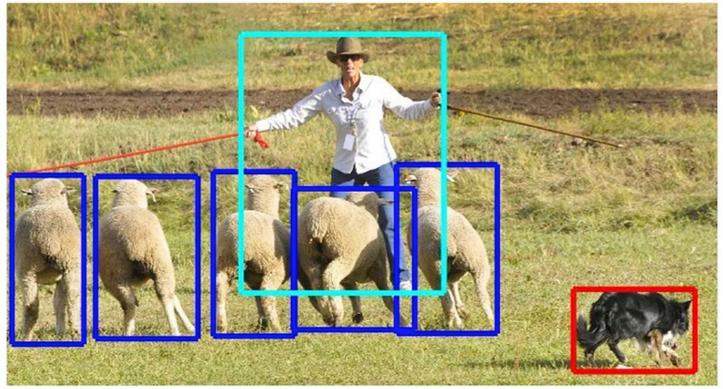
Semantic segmentation

Microsoft COCO: Common Objects in Context

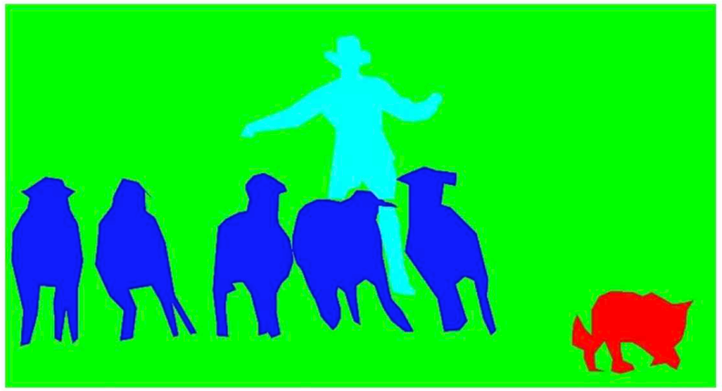
Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, Piotr Dollár

arXiv:1405.0312v3 [cs.CV] 21 Feb 2015

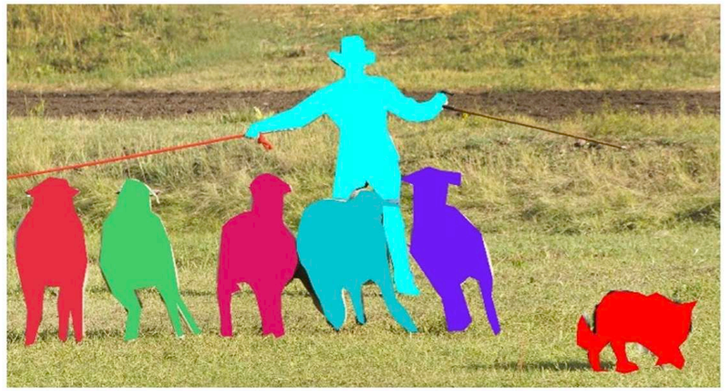
(a) Image classification



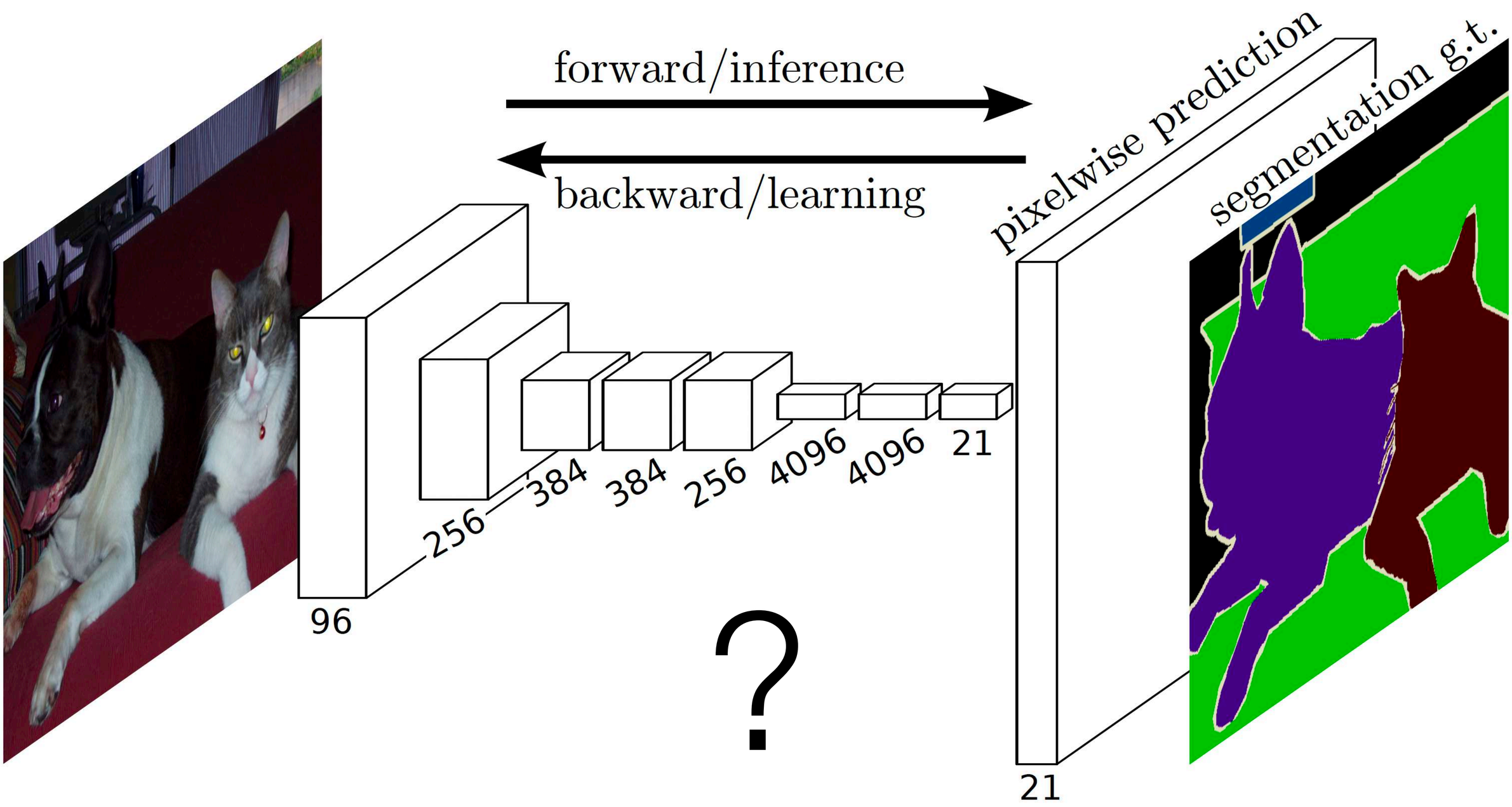
(b) Object localization



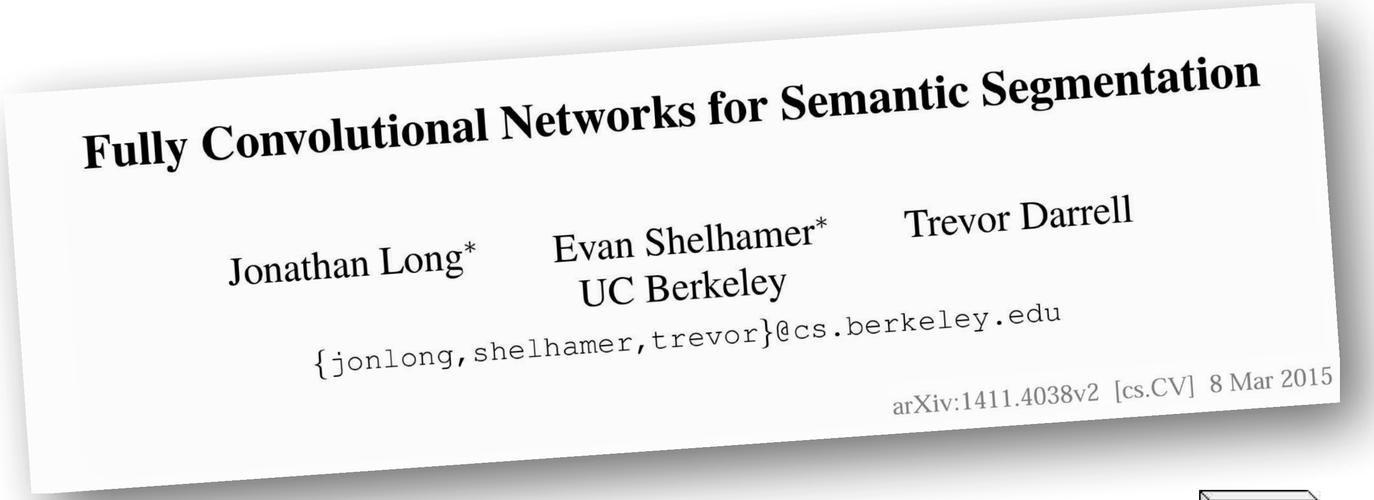
(c) Semantic segmentation



(d) This work

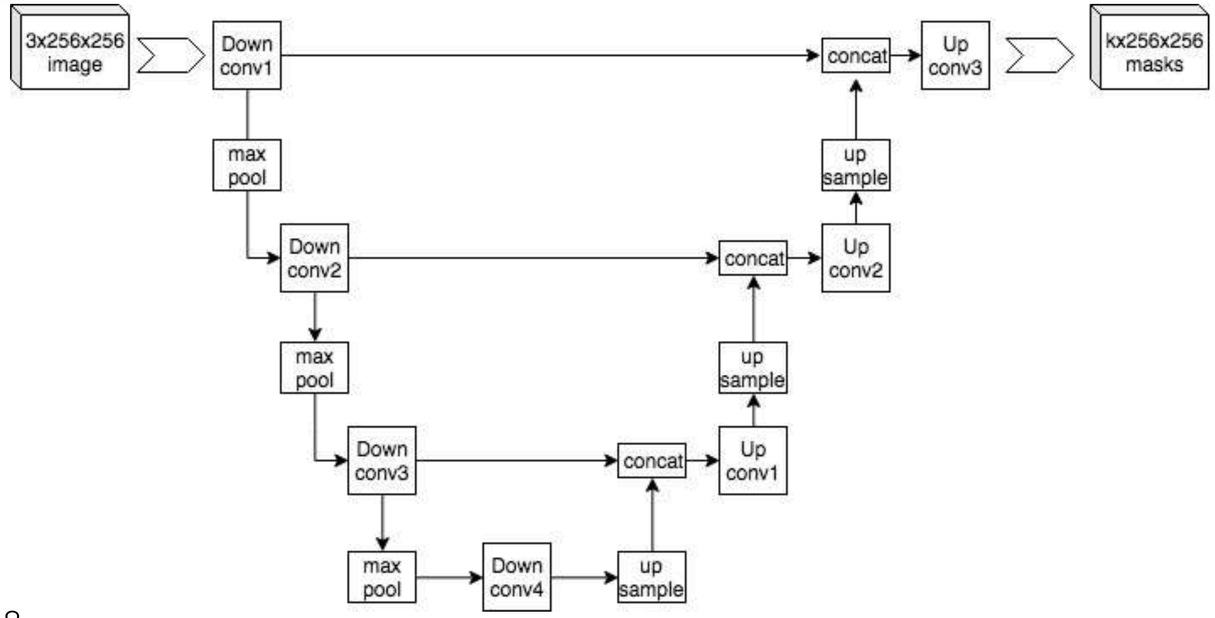

forward/inference
backward/learning
pixelwise prediction
segmentation g.t.
96
256
384
384
256
4096
4096
21
?
21

U-net

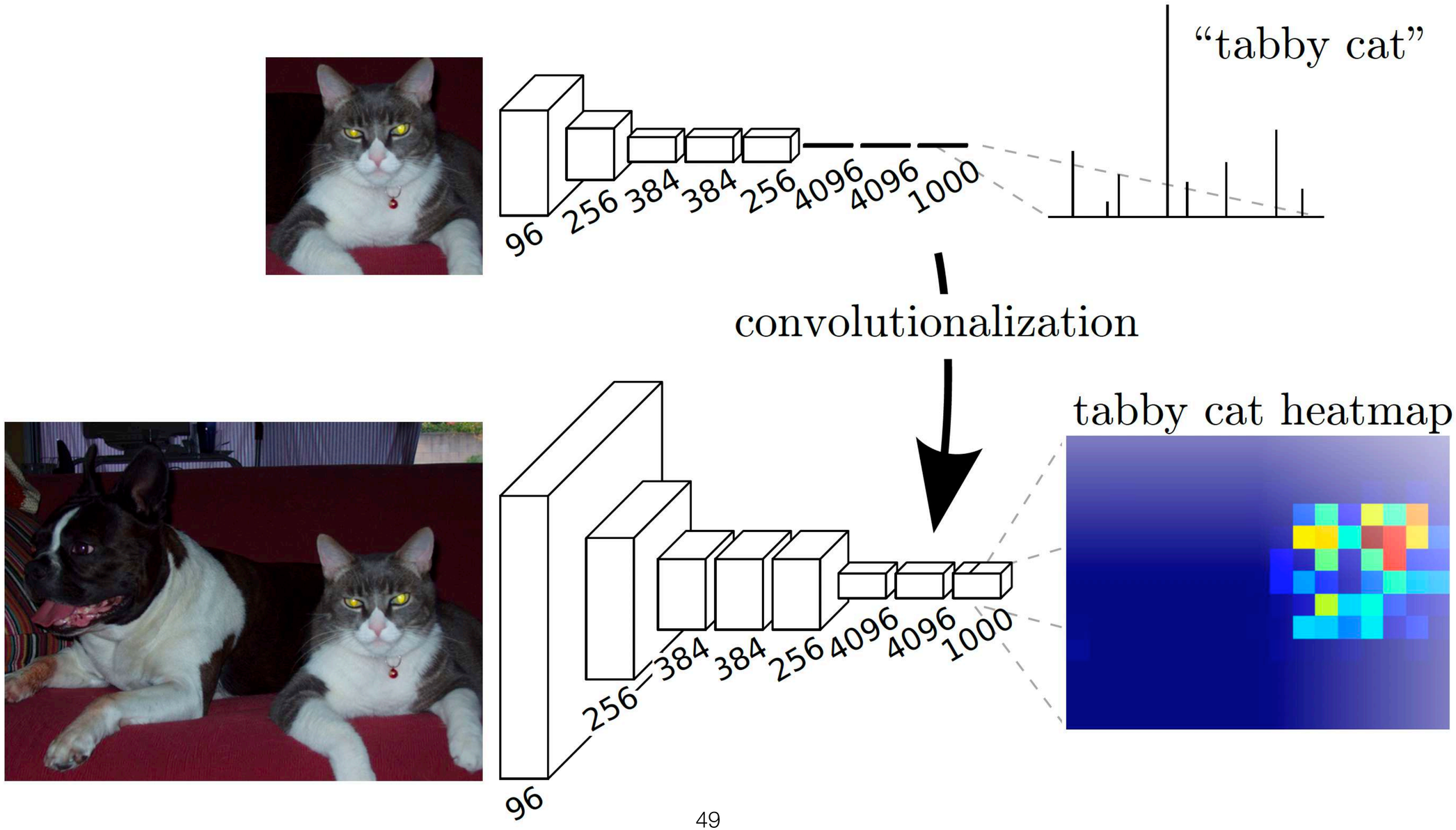

Fully Convolutional Networks for Semantic Segmentation

Jonathan Long* Evan Shelhamer* Trevor Darrell
UC Berkeley
{jonlong,shelhamer,trevor}@cs.berkeley.edu

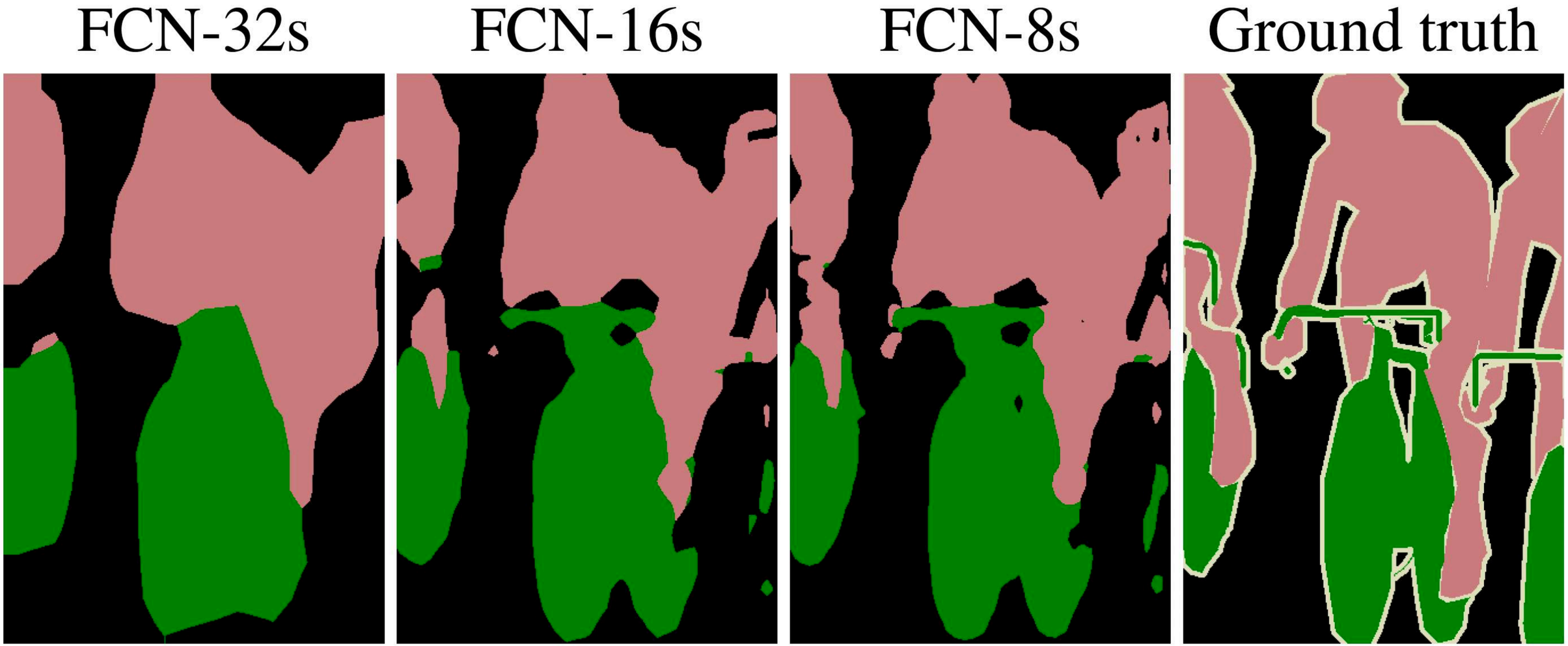
arXiv:1411.4038v2 [cs.CV] 8 Mar 2015



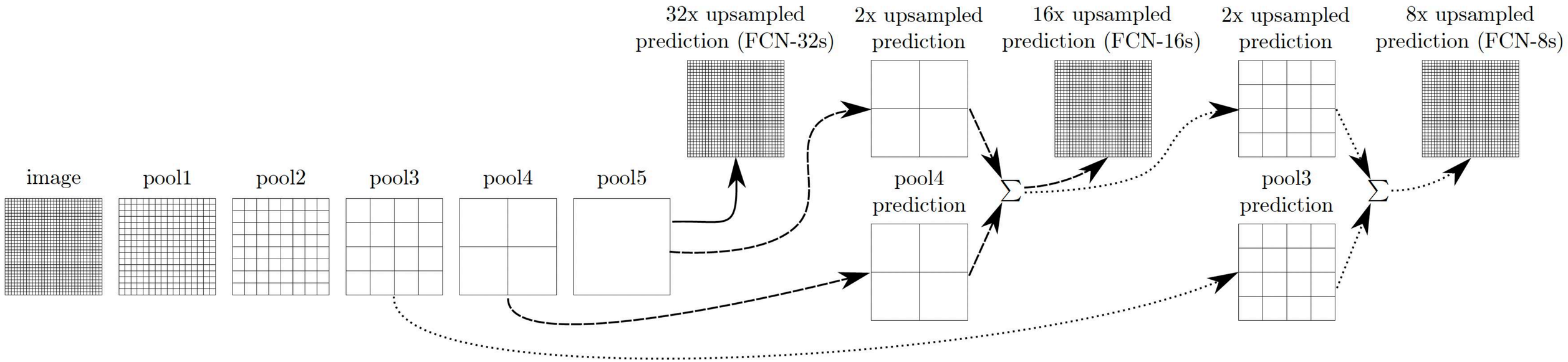
Idea 1: pixel-level classification

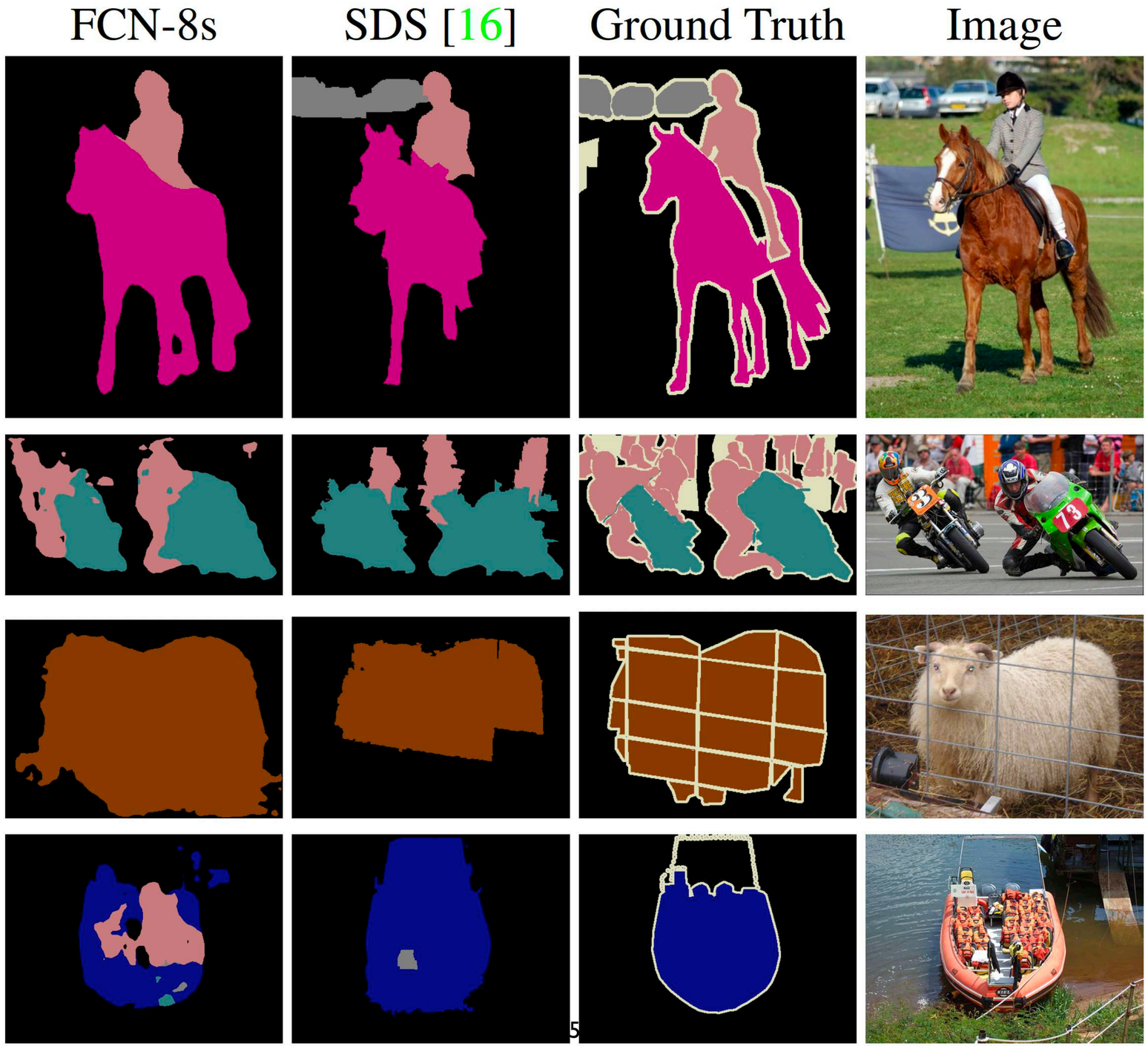


Idea 2: multiscale



Combining scales




FCN-8s
SDS [16]
Ground Truth
Image

Summary

- Fully connected networks FCNs
- Estimating n of parameters
- Convolutional neural networks CNNs
- CNN-FCN for image classification
- resNet
- Semantic segmentation
- U-net